Use Case No:13

Jenkins Availability Monitoring and Alerting on AWS EC2

Problem Statement:

Ensure Jenkins service running on an EC2 instance is continuously monitored for downtime and restarts, generate reports for audit, and trigger automated alerts using AWS services.

Trigger Condition:

Jenkins service goes DOWN or restarts unexpectedly.

Workflow:

1. **Install Jenkins on EC2:**
   - Deploy Jenkins using package manager and enable systemd service.
2. **Monitor Jenkins Status:**
   - Use a bash script scheduled via cron to check systemctl is-active jenkins.
   - Log events (UP/DOWN) with timestamps in /var/log/jenkins_monitor.log.
3. **Generate Report:**
   - Parse log file to count downtime and uptime occurrences
4. **Configure AWS Alerts:**
   - Enable CloudWatch Agent to monitor Jenkins process or custom metric.
   - Create CloudWatch Alarm for Jenkins DOWN or restart events.
   - Attach SNS Topic for email/SMS notifications.

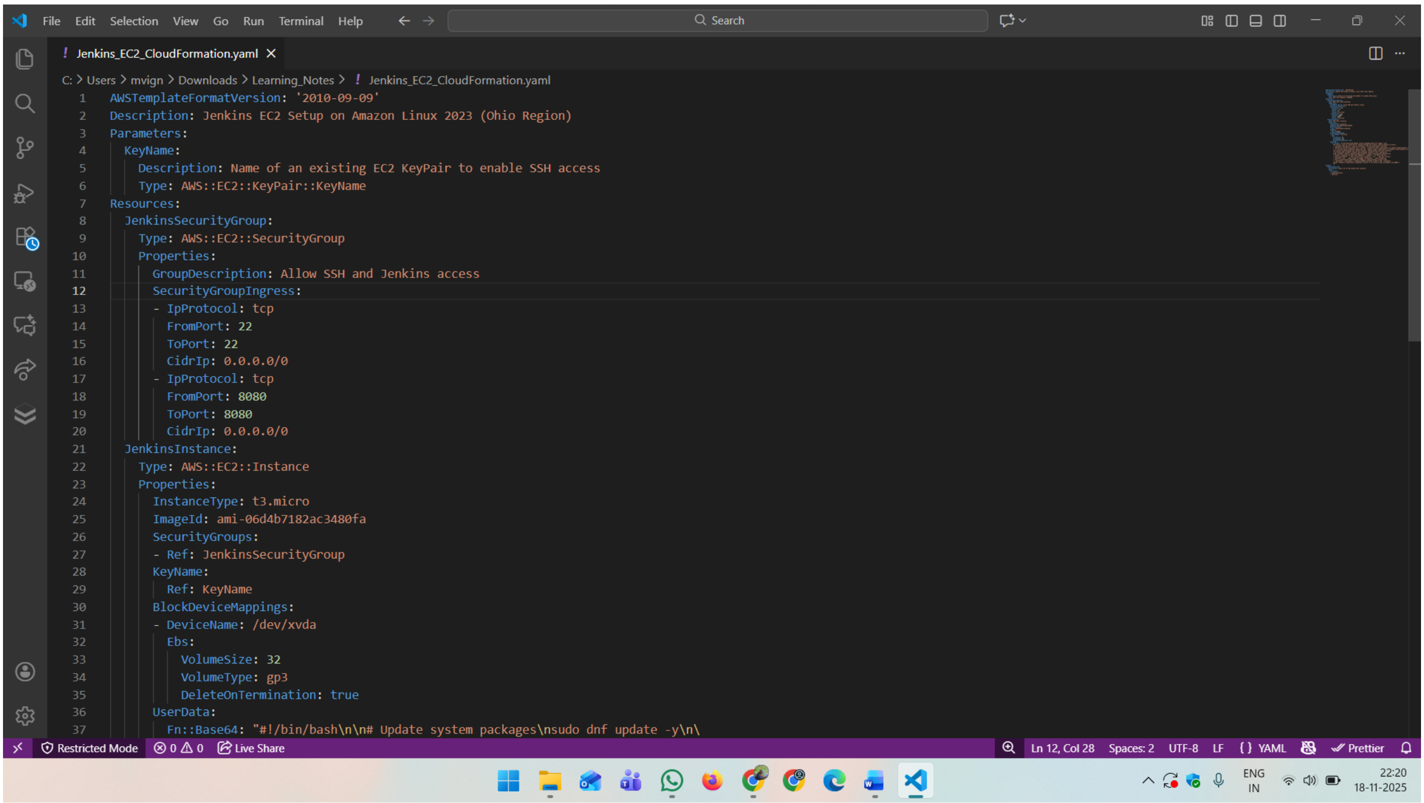
Expected Outcome:

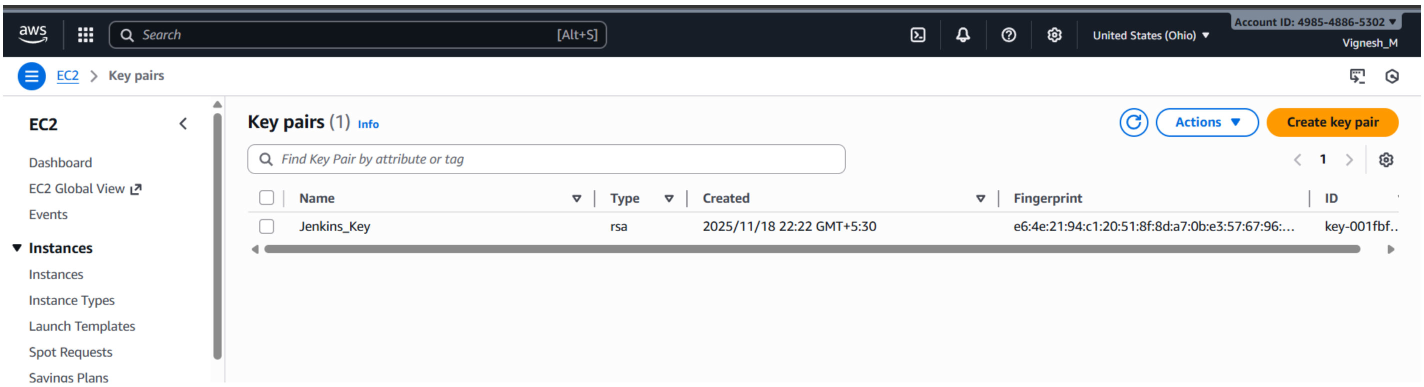
- Jenkins downtime and restart events tracked automatically.
- Alerts triggered instantly when Jenkins goes down or restarts.

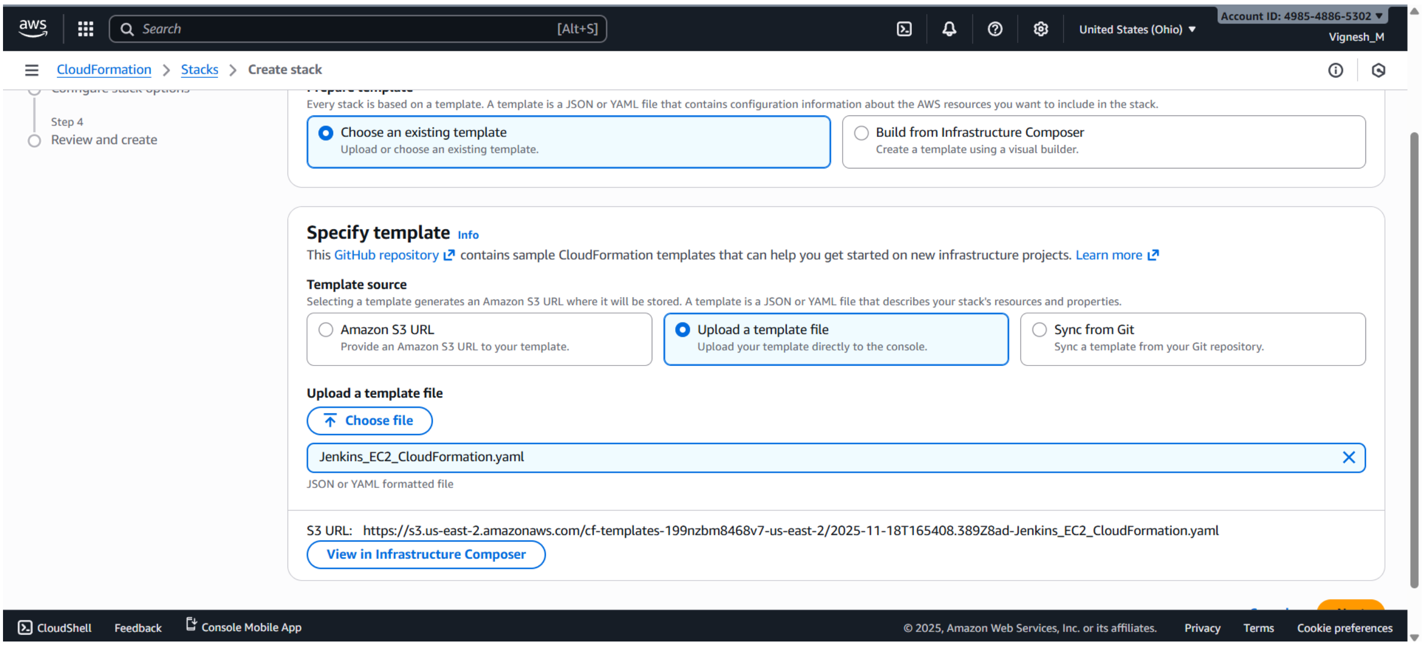
Step 1: Creating EC2 Instance and installing Jenkins using Cloud Formation:

Procedure:

- Create a Cloud Formation yaml file and store it in local system.
  - Details about EC2 Instance type and AMI Image ID.
  - Details about the Inbound and Outbound rule.
  - Details about the application and installation instructions.
- Create a Key-Pair and download the private key.
- Cloud Formation Console → Create Stack → Upload the template cloud formation yaml file.
- Give the Stack name and Key-Pair name which we created. → Create Stack.
- Instance created with all the configuration which we mentioned in the yaml file.
- Open browser and <EC2_Public_IP>:8080. Jenkins Interface will open.
- **Result:** EC2 Instance created, and Jenkins installed, and Jenkins server is Up.






Search
[Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
CloudFormation > Stacks > Create stack
Step 4
Review and create
Every stack is based on a template. A template is a JSON or YAML file that contains configuration information about the AWS resources you want to include in the stack.
Choose an existing template
Upload or choose an existing template.
Build from Infrastructure Composer
Create a template using a visual builder.
Specify template Info
This GitHub repository contains sample CloudFormation templates that can help you get started on new infrastructure projects. Learn more
Template source
Selecting a template generates an Amazon S3 URL where it will be stored. A template is a JSON or YAML file that describes your stack's resources and properties.
Amazon S3 URL
Provide an Amazon S3 URL to your template.
Upload a template file
Upload your template directly to the console.
Sync from Git
Sync a template from your Git repository.
Upload a template file
Choose file
Jenkins_EC2_CloudFormation.yaml
JSON or YAML formatted file
S3 URL: https://s3.us-east-2.amazonaws.com/cf-templates-199nzbm8468v7-us-east-2/2025-11-18T165408.389Z8ad-Jenkins_EC2_CloudFormation.yaml
View in Infrastructure Composer
CloudShell
Feedback
Console Mobile App
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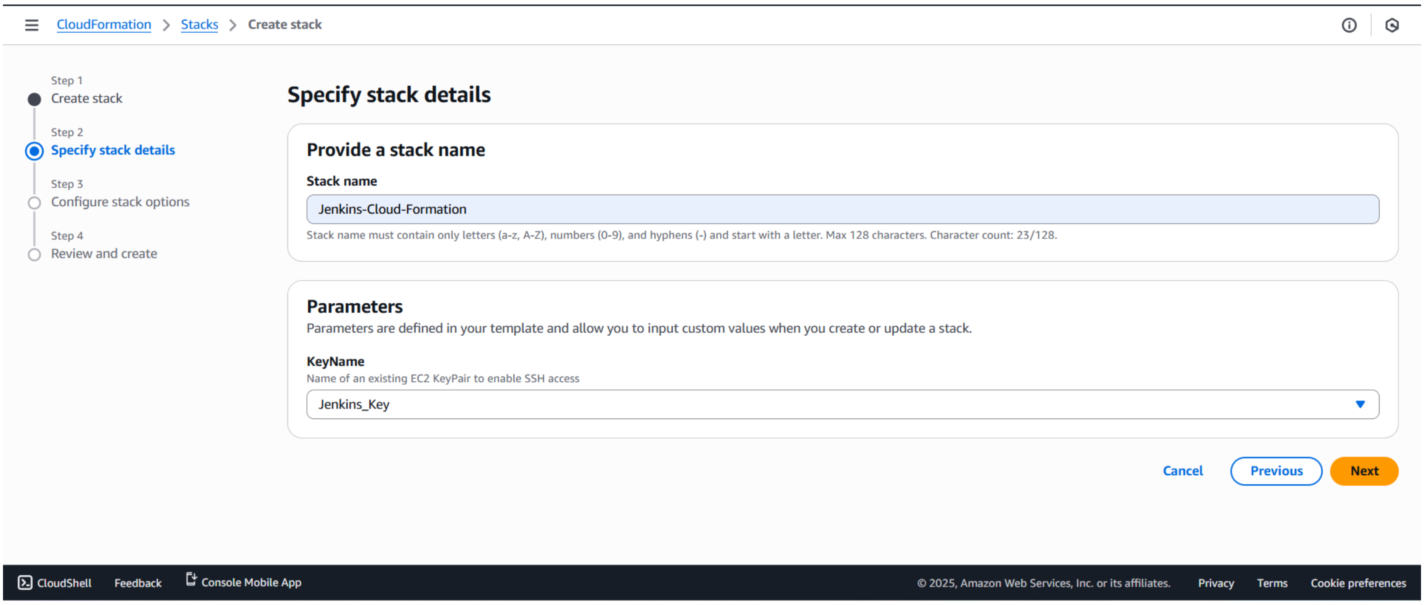
CloudFormation > Stacks > Create stack
Step 1
Create stack
Step 2
Specify stack details
Step 3
Configure stack options
Step 4
Review and create
Specify stack details
Provide a stack name
Stack name
Jenkins-Cloud-Formation
Stack name must contain only letters (a-z, A-Z), numbers (0-9), and hyphens (-) and start with a letter. Max 128 characters. Character count: 23/128.
Parameters
Parameters are defined in your template and allow you to input custom values when you create or update a stack.
KeyName
Name of an existing EC2 KeyPair to enable SSH access
Jenkins_Key
Cancel
Previous
Next
CloudShell
Feedback
Console Mobile App
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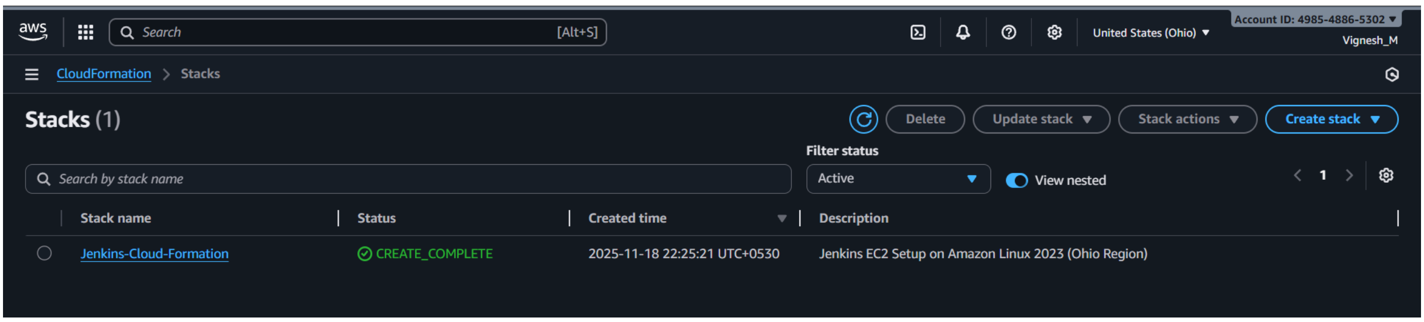
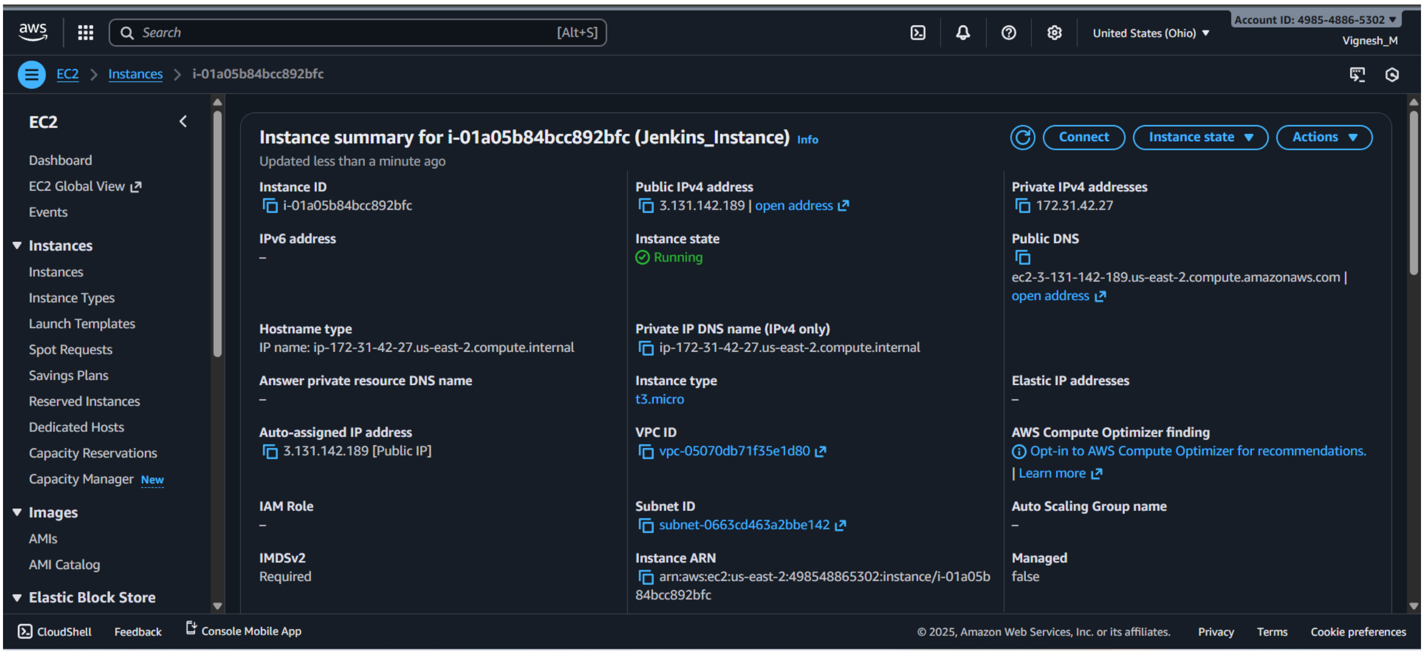

Search
[Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
CloudFormation > Stacks
Stacks (1)
Delete
Update stack
Stack actions
Create stack
Filter status
Search by stack name
Active
View nested
Stack name
Status
Created time
Description
Jenkins-Cloud-Formation
CREATE_COMPLETE
2025-11-18 22:25:21 UTC+0530
Jenkins EC2 Setup on Amazon Linux 2023 (Ohio Region)


Search [Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
EC2 > Instances > i-01a05b84bcc892bfc
EC2
Dashboard
EC2 Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Instance summary for i-01a05b84bcc892bfc (Jenkins_Instance) Info
Updated less than a minute ago
Connect
Instance state
Actions
Instance ID
i-01a05b84bcc892bfc
Public IPv4 address
3.131.142.189 | open address
Private IPv4 addresses
172.31.42.27
IPv6 address
–
Instance state
Running
Public DNS
ec2-3-131-142-189.us-east-2.compute.amazonaws.com | open address
Hostname type
IP name: ip-172-31-42-27.us-east-2.compute.internal
Private IP DNS name (IPv4 only)
ip-172-31-42-27.us-east-2.compute.internal
Answer private resource DNS name
–
Instance type
t3.micro
Elastic IP addresses
–
Auto-assigned IP address
3.131.142.189 [Public IP]
VPC ID
vpc-05070db71f35e1d80
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more
IAM Role
–
Subnet ID
subnet-0663cd463a2bbe142
Auto Scaling Group name
–
IMDSv2
Required
Instance ARN
arn:aws:ec2:us-east-2:498548865302:instance/i-01a05b84bcc892bfc
Managed
false
CloudShell
Feedback
Console Mobile App
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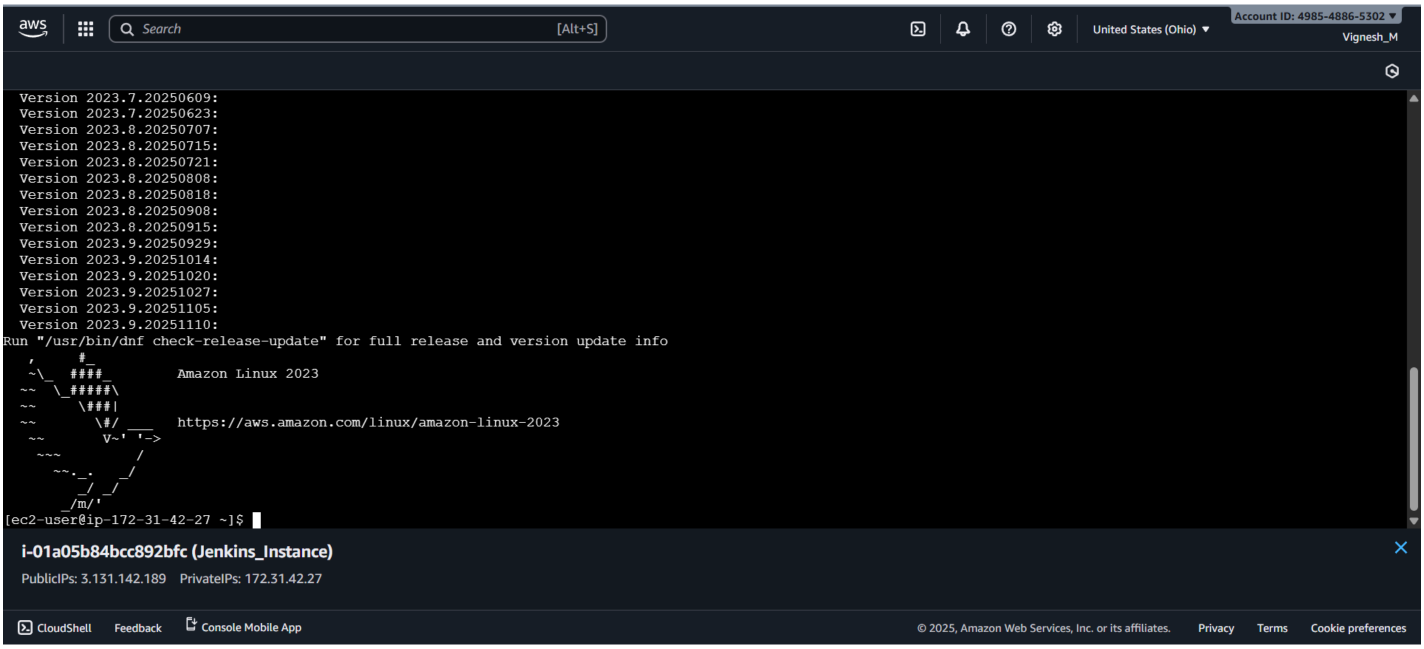
Search [Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
Version 2023.7.20250609:
Version 2023.7.20250623:
Version 2023.8.20250707:
Version 2023.8.20250715:
Version 2023.8.20250721:
Version 2023.8.20250808:
Version 2023.8.20250818:
Version 2023.8.20250908:
Version 2023.8.20250915:
Version 2023.9.20250929:
Version 2023.9.20251014:
Version 2023.9.20251020:
Version 2023.9.20251027:
Version 2023.9.20251105:
Version 2023.9.20251110:
Run "/usr/bin/dnf check-release-update" for full release and version update info
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-172-31-42-27 ~]$
i-01a05b84bcc892bfc (Jenkins_Instance)
PublicIPs: 3.131.142.189 PrivateIPs: 172.31.42.27
CloudShell
Feedback
Console Mobile App
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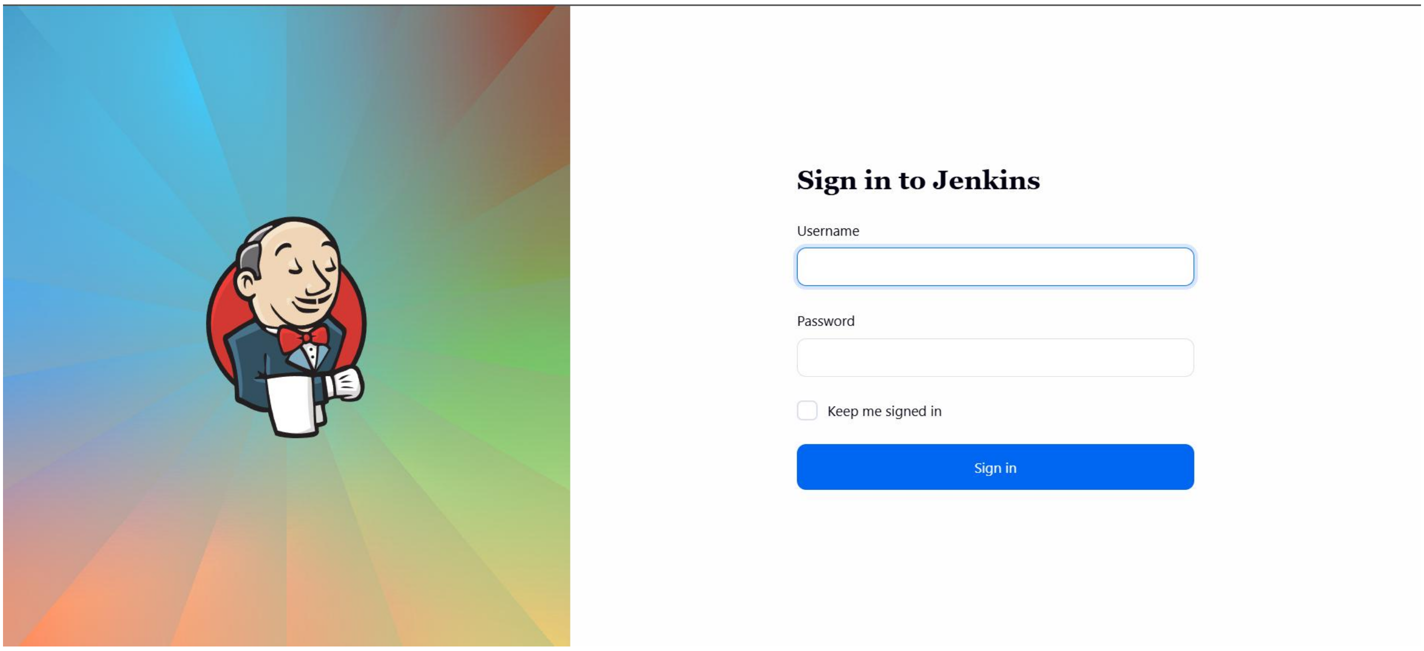

Sign in to Jenkins
Username
Password
Keep me signed in
Sign in

Step 2: Create Shell Script file in /usr/local/bin location:

Procedure:

- Inside EC2 terminal create jenkins_monitor.sh and Jenkins_monitor_report.sh in /usr/local/bin location.
- Change the permission of the file and make it executable.
  - sudo chmod +x <File_Name>
- Install crontab for cronjob scheduling if it is not there.
  - Command 1: sudo yum install cronie -y
  - Command 2: sudo systemctl enable crond
  - Command 3: sudo systemctl start crond
  - Command 4: sudo systemctl status crond
- Set the cronjob for executing the Jenkins_monitor.sh script every minute to check the Jenkins status.
  - sudo crontab -e
  - sudo crontab -l
- For every one minute the cronjob will run and the log will store in /var/log/Jenkins_monitor.sh
- If we run the jenkins_monitor_report.sh, it will parse the log file and show the number of occurrences of UP, DOWN and RESTART and last 10 log events.
- **Result:**
  - Jenkins_monitor.sh and Jenkins_monitor_report.sh file create.
  - Cronjob set successfully for Jenkins_monitor.sh.
  - /var/log/Jenkins_monitor.log file created and log are fetching correctly.
  - jenkins_monitor_report.sh is executing and report is generations correctly.

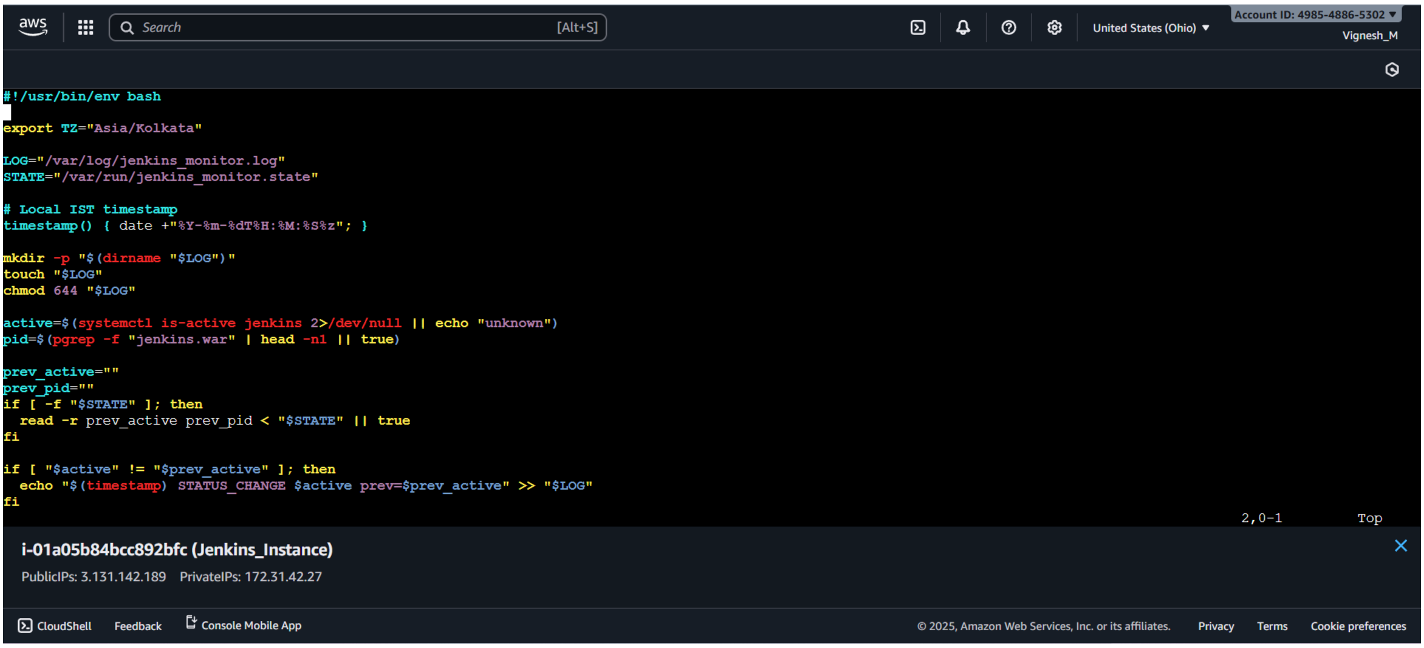
Jenkins_monitor.sh code:

```
#!/usr/bin/env bash
export TZ="Asia/Kolkata"
LOG="/var/log/jenkins_monitor.log"
STATE="/var/run/jenkins_monitor.state"
# Local IST timestamp
timestamp() { date +"%Y-%m-%dT%H:%M:%S%z"; }
mkdir -p "$(dirname "$LOG")"
touch "$LOG"
chmod 644 "$LOG"
active=$(systemctl is-active jenkins 2>/dev/null || echo "unknown")
pid=$(pgrep -f "jenkins.war" | head -n1 || true)
prev_active=""
prev_pid=""
if [ -f "$STATE" ]; then
  read -r prev_active prev_pid < "$STATE" || true
fi
if [ "$active" != "$prev_active" ]; then
```

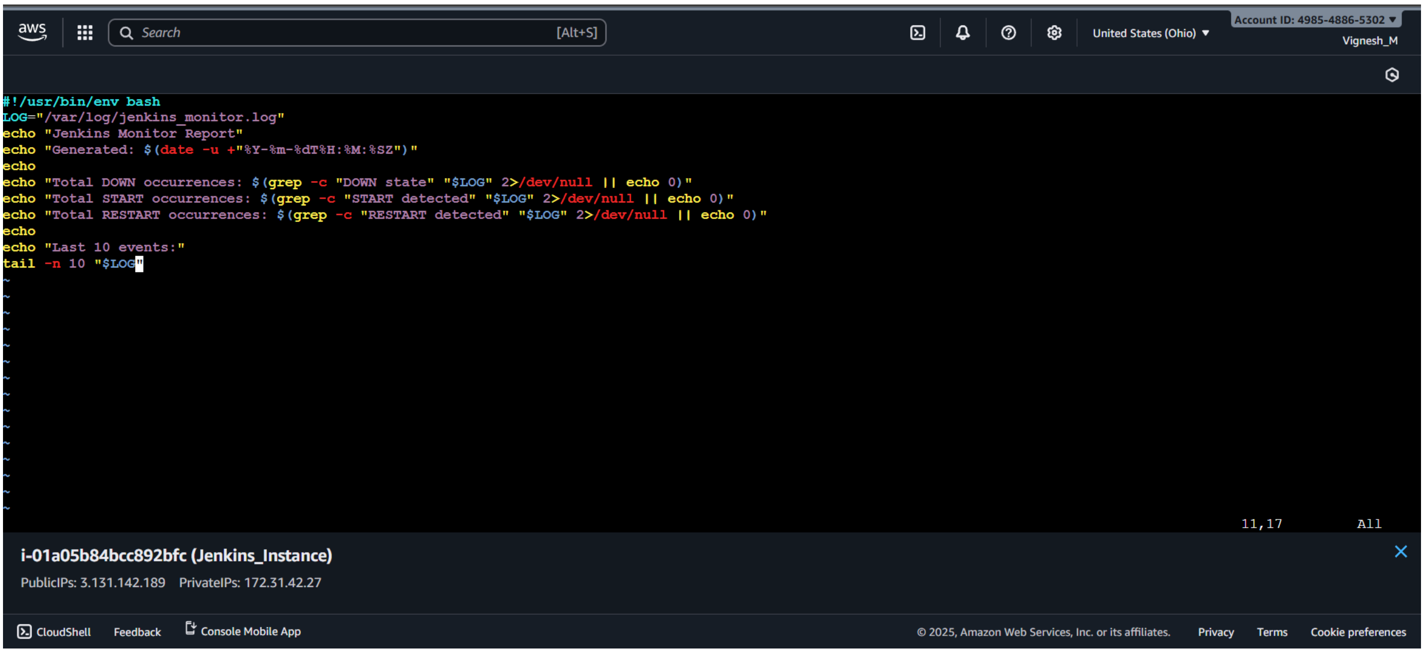
```
  echo "$(timestamp) STATUS_CHANGE $active prev=$prev_active" >> "$LOG"
fi
if [ -n "$prev_pid" ] && [ -n "$pid" ] && [ "$pid" != "$prev_pid" ]; then
  echo "$(timestamp) RESTART detected new_pid=$pid old_pid=$prev_pid" >> "$LOG"
fi
if [ -z "$prev_pid" ] && [ -n "$pid" ] && [ "$active" = "active" ]; then
  echo "$(timestamp) START detected pid=$pid" >> "$LOG"
fi
if [ "$active" != "active" ]; then
  echo "$(timestamp) DOWN state=$active" >> "$LOG"
fi
echo "$active $pid" > "$STATE"
```

Jenkins_monitor_report.sh code:

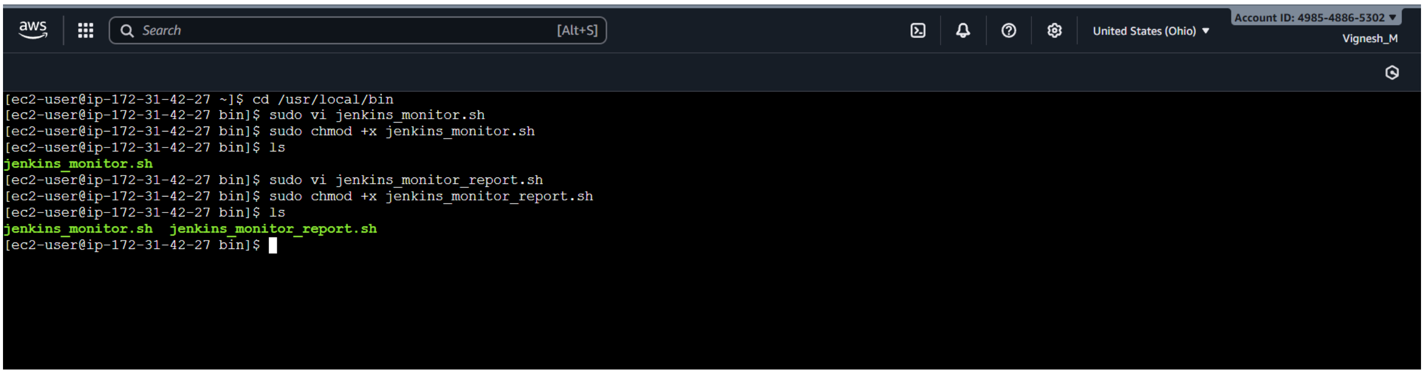
```
#!/usr/bin/env bash
LOG="/var/log/jenkins_monitor.log"
echo "Jenkins Monitor Report"
echo "Generated: $(date -u +"%Y-%m-%dT%H:%M:%SZ")"
echo
echo "Total DOWN occurrences: $(grep -c "DOWN state" "$LOG" 2>/dev/null || echo 0)"
echo "Total START occurrences: $(grep -c "START detected" "$LOG" 2>/dev/null || echo 0)"
echo "Total RESTART occurrences: $(grep -c "RESTART detected" "$LOG" 2>/dev/null || echo 0)"
echo
echo "Last 10 events:"
tail -n 10 "$LOG"
```



```
#!/usr/bin/env bash

export TZ="Asia/Kolkata"

LOG="/var/log/jenkins_monitor.log"
STATE="/var/run/jenkins_monitor.state"

# Local IST timestamp
timestamp() { date +"%Y-%m-%dT%H:%M:%S%z"; }

mkdir -p "$(dirname "$LOG")"
touch "$LOG"
chmod 644 "$LOG"

active=$(systemctl is-active jenkins 2>/dev/null || echo "unknown")
pid=$(pgrep -f "jenkins.war" | head -n1 || true)

prev_active=""
prev_pid=""
if [ -f "$STATE" ]; then
  read -r prev_active prev_pid < "$STATE" || true
fi

if [ "$active" != "$prev_active" ]; then
  echo "$(timestamp) STATUS_CHANGE $active prev=$prev_active" >> "$LOG"
fi
                                                  2,0-1         Top
```

```
#!/usr/bin/env bash
LOG="/var/log/jenkins_monitor.log"
echo "Jenkins Monitor Report"
echo "Generated: $(date -u +"%Y-%m-%dT%H:%M:%SZ")"
echo
echo "Total DOWN occurrences: $(grep -c "DOWN state" "$LOG" 2>/dev/null || echo 0)"
echo "Total START occurrences: $(grep -c "START detected" "$LOG" 2>/dev/null || echo 0)"
echo "Total RESTART occurrences: $(grep -c "RESTART detected" "$LOG" 2>/dev/null || echo 0)"
echo
echo "Last 10 events:"
tail -n 10 "$LOG"
~
                                                  11,17         All
```

```
[ec2-user@ip-172-31-42-27 ~]$ cd /usr/local/bin
[ec2-user@ip-172-31-42-27 bin]$ sudo vi jenkins_monitor.sh
[ec2-user@ip-172-31-42-27 bin]$ sudo chmod +x jenkins_monitor.sh
[ec2-user@ip-172-31-42-27 bin]$ ls
jenkins_monitor.sh
[ec2-user@ip-172-31-42-27 bin]$ sudo vi jenkins_monitor_report.sh
[ec2-user@ip-172-31-42-27 bin]$ sudo chmod +x jenkins_monitor_report.sh
[ec2-user@ip-172-31-42-27 bin]$ ls
jenkins_monitor.sh  jenkins_monitor_report.sh
[ec2-user@ip-172-31-42-27 bin]$
```

```
* * * * * /usr/local/bin/jenkins_monitor.sh
~
```

```
[ec2-user@ip-172-31-42-27 /]$ crontab -e
no crontab for ec2-user - using an empty one
crontab: installing new crontab
[ec2-user@ip-172-31-42-27 /]$ crontab -l
* * * * * /usr/local/bin/jenkins_monitor.sh
[ec2-user@ip-172-31-42-27 /]$
```

```
[ec2-user@ip-172-31-42-27 /]$ cd var/log
[ec2-user@ip-172-31-42-27 log]$ ls
README  audit  chrony                 cloud-init.log    dnf.log      firewalld   jenkins_monitor.log  lastlog  sa    tallylog
amazon  btmp   cloud-init-output.log  dnf.librepo.log   dnf.rpm.log  hawkey.log  journal              private  sssd  wtmp
[ec2-user@ip-172-31-42-27 log]$ cat jenkins_monitor.log
2025-11-18T22:44:02+0530 STATUS_CHANGE inactive
unknown prev=
2025-11-18T22:44:02+0530 DOWN state=inactive
unknown
2025-11-18T22:45:02+0530 STATUS_CHANGE active prev=inactive
2025-11-18T22:45:02+0530 START detected pid=27351
[ec2-user@ip-172-31-42-27 log]$
```

i-01a05b84bcc892bfc (Jenkins_Instance)

PublicIPs: 3.131.142.189 PrivateIPs: 172.31.42.27

```
[ec2-user@ip-172-31-42-27 /]$ cd /usr/local/bin
[ec2-user@ip-172-31-42-27 bin]$ sudo jenkins_monitor_report.sh
Jenkins Monitor Report
Generated: 2025-11-18T19:40:18Z

Total DOWN occurrences: 29
Total START occurrences: 13
Total RESTART occurrences: 7

Last 10 events:
unknown
2025-11-19T00:55:01+0530 STATUS_CHANGE inactive
unknown prev=inactive
2025-11-19T00:55:01+0530 DOWN state=inactive
unknown
2025-11-19T00:56:02+0530 STATUS_CHANGE active prev=inactive
2025-11-19T00:56:02+0530 START detected pid=38264
2025-11-19T00:59:02+0530 RESTART detected new_pid=38594 old_pid=38264
2025-11-19T01:03:02+0530 RESTART detected new_pid=39032 old_pid=38594
2025-11-19T01:06:02+0530 RESTART detected new_pid=39339 old_pid=39032
[ec2-user@ip-172-31-42-27 bin]$
```

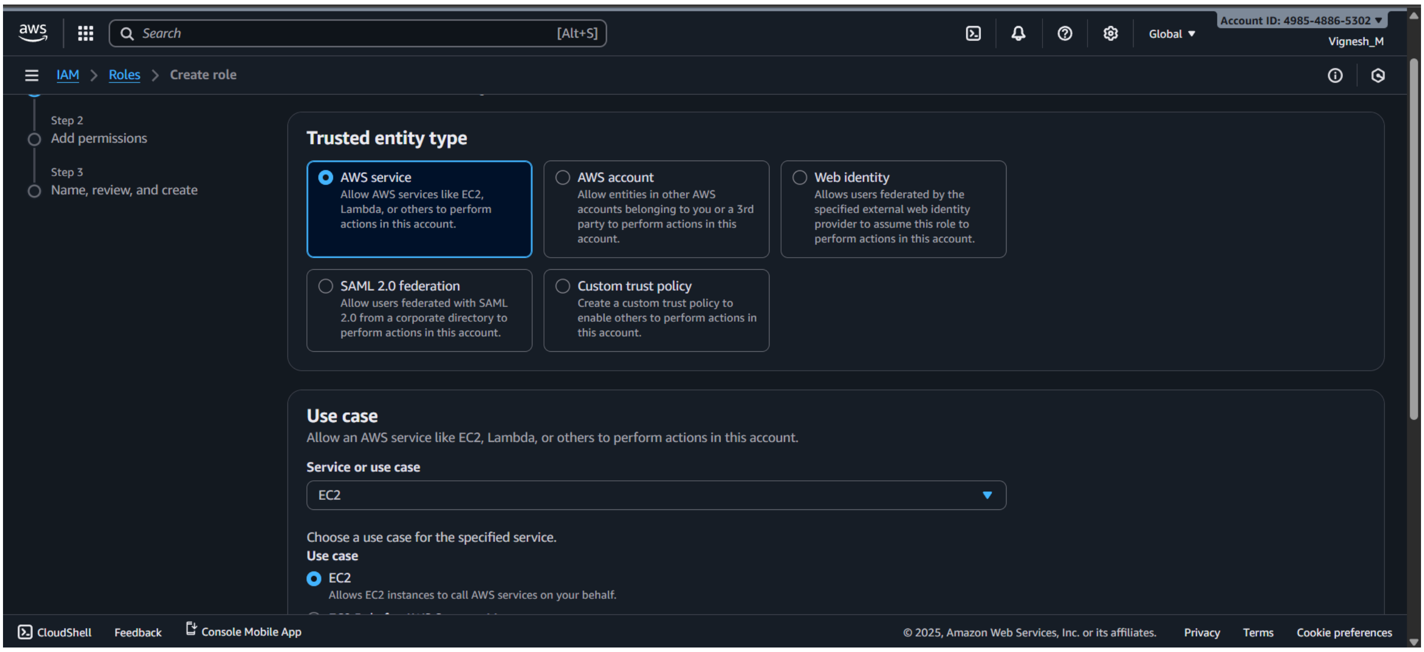
i-01a05b84bcc892bfc (Jenkins_Instance)

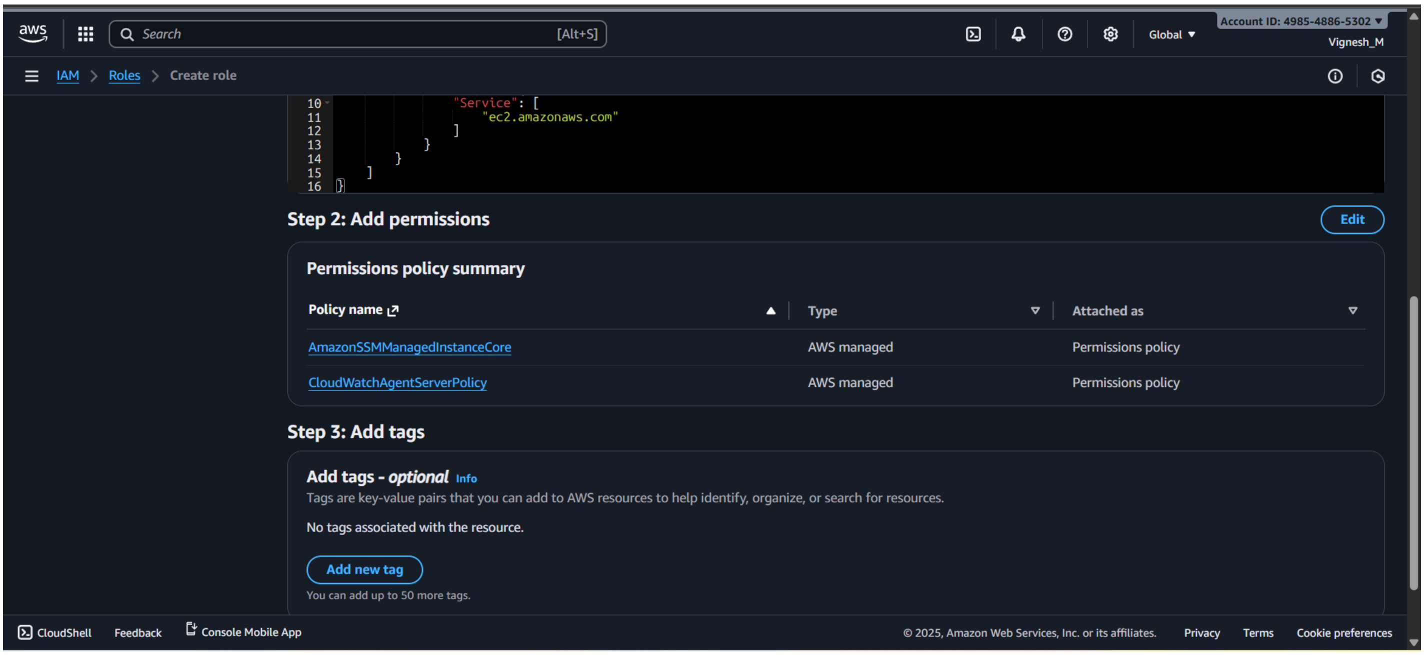
PublicIPs: 3.131.142.189 PrivateIPs: 172.31.42.27

Step 3: Install and Configure the CloudWatch Agent:

Procedure:

- Create a role and attach it to EC2 instances for configuring the cloud watch agent in the EC2 instance.
  - **Policy 1:** AmazonSSMManagedInstanceCore
  - **Policy 2:** CloudWatchServerPolicy
  - **Role Name:** EC2CloudWatchAgentRole
- Attach the create role to the EC2 instance.
- Install the cloud watch agent and configure that with our EC2 Instance.
  - **Command 1:** sudo yum install amazon-cloudwatch-agent -y
  - **Command 2:** sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-config-wizard
  - **Command 3:** sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -a fetch-config -m ec2 -c file:/opt/aws/amazon-cloudwatch-agent/bin/config.json
  - **Command 4:** sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -a start
  - **Command 5:** sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -a status
- After executing the commands, config file will create, and it will have all the metrics and log file details which we are going to monitor.
- **Result:**
  - Role assigned to EC2 instance.
  - Cloud watch agent installed and configured in EC2 instance successfully.
  - Jenkins_monitor.log file log fetch by cloud watch agent and send it to cloud watch console correctly.




Search [Alt+S]
Global
Account ID: 4985-4886-5302
Vignesh_M
IAM > Roles > Create role
"Service": [
"ec2.amazonaws.com"
]
}
}
]
}
Step 2: Add permissions
Edit
Permissions policy summary
Policy name
Type
Attached as
AmazonSSMManagedInstanceCore
AWS managed
Permissions policy
CloudWatchAgentServerPolicy
AWS managed
Permissions policy
Step 3: Add tags
Add tags - optional Info
Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.
No tags associated with the resource.
Add new tag
You can add up to 50 more tags.
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Feedback
Console Mobile App
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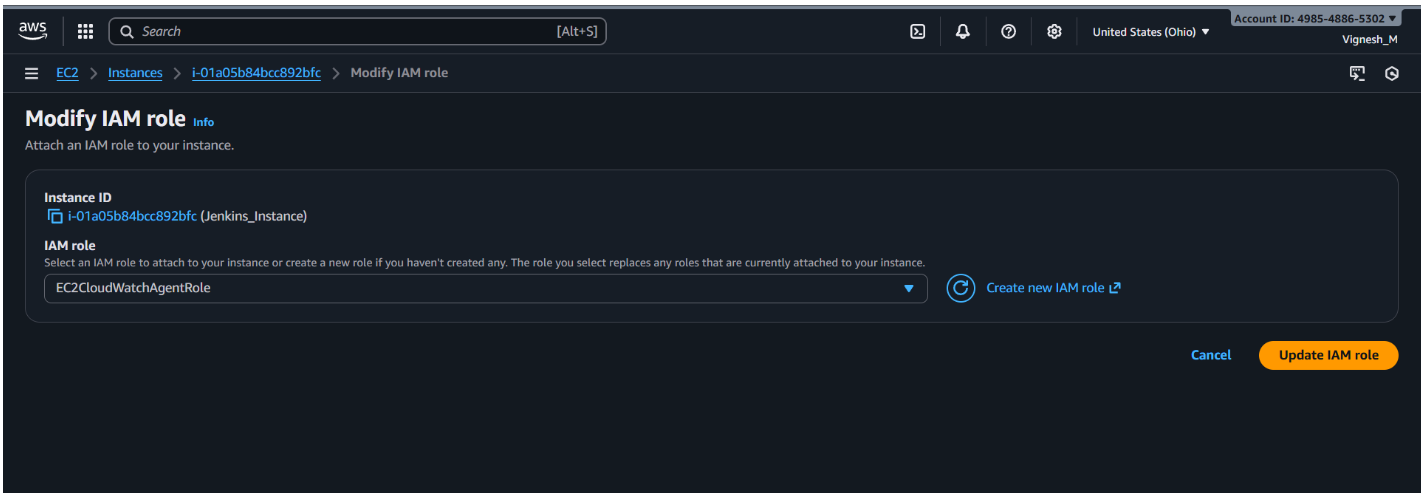
Search [Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
EC2 > Instances > i-01a05b84bcc892bfc > Modify IAM role
Modify IAM role Info
Attach an IAM role to your instance.
Instance ID
i-01a05b84bcc892bfc (Jenkins_Instance)
IAM role
Select an IAM role to attach to your instance or create a new role if you haven't created any. The role you select replaces any roles that are currently attached to your instance.
EC2CloudWatchAgentRole
Create new IAM role
Cancel
Update IAM role


Search [Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
[ec2-user@ip-172-31-42-27 /]$ sudo yum install amazon-cloudwatch-agent -y
Last metadata expiration check: 0:19:49 ago on Tue Nov 18 16:56:18 2025.
Dependencies resolved.
Package Architecture Version Repository Size
Installing:
amazon-cloudwatch-agent x86_64 1.300028.1-1.amzn2023 amazonlinux 77 M
Transaction Summary
Install 1 Package
Total download size: 77 M
Installed size: 372 M
Downloading Packages:
amazon-cloudwatch-agent-1.300028.1-1.amzn2023.x86_64.rpm 58 MB/s | 77 MB 00:01
Total 56 MB/s | 77 MB 00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
i-01a05b84bcc892bfc (Jenkins_Instance)
PublicIPs: 3.131.142.189 PrivateIPs: 172.31.42.27
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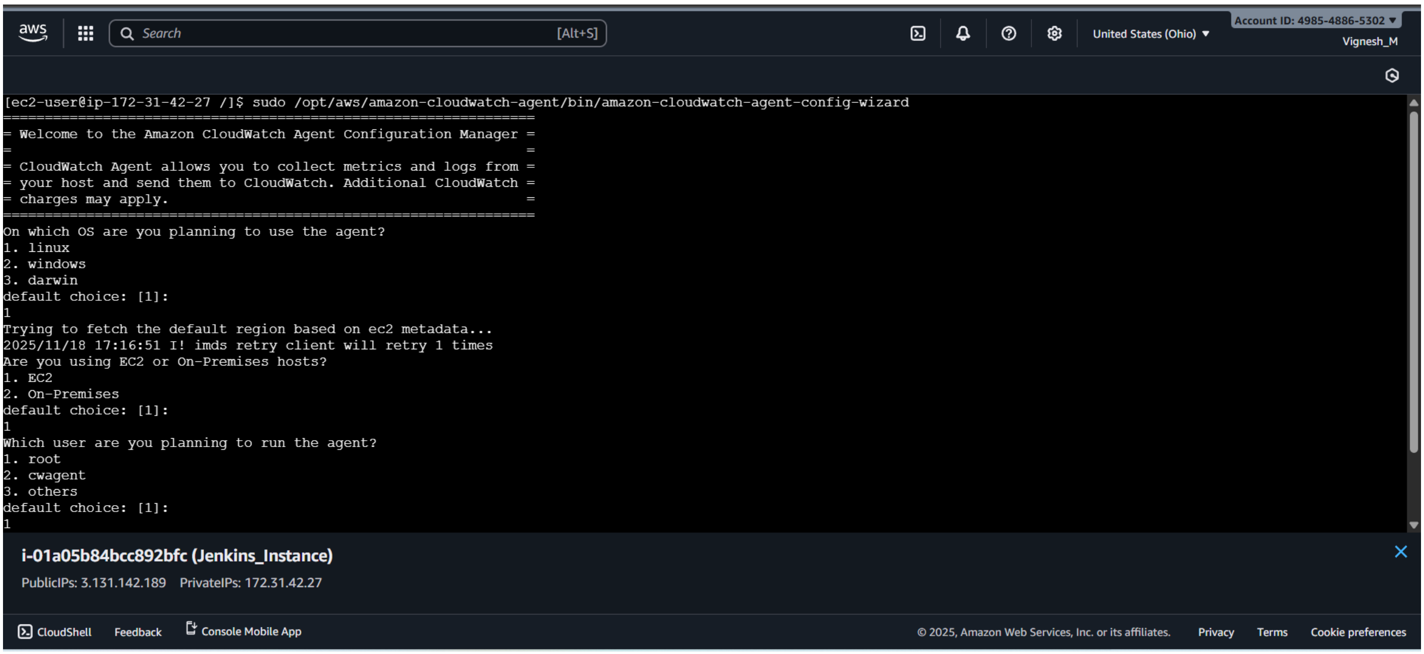
aws
Search [Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
[ec2-user@ip-172-31-42-27 /]$ sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-config-wizard
==
= Welcome to the Amazon CloudWatch Agent Configuration Manager =
=   =
= CloudWatch Agent allows you to collect metrics and logs from =
= your host and send them to CloudWatch. Additional CloudWatch =
= charges may apply.   =
==
On which OS are you planning to use the agent?
1. linux
2. windows
3. darwin
default choice: [1]:
1
Trying to fetch the default region based on ec2 metadata...
2025/11/18 17:16:51 I! imds retry client will retry 1 times
Are you using EC2 or On-Premises hosts?
1. EC2
2. On-Premises
default choice: [1]:
1
Which user are you planning to run the agent?
1. root
2. cwagent
3. others
default choice: [1]:
1
i-01a05b84bcc892bfc (Jenkins_Instance)
PublicIPs: 3.131.142.189 PrivateIPs: 172.31.42.27
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Console Mobile App
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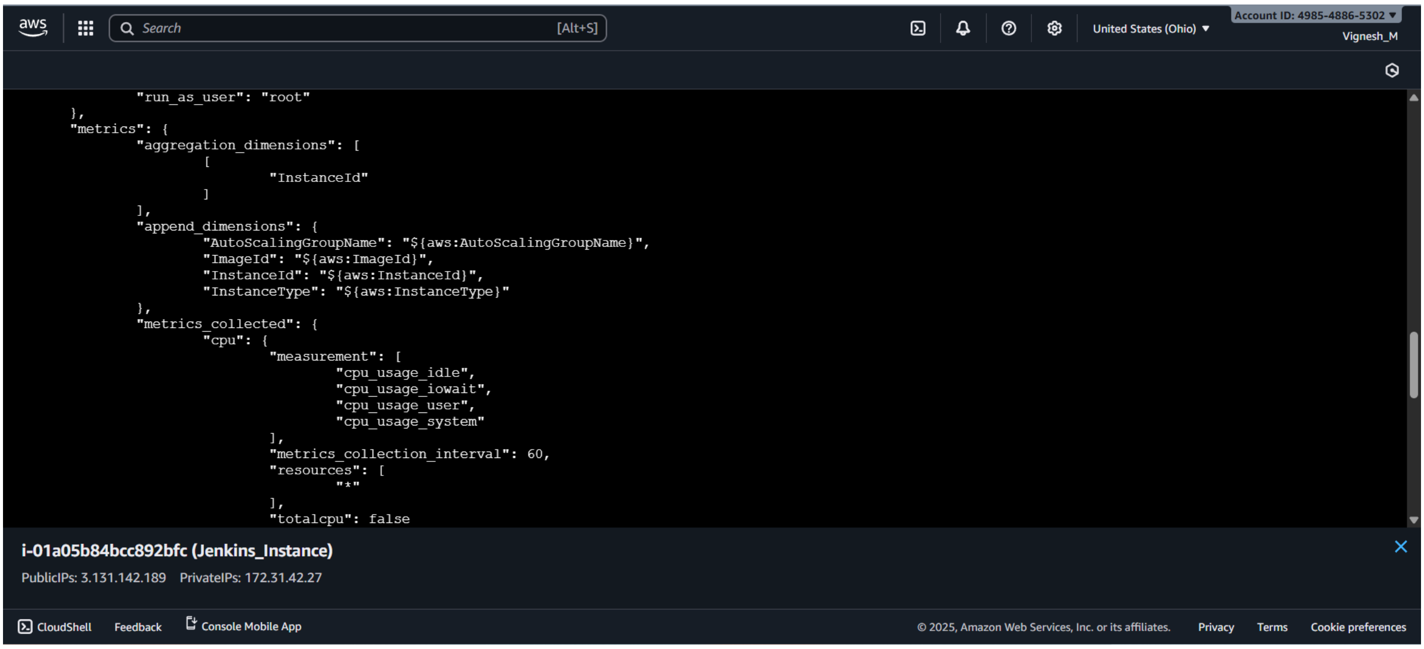
aws
Search [Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
"run_as_user": "root"
},
"metrics": {
"aggregation_dimensions": [
[
"InstanceId"
]
],
"append_dimensions": {
"AutoScalingGroupName": "${aws:AutoScalingGroupName}",
"ImageId": "${aws:ImageId}",
"InstanceId": "${aws:InstanceId}",
"InstanceType": "${aws:InstanceType}"
},
"metrics_collected": {
"cpu": {
"measurement": [
"cpu_usage_idle",
"cpu_usage_iowait",
"cpu_usage_user",
"cpu_usage_system"
],
"metrics_collection_interval": 60,
"resources": [
"*"
],
"totalcpu": false
i-01a05b84bcc892bfc (Jenkins_Instance)
PublicIPs: 3.131.142.189 PrivateIPs: 172.31.42.27
CloudShell
Feedback
Console Mobile App
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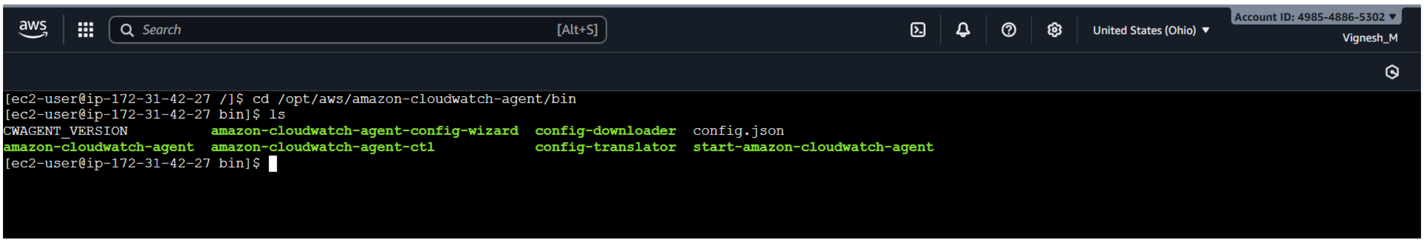
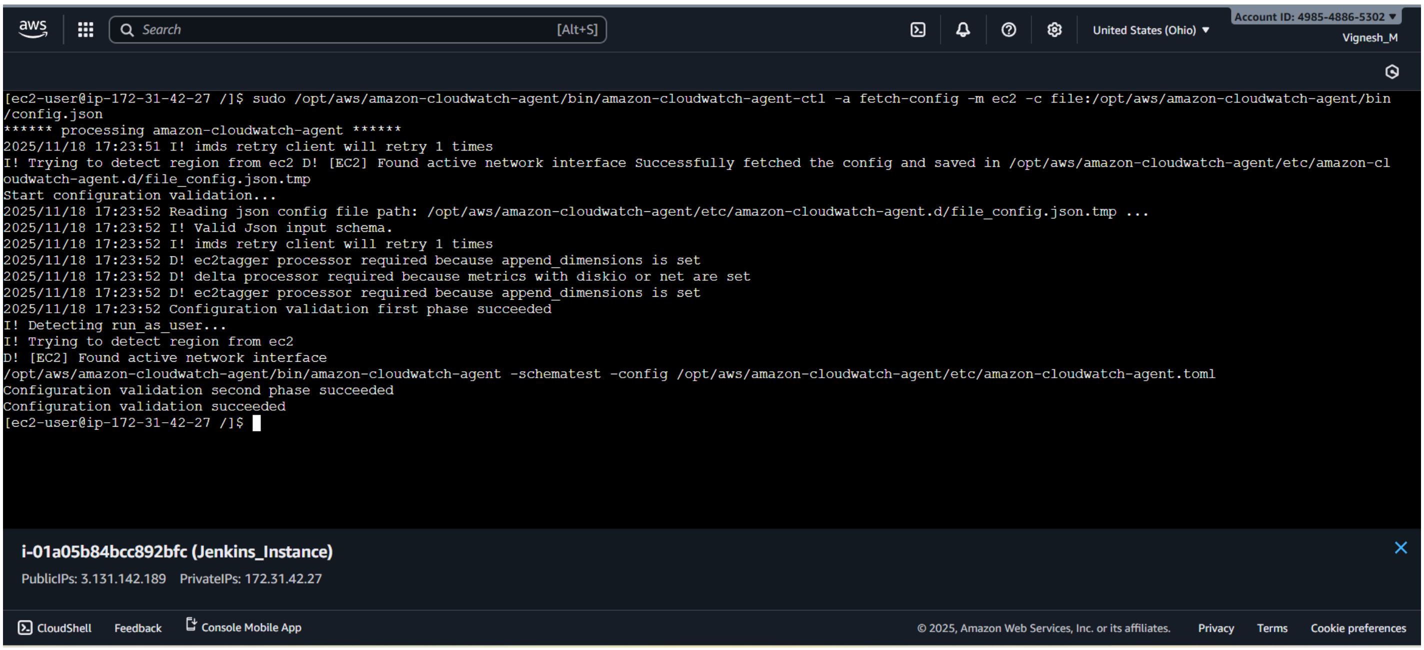
Cookie preferences


aws
Search [Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
[ec2-user@ip-172-31-42-27 /]$ cd /opt/aws/amazon-cloudwatch-agent/bin
[ec2-user@ip-172-31-42-27 bin]$ ls
CWAGENT_VERSION amazon-cloudwatch-agent-config-wizard config-downloader config.json
amazon-cloudwatch-agent amazon-cloudwatch-agent-ctl config-translator start-amazon-cloudwatch-agent
[ec2-user@ip-172-31-42-27 bin]$


aws
Search
[Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
[ec2-user@ip-172-31-42-27 /]$ sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -a fetch-config -m ec2 -c file:/opt/aws/amazon-cloudwatch-agent/bin
/config.json
****** processing amazon-cloudwatch-agent ******
2025/11/18 17:23:51 I! imds retry client will retry 1 times
I! Trying to detect region from ec2 D! [EC2] Found active network interface Successfully fetched the config and saved in /opt/aws/amazon-cloudwatch-agent/etc/amazon-cl
oudwatch-agent.d/file_config.json.tmp
Start configuration validation...
2025/11/18 17:23:52 Reading json config file path: /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.d/file_config.json.tmp ...
2025/11/18 17:23:52 I! Valid Json input schema.
2025/11/18 17:23:52 I! imds retry client will retry 1 times
2025/11/18 17:23:52 D! ec2tagger processor required because append_dimensions is set
2025/11/18 17:23:52 D! delta processor required because metrics with diskio or net are set
2025/11/18 17:23:52 D! ec2tagger processor required because append_dimensions is set
2025/11/18 17:23:52 Configuration validation first phase succeeded
I! Detecting run_as_user...
I! Trying to detect region from ec2
D! [EC2] Found active network interface
/opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent -schematest -config /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.toml
Configuration validation second phase succeeded
Configuration validation succeeded
[ec2-user@ip-172-31-42-27 /]$
i-01a05b84bcc892bfc (Jenkins_Instance)
PublicIPs: 3.131.142.189 PrivateIPs: 172.31.42.27
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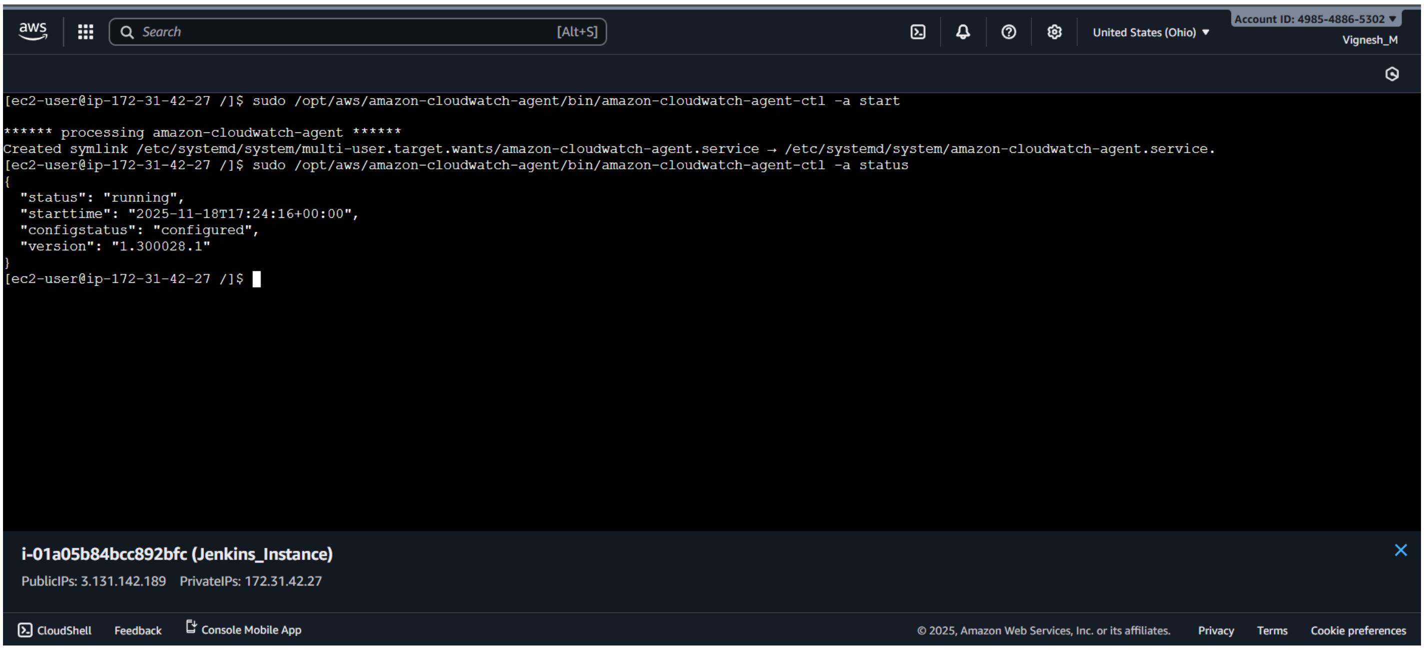
aws
Search
[Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
[ec2-user@ip-172-31-42-27 /]$ sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -a start
****** processing amazon-cloudwatch-agent ******
Created symlink /etc/systemd/system/multi-user.target.wants/amazon-cloudwatch-agent.service → /etc/systemd/system/amazon-cloudwatch-agent.service.
[ec2-user@ip-172-31-42-27 /]$ sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -a status
{
"status": "running",
"starttime": "2025-11-18T17:24:16+00:00",
"configstatus": "configured",
"version": "1.300028.1"
}
[ec2-user@ip-172-31-42-27 /]$
i-01a05b84bcc892bfc (Jenkins_Instance)
PublicIPs: 3.131.142.189 PrivateIPs: 172.31.42.27
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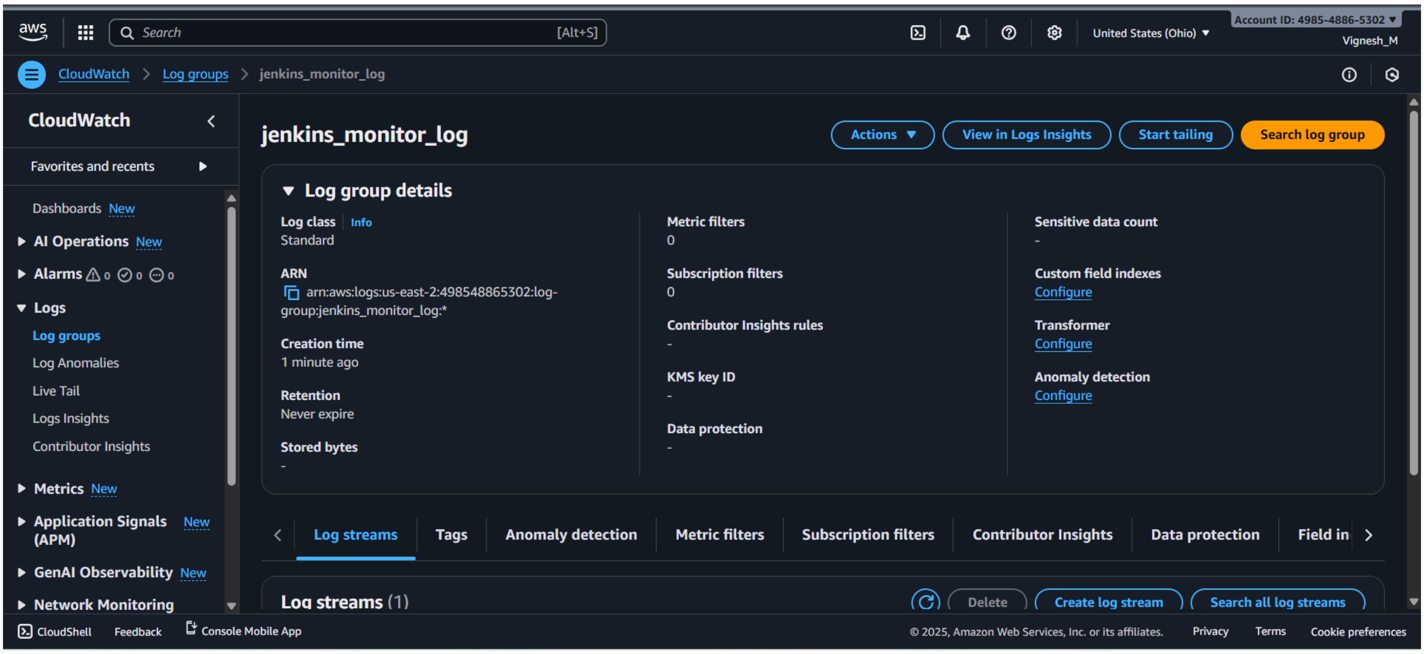
aws
Search
[Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
CloudWatch
Log groups
jenkins_monitor_log
CloudWatch
Favorites and recents
Dashboards New
AI Operations New
Alarms 0 0 0
Logs
Log groups
Log Anomalies
Live Tail
Logs Insights
Contributor Insights
Metrics New
Application Signals (APM) New
GenAI Observability New
Network Monitoring
jenkins_monitor_log
Actions
View in Logs Insights
Start tailing
Search log group
Log group details
Log class Info
Standard
ARN
arn:aws:logs:us-east-2:498548865302:log-group:jenkins_monitor_log:*
Creation time
1 minute ago
Retention
Never expire
Stored bytes
-
Metric filters
0
Subscription filters
0
Contributor Insights rules
-
KMS key ID
-
Data protection
-
Sensitive data count
-
Custom field indexes
Configure
Transformer
Configure
Anomaly detection
Configure
Log streams
Tags
Anomaly detection
Metric filters
Subscription filters
Contributor Insights
Data protection
Field in
Log streams (1)
Delete
Create log stream
Search all log streams
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Feedback
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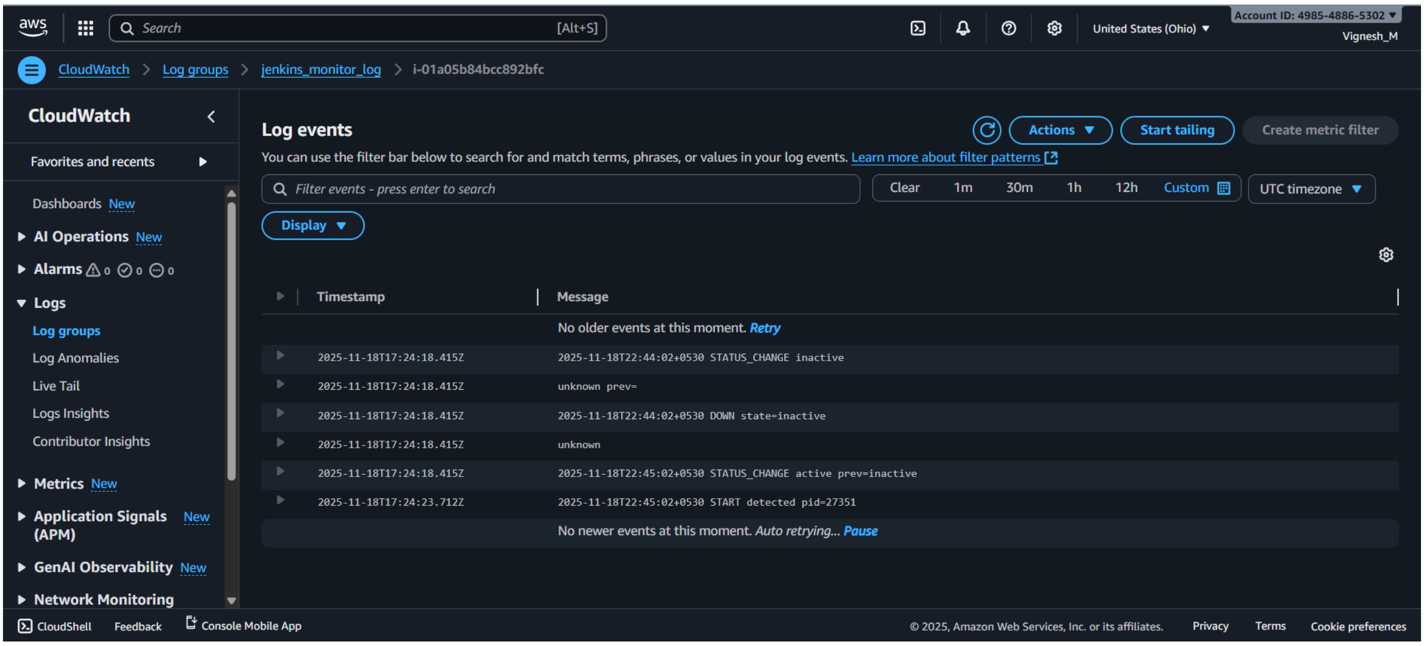
aws
Search
[Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
CloudWatch
Log groups
jenkins_monitor_log
i-01a05b84bcc892bfc
CloudWatch
Favorites and recents
Dashboards New
AI Operations New
Alarms 0 0 0
Logs
Log groups
Log Anomalies
Live Tail
Logs Insights
Contributor Insights
Metrics New
Application Signals (APM) New
GenAI Observability New
Network Monitoring
Log events
Actions
Start tailing
Create metric filter
You can use the filter bar below to search for and match terms, phrases, or values in your log events. Learn more about filter patterns
Filter events - press enter to search
Clear
1m
30m
1h
12h
Custom
UTC timezone
Display
Timestamp
Message
No older events at this moment. Retry
2025-11-18T17:24:18.415Z
2025-11-18T22:44:02+0530 STATUS_CHANGE inactive
2025-11-18T17:24:18.415Z
unknown prev=
2025-11-18T17:24:18.415Z
2025-11-18T22:44:02+0530 DOWN state=inactive
2025-11-18T17:24:18.415Z
unknown
2025-11-18T17:24:18.415Z
2025-11-18T22:45:02+0530 STATUS_CHANGE active prev=inactive
2025-11-18T17:24:23.712Z
2025-11-18T22:45:02+0530 START detected pid=27351
No newer events at this moment. Auto retrying... Pause
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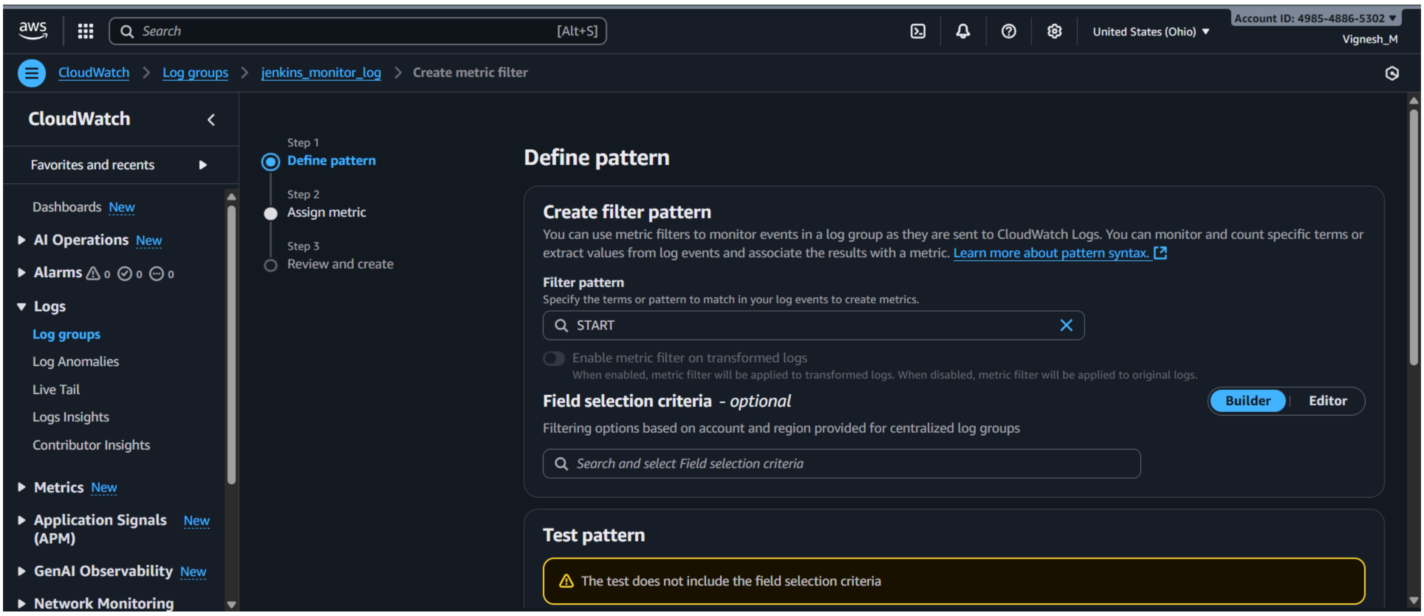
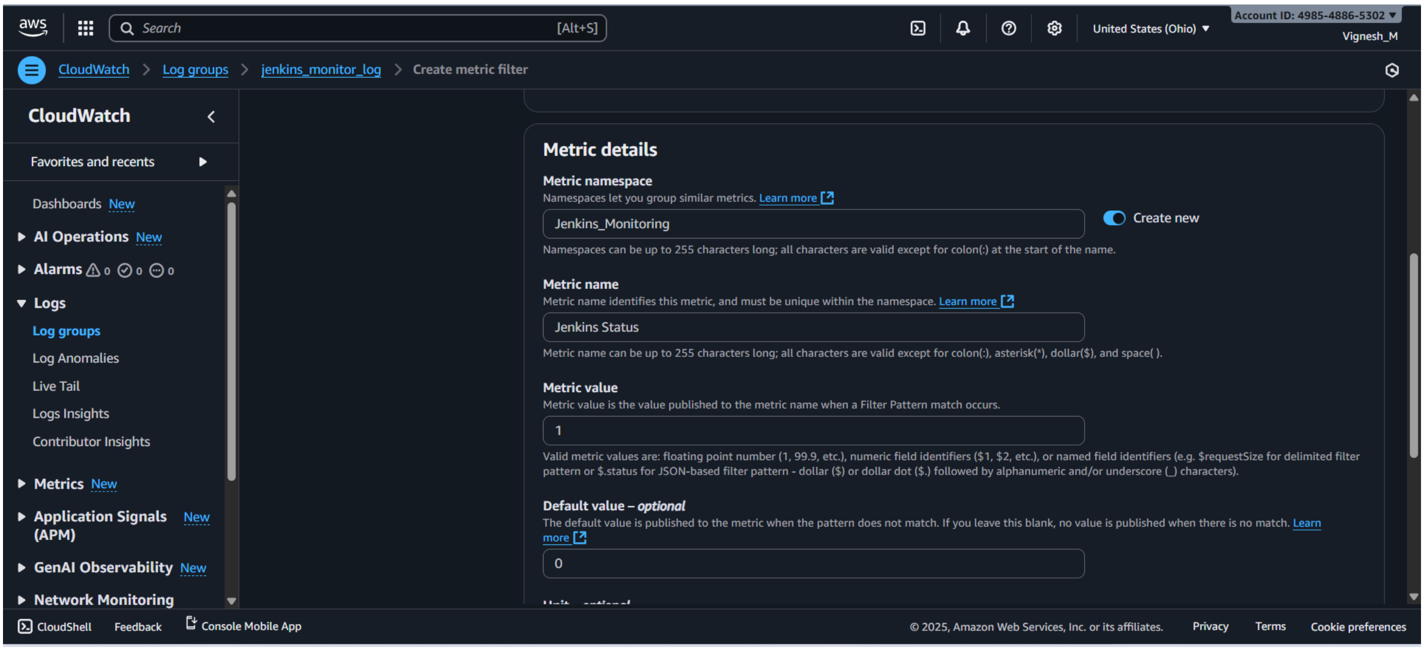
Step 4: Create Metric Filter for Jenkins Down and Restart Events:

Procedure:

- Create a custom metric filter to create alarm for Jenkins down and restart events.
  - Log Groups → <Monitorign_Log_Name> → Create metric filter → Create metric → Enter metric details → Create metric.
- **Metric for Jenkins Up Events:**
  - **Filter Pattern:** START
  - **Filter Name:** Jenkins_Up_Event
  - **Metric Namespace:** Jenkins_Monitoring
  - **Metric Name:** Jenkins Status
  - **Metric Value:** 1
  - **Default Value:** 0
- **Metric for Jenkins Restart Events:**
  - **Filter Pattern:** RESTART detected
  - **Filter Name:** Jenkins_Restart_Event
  - **Metric Namespace:** Jenkins_Monitoring
  - **Metric Name:** Jenkins Restart
  - **Metric Value:** 1
- **Result:** Metric for Jenkins down and Jenkins restart event created successfully.




aws
Search
[Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
CloudWatch > Log groups > jenkins_monitor_log > Create metric filter
CloudWatch
Favorites and recents
Dashboards New
AI Operations New
Alarms 0 0 0
Logs
Log groups
Log Anomalies
Live Tail
Logs Insights
Contributor Insights
Metrics New
Application Signals (APM) New
GenAI Observability New
Network Monitoring
Metric details
Metric namespace
Namespaces let you group similar metrics. Learn more
Jenkins_Monitoring
Create new
Namespaces can be up to 255 characters long; all characters are valid except for colon(:) at the start of the name.
Metric name
Metric name identifies this metric, and must be unique within the namespace. Learn more
Jenkins Status
Metric name can be up to 255 characters long; all characters are valid except for colon(:), asterisk(*), dollar($), and space( ).
Metric value
Metric value is the value published to the metric name when a Filter Pattern match occurs.
1
Valid metric values are: floating point number (1, 99.9, etc.), numeric field identifiers ($1, $2, etc.), or named field identifiers (e.g. $requestSize for delimited filter pattern or $.status for JSON-based filter pattern - dollar ($) or dollar dot ($.) followed by alphanumeric and/or underscore (_) characters).
Default value – optional
The default value is published to the metric when the pattern does not match. If you leave this blank, no value is published when there is no match. Learn more
0
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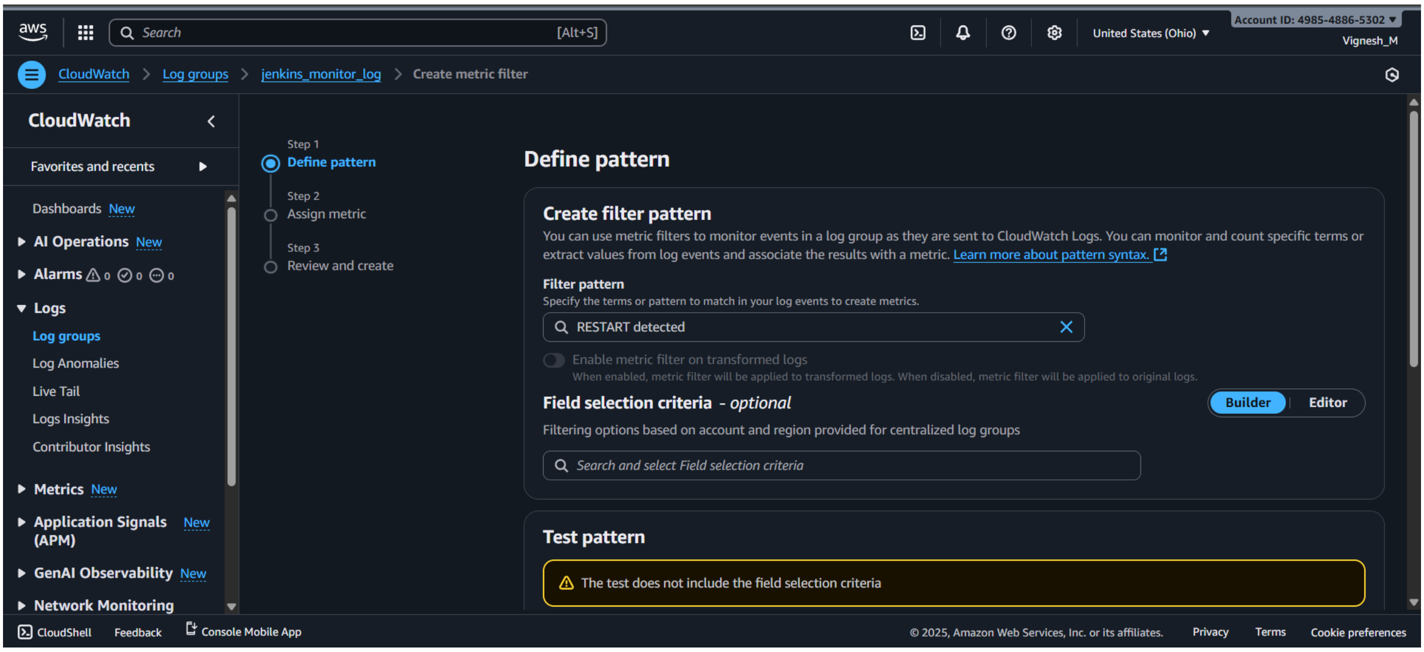
aws
Search
[Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
CloudWatch > Log groups > jenkins_monitor_log > Create metric filter
CloudWatch
Favorites and recents
Dashboards New
AI Operations New
Alarms 0 0 0
Logs
Log groups
Log Anomalies
Live Tail
Logs Insights
Contributor Insights
Metrics New
Application Signals (APM) New
GenAI Observability New
Network Monitoring
Step 1
Define pattern
Step 2
Assign metric
Step 3
Review and create
Define pattern
Create filter pattern
You can use metric filters to monitor events in a log group as they are sent to CloudWatch Logs. You can monitor and count specific terms or extract values from log events and associate the results with a metric. Learn more about pattern syntax.
Filter pattern
Specify the terms or pattern to match in your log events to create metrics.
RESTART detected
Enable metric filter on transformed logs
When enabled, metric filter will be applied to transformed logs. When disabled, metric filter will be applied to original logs.
Field selection criteria - optional
Builder
Editor
Filtering options based on account and region provided for centralized log groups
Search and select Field selection criteria
Test pattern
The test does not include the field selection criteria
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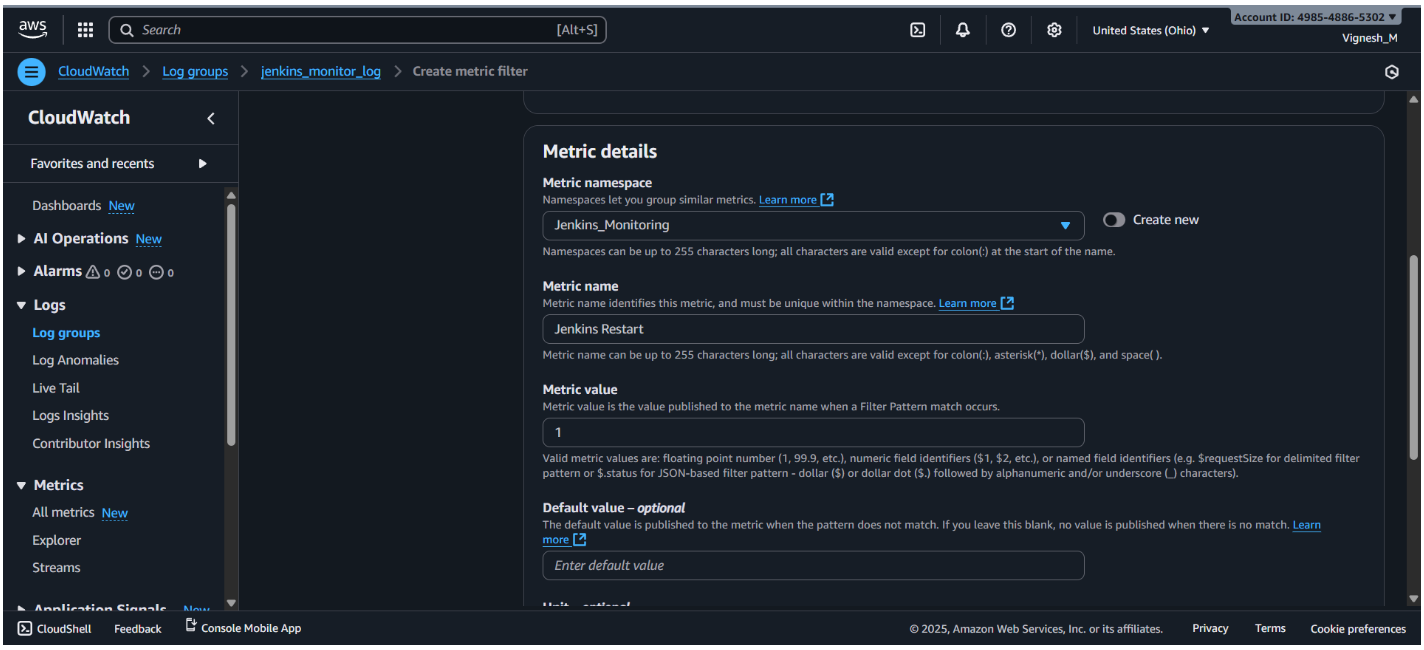
CloudWatch > Log groups > jenkins_monitor_log > Create metric filter
CloudWatch
Favorites and recents
Dashboards New
AI Operations New
Alarms 0 0 0
Logs
Log groups
Log Anomalies
Live Tail
Logs Insights
Contributor Insights
Metrics
All metrics New
Explorer
Streams
Metric details
Metric namespace
Namespaces let you group similar metrics. Learn more
Jenkins_Monitoring
Create new
Namespaces can be up to 255 characters long; all characters are valid except for colon(:) at the start of the name.
Metric name
Metric name identifies this metric, and must be unique within the namespace. Learn more
Jenkins Restart
Metric name can be up to 255 characters long; all characters are valid except for colon(:), asterisk(*), dollar($), and space( ).
Metric value
Metric value is the value published to the metric name when a Filter Pattern match occurs.
1
Valid metric values are: floating point number (1, 99.9, etc.), numeric field identifiers ($1, $2, etc.), or named field identifiers (e.g. $requestSize for delimited filter pattern or $.status for JSON-based filter pattern - dollar ($) or dollar dot ($.) followed by alphanumeric and/or underscore (_) characters).
Default value – optional
The default value is published to the metric when the pattern does not match. If you leave this blank, no value is published when there is no match. Learn more
Enter default value
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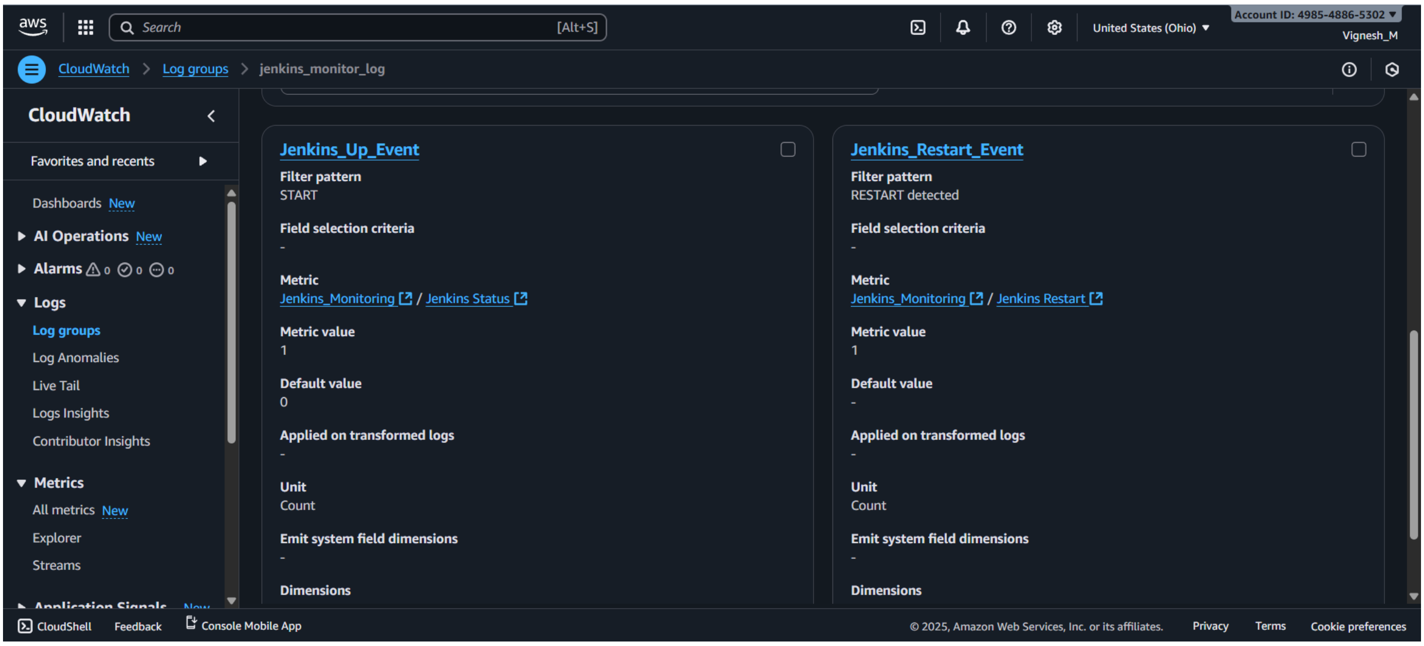
CloudWatch > Log groups > jenkins_monitor_log
CloudWatch
Favorites and recents
Dashboards New
AI Operations New
Alarms 0 0 0
Logs
Log groups
Log Anomalies
Live Tail
Logs Insights
Contributor Insights
Metrics
All metrics New
Explorer
Streams
Jenkins_Up_Event
Filter pattern
START
Field selection criteria
-
Metric
Jenkins_Monitoring / Jenkins Status
Metric value
1
Default value
0
Applied on transformed logs
-
Unit
Count
Emit system field dimensions
-
Dimensions
Jenkins_Restart_Event
Filter pattern
RESTART detected
Field selection criteria
-
Metric
Jenkins_Monitoring / Jenkins Restart
Metric value
1
Default value
-
Applied on transformed logs
-
Unit
Count
Emit system field dimensions
-
Dimensions
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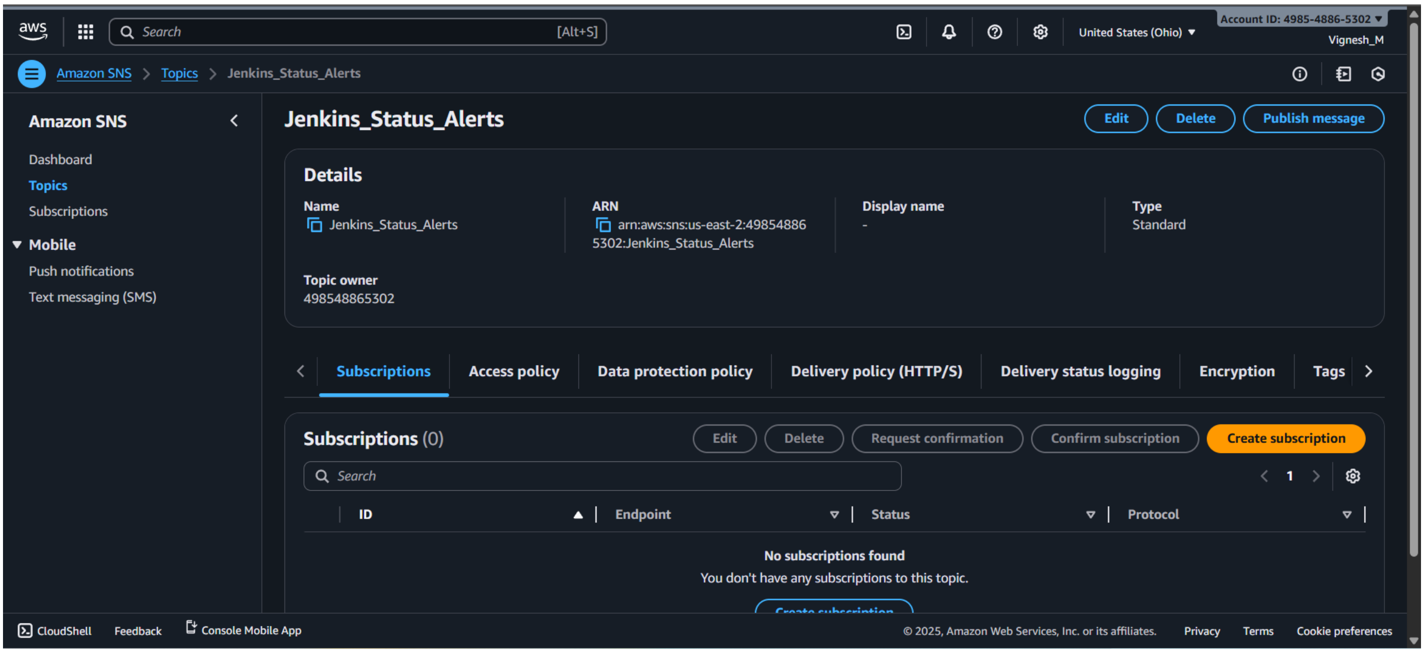
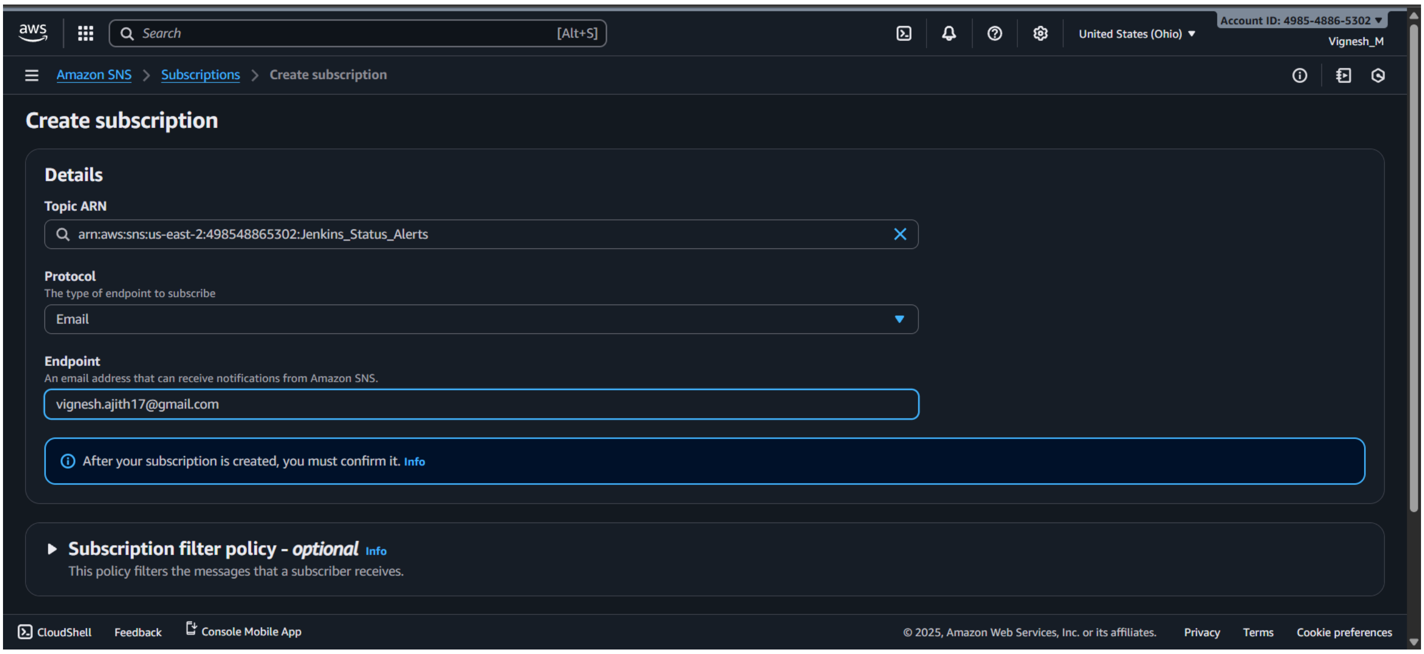
Step 5: Create SNS Topics for Jenkins Alerts:

Procedure:

- Create a SNS topics and subscriptions for Jenkins Alerts
  - Amazon SNS → Create Alert → <Alert_Name> → Create Subscription → Enter Topic ARN → Protocol (Email) → Enter mail ID. → create Subscription → Verify the email address.
- **SNS for Jenkins Up Events Alert:**
  - **Topic Name:** Jenkins_Status_Alert
  - **Topic ARN:** ARN_Jenkins_Status_Alert (Automatically fetch)
  - **Protocol:** Email
  - **Email Id:** Vignesh.ajith17@gmail.com
- **SNS for Jenkins Restart Events Alert:**
  - **Topic Name:** Jenkins_Restart_Alert
  - **Topic ARN:** ARN_Jenkins_Restart_Alert (Automatically fetch)
  - **Protocol:** Email
  - **Email Id:** Vignesh.ajith17@gmail.com
- Verification link will be sent to email. Verify it.
- **Result:** SNS alert for Jenkins Down and Jenkins Restart events created successfully.




Search [Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
Amazon SNS > Subscriptions > Create subscription
Create subscription
Details
Topic ARN
arn:aws:sns:us-east-2:498548865302:Jenkins_Status_Alerts
Protocol
The type of endpoint to subscribe
Email
Endpoint
An email address that can receive notifications from Amazon SNS.
vignesh.ajith17@gmail.com
After your subscription is created, you must confirm it. Info
Subscription filter policy - optional Info
This policy filters the messages that a subscriber receives.
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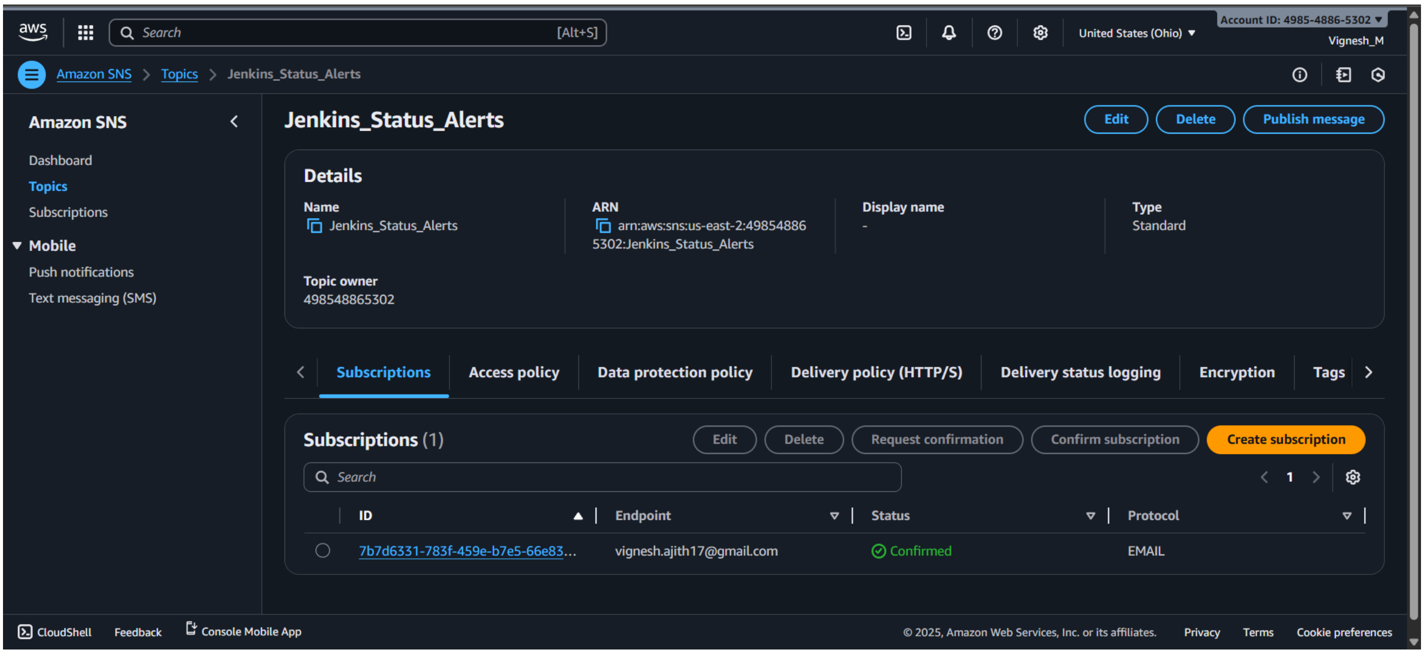
Search [Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
Amazon SNS > Topics > Jenkins_Status_Alerts
Amazon SNS
Dashboard
Topics
Subscriptions
Mobile
Push notifications
Text messaging (SMS)
Jenkins_Status_Alerts
Edit
Delete
Publish message
Details
Name
Jenkins_Status_Alerts
ARN
arn:aws:sns:us-east-2:49854886
5302:Jenkins_Status_Alerts
Display name
-
Type
Standard
Topic owner
498548865302
Subscriptions
Access policy
Data protection policy
Delivery policy (HTTP/S)
Delivery status logging
Encryption
Tags
Subscriptions (1)
Edit
Delete
Request confirmation
Confirm subscription
Create subscription
Search
ID
Endpoint
Status
Protocol
7b7d6331-783f-459e-b7e5-66e83...
vignesh.ajith17@gmail.com
Confirmed
EMAIL
CloudShell
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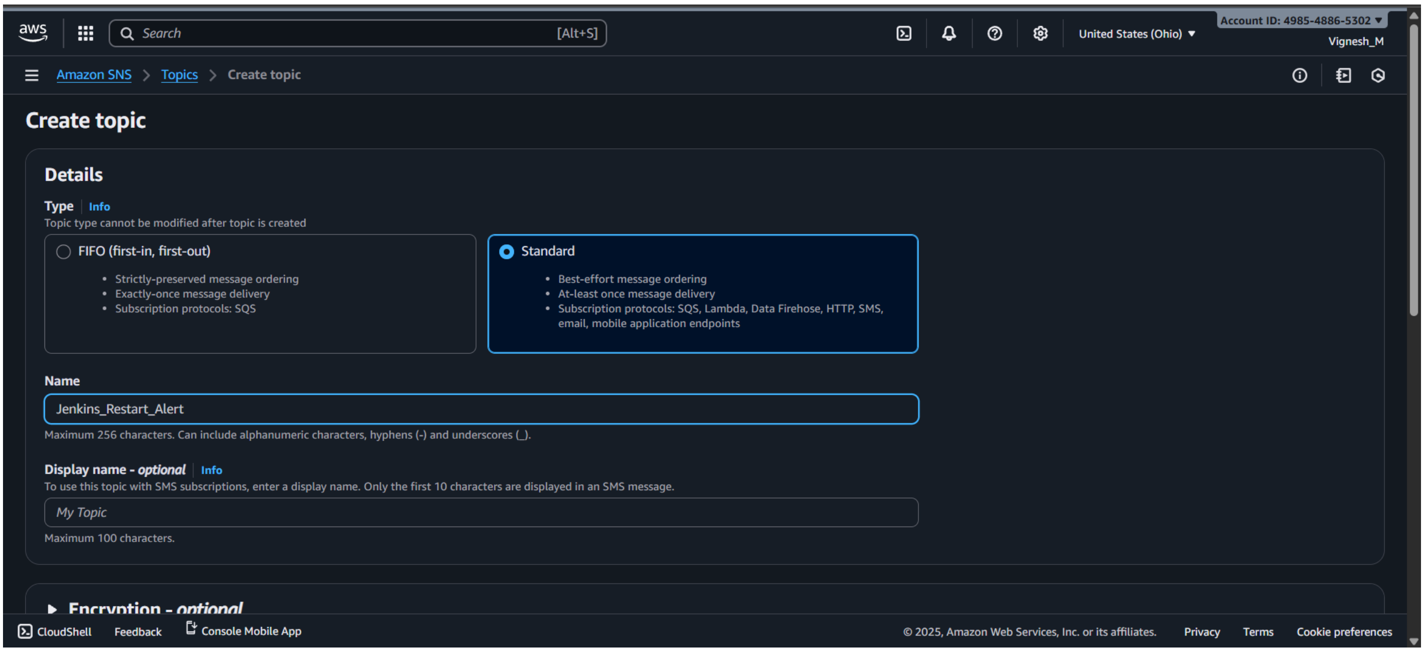
Search [Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
Amazon SNS > Topics > Create topic
Create topic
Details
Type Info
Topic type cannot be modified after topic is created
FIFO (first-in, first-out)
Strictly-preserved message ordering
Exactly-once message delivery
Subscription protocols: SQS
Standard
Best-effort message ordering
At-least once message delivery
Subscription protocols: SQS, Lambda, Data Firehose, HTTP, SMS, email, mobile application endpoints
Name
Jenkins_Restart_Alert
Maximum 256 characters. Can include alphanumeric characters, hyphens (-) and underscores (_).
Display name - optional Info
To use this topic with SMS subscriptions, enter a display name. Only the first 10 characters are displayed in an SMS message.
My Topic
Maximum 100 characters.
Encryption - optional
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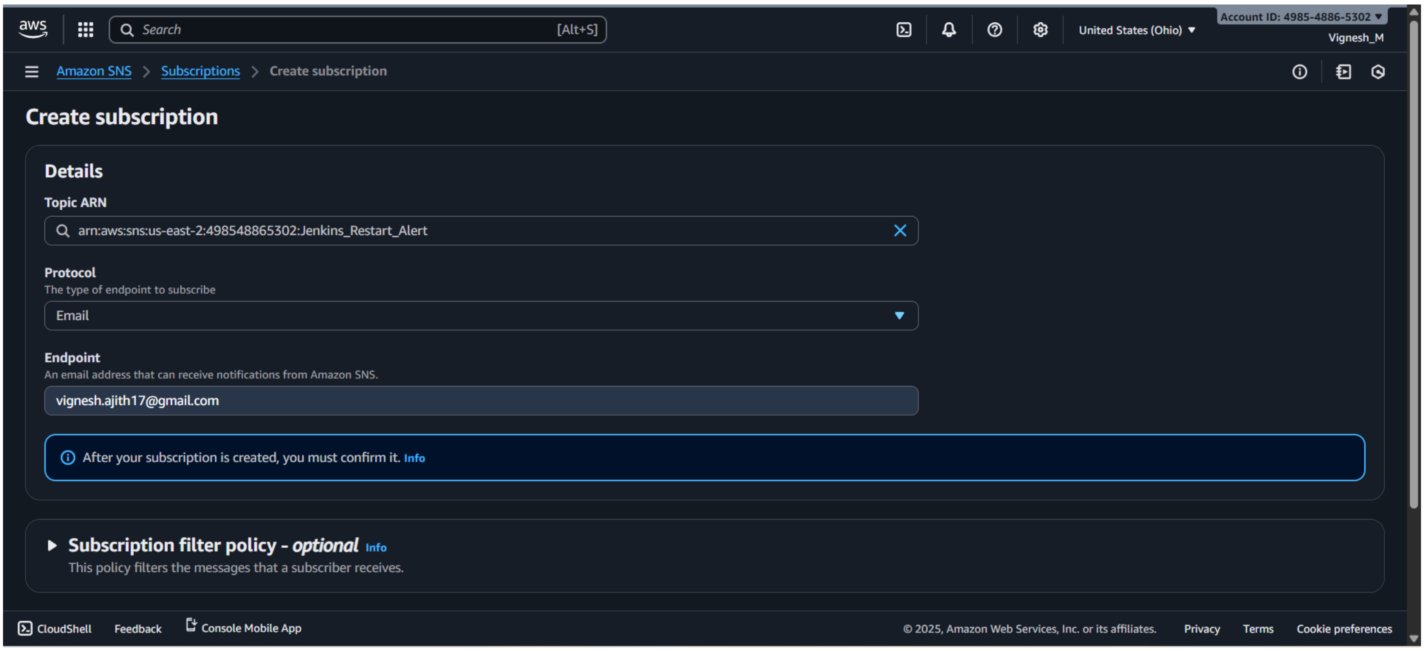
Search [Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
Amazon SNS > Subscriptions > Create subscription
Create subscription
Details
Topic ARN
arn:aws:sns:us-east-2:498548865302:Jenkins_Restart_Alert
Protocol
The type of endpoint to subscribe
Email
Endpoint
An email address that can receive notifications from Amazon SNS.
vignesh.ajith17@gmail.com
After your subscription is created, you must confirm it. Info
Subscription filter policy - optional Info
This policy filters the messages that a subscriber receives.
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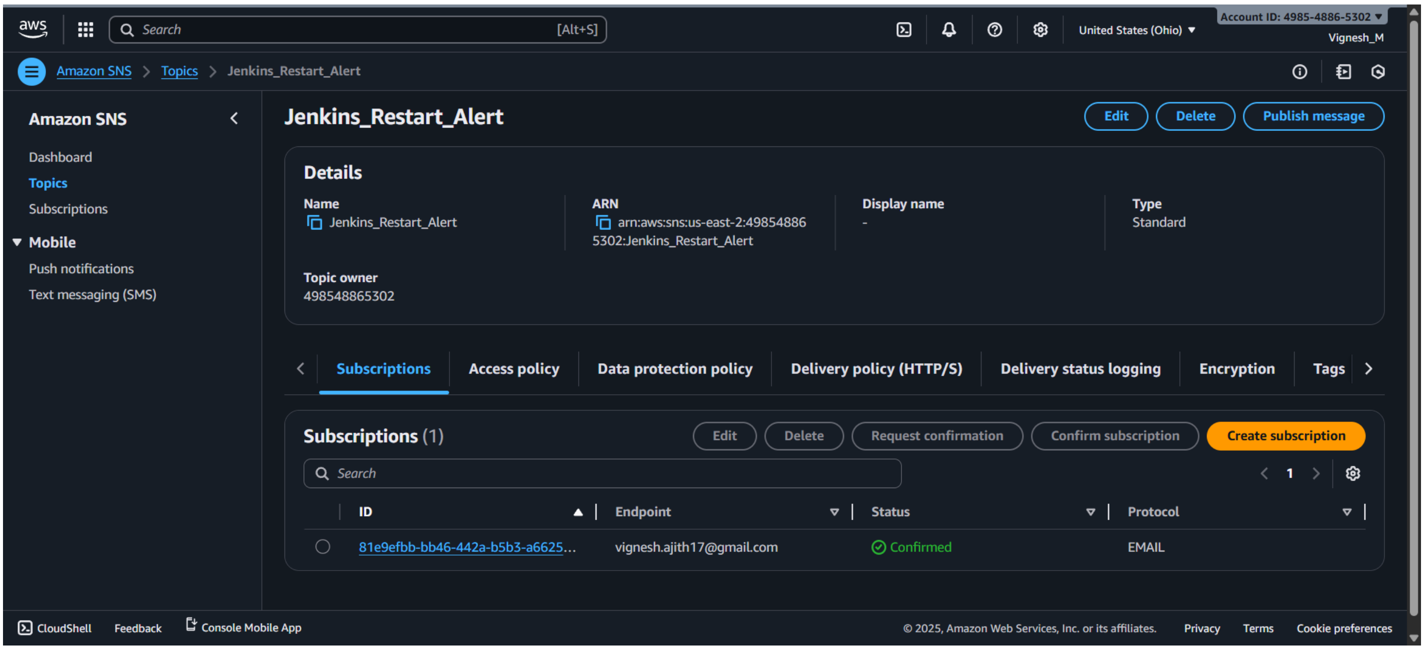
Search [Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
Amazon SNS > Topics > Jenkins_Restart_Alert
Amazon SNS
Dashboard
Topics
Subscriptions
Mobile
Push notifications
Text messaging (SMS)
Jenkins_Restart_Alert
Edit Delete Publish message
Details
Name
Jenkins_Restart_Alert
ARN
arn:aws:sns:us-east-2:498548865302:Jenkins_Restart_Alert
Display name
-
Type
Standard
Topic owner
498548865302
Subscriptions Access policy Data protection policy Delivery policy (HTTP/S) Delivery status logging Encryption Tags
Subscriptions (1)
Edit Delete Request confirmation Confirm subscription Create subscription
Search
ID Endpoint Status Protocol
81e9efbb-bb46-442a-b5b3-a6625... vignesh.ajith17@gmail.com Confirmed EMAIL
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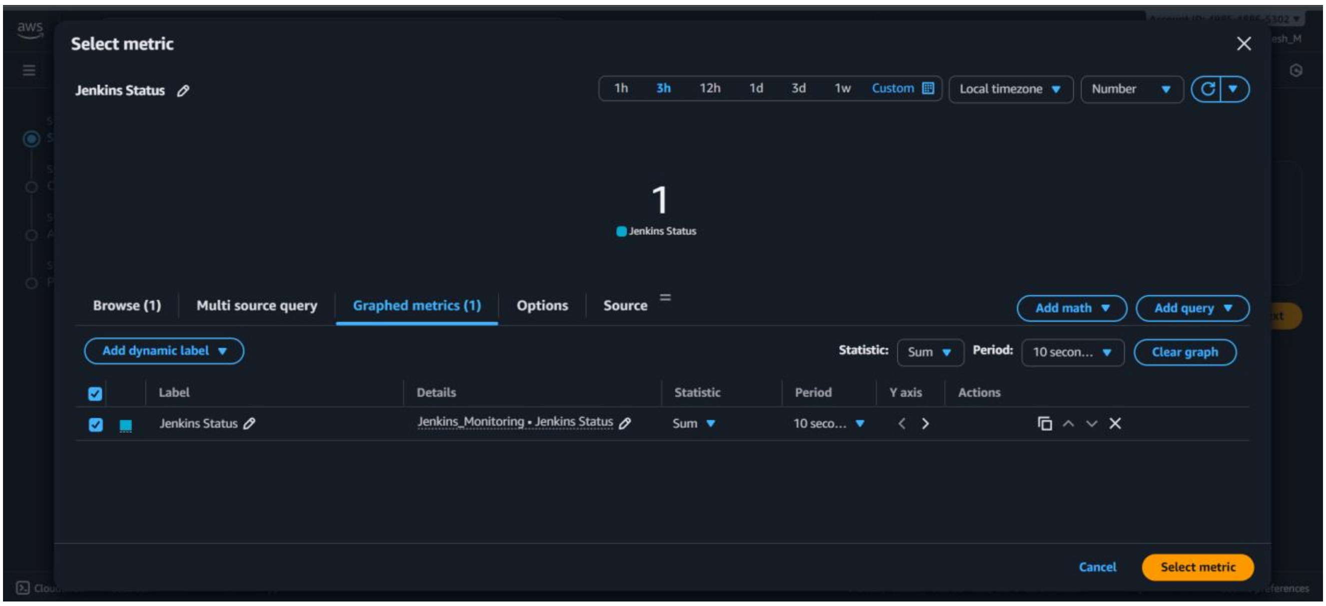
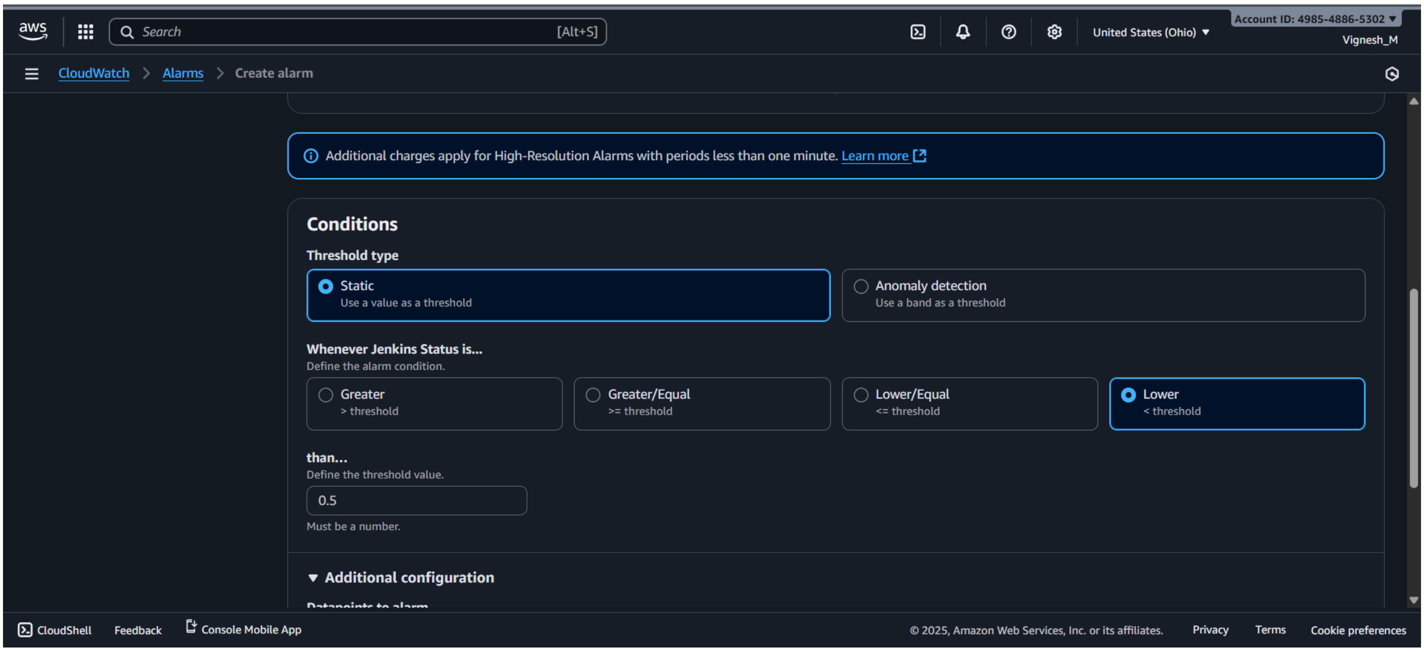
Step 6: Create CloudWatch Alarms for Jenkins Down and Restart Events:

Procedure:

- Create an alarm for Jenkins Down and Jenkins Restart Events.
  - Amazon cloud watch → Create Alaram → Select namespace → Select namespace → Select metrics → Set Statistics and Period → select threshold type → Select Jenkins status is → Set threshold value → Set Additional configuration → Select SNS Topic → Finally Create Alarm
- **Alarm for Jenkins Down Events:**
  - **Metric Namespace:** Jenkins_Monitoring
  - **Metric Name:** Jenkins Status
  - **Statistics:** Sum
  - **Periods:** 10 Seconds
  - **Threshold Type:** Static
  - **Jenkins Status is:** Lower
  - **Threshold Value:** 0.5
  - **Missing Data Treatment:** Treat missing data as ignore (maintain the alarm state)
  - **Alarm State Tigger:** In Alarm
  - **Select SNS Topic:** Jenkins_Status_Alerts
- **Alarm for Jenkins Restart Events:**
  - **Metric Namespace:** Jenkins_Monitoring
  - **Metric Name:** Jenkins Restart
  - **Statistics:** Sum
  - **Periods:** 10 Seconds
  - **Threshold Type:** Static
  - **Jenkins Status is:** Greater/Equal
  - **Threshold Value:** 1
  - **Missing Data Treatment:** Treat missing data as good (not breaching threshold)
  - **Alarm State Tigger:** In Alarm
  - **Select SNS Topic:** Jenkins_Restart_Alerts
- Create a Dashboard with all the necessary metrics and alarms.
- **Result:**
  - Alarm for Jenkins Down and Jenkins Restart was created successfully.
  - Dashboard created with necessary metrics and alarms.




CloudWatch > Alarms > Create alarm
Additional charges apply for High-Resolution Alarms with periods less than one minute. Learn more
Conditions
Threshold type
Static
Use a value as a threshold
Anomaly detection
Use a band as a threshold
Whenever Jenkins Status is...
Define the alarm condition.
Greater
> threshold
Greater/Equal
>= threshold
Lower/Equal
<= threshold
Lower
< threshold
than...
Define the threshold value.
0.5
Must be a number.
Additional configuration
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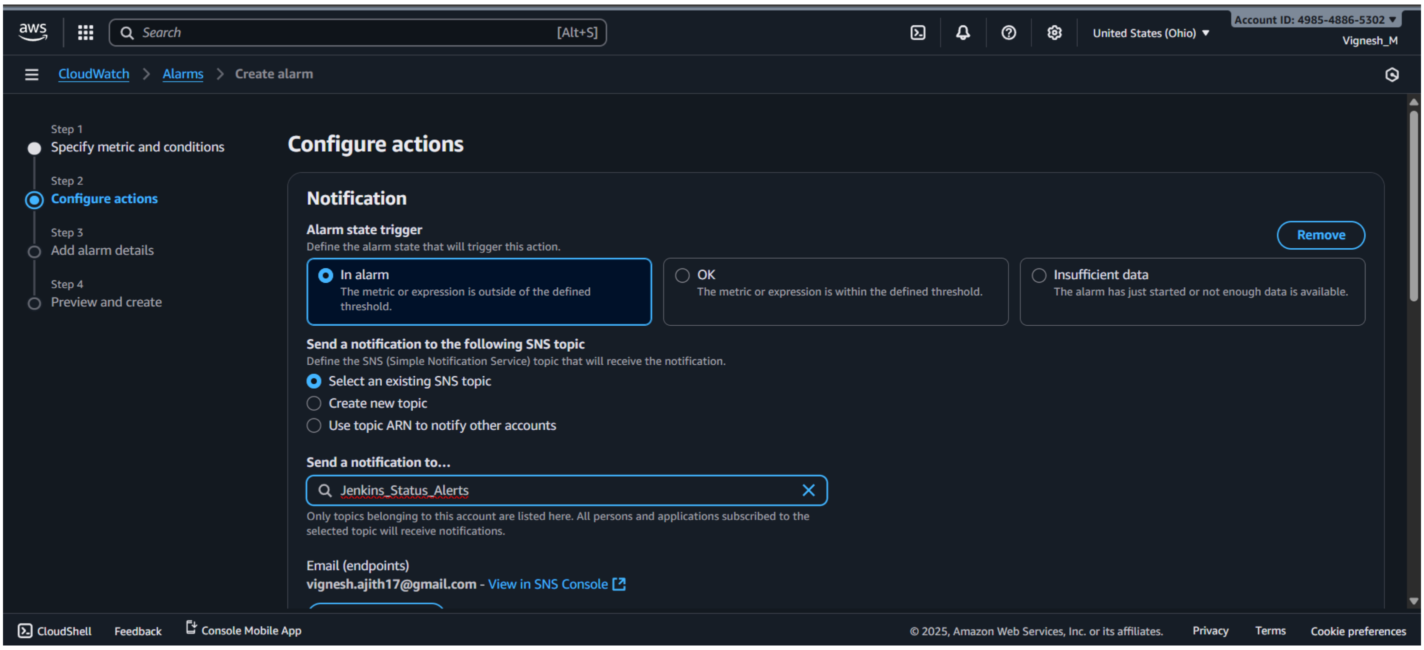
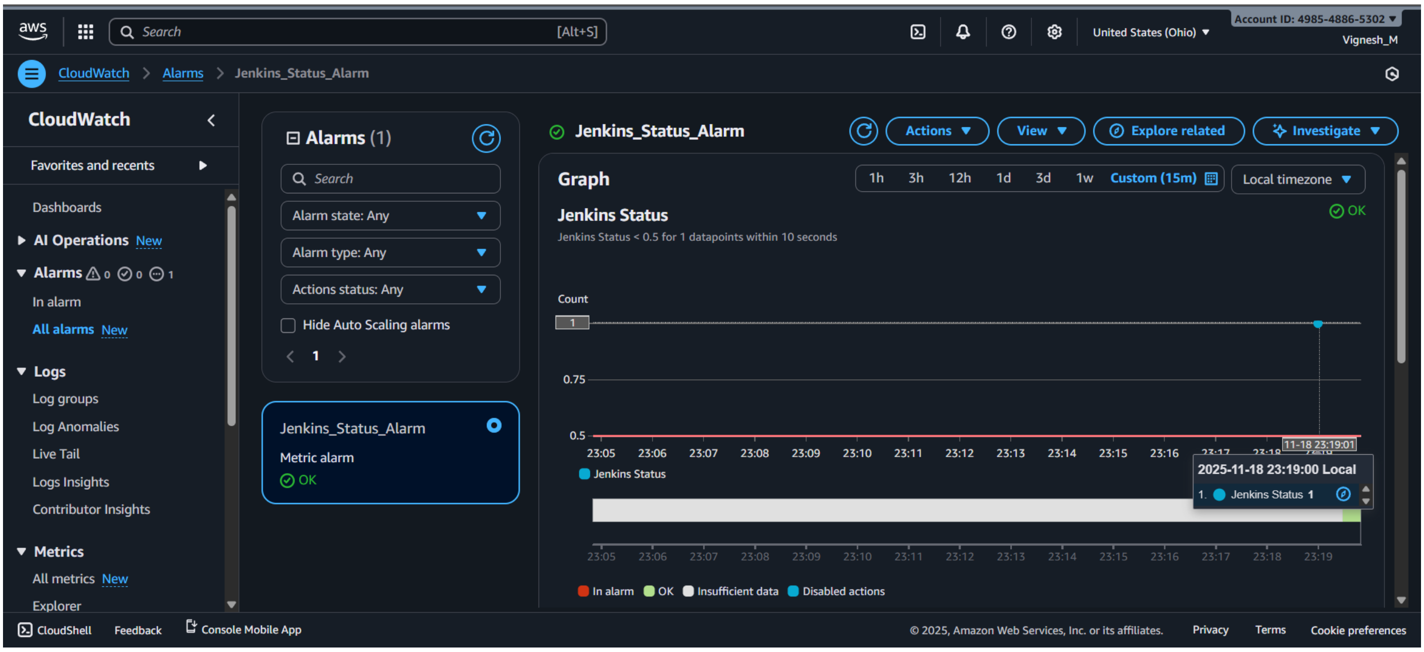
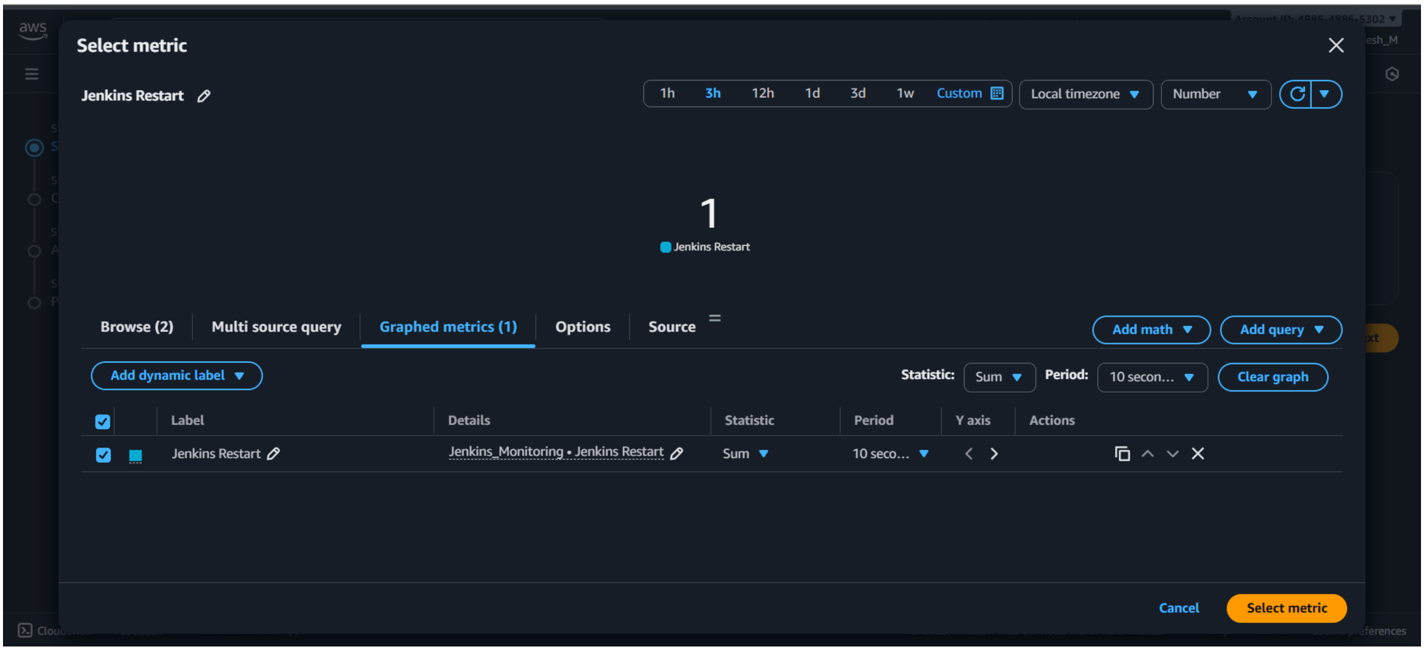
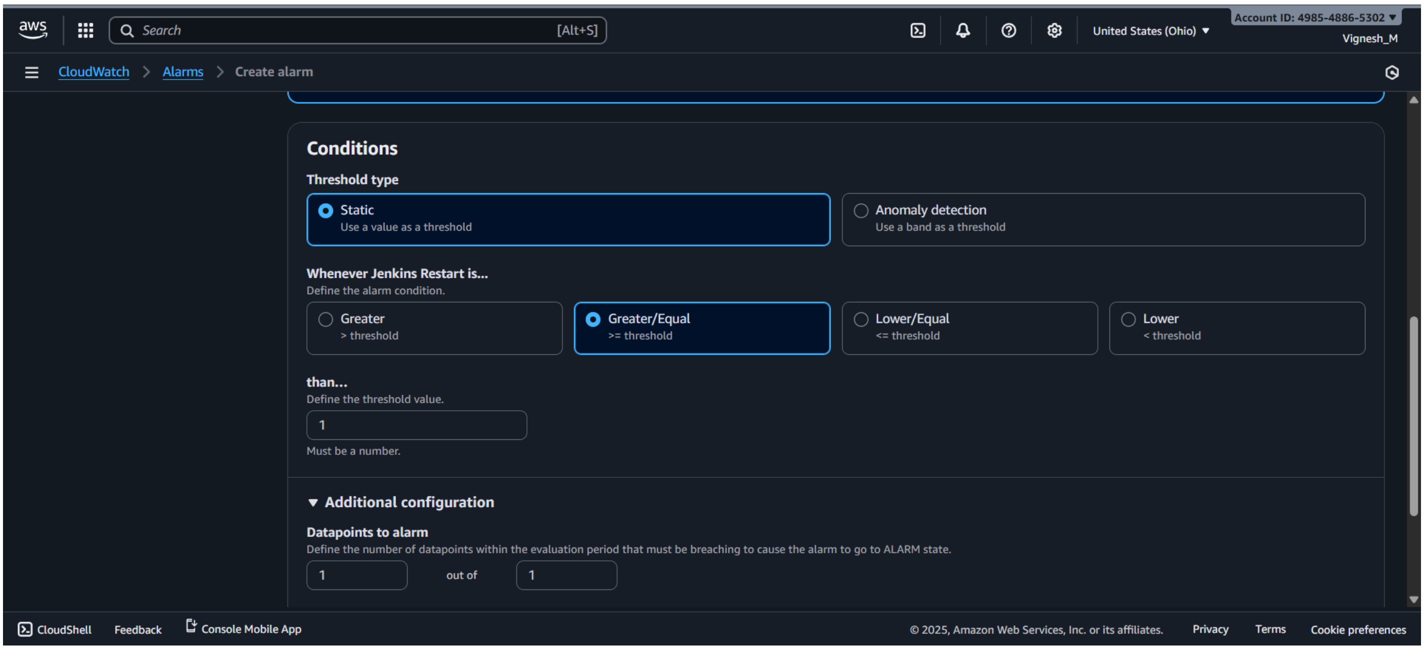

CloudWatch > Alarms > Create alarm
Step 1
Specify metric and conditions
Step 2
Configure actions
Step 3
Add alarm details
Step 4
Preview and create
Configure actions
Notification
Alarm state trigger
Define the alarm state that will trigger this action.
Remove
In alarm
The metric or expression is outside of the defined threshold.
OK
The metric or expression is within the defined threshold.
Insufficient data
The alarm has just started or not enough data is available.
Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.
Select an existing SNS topic
Create new topic
Use topic ARN to notify other accounts
Send a notification to...
Jenkins_Status_Alerts
Only topics belonging to this account are listed here. All persons and applications subscribed to the selected topic will receive notifications.
Email (endpoints)
vignesh.ajith17@gmail.com - View in SNS Console


CloudWatch > Alarms > Jenkins_Status_Alarm
CloudWatch
Favorites and recents
Dashboards
AI Operations New
Alarms
In alarm
All alarms New
Logs
Log groups
Log Anomalies
Live Tail
Logs Insights
Contributor Insights
Metrics
All metrics New
Explorer
Alarms (1)
Alarm state: Any
Alarm type: Any
Actions status: Any
Hide Auto Scaling alarms
Jenkins_Status_Alarm
Metric alarm
OK
Jenkins_Status_Alarm
Actions
View
Explore related
Investigate
Graph
1h 3h 12h 1d 3d 1w Custom (15m)
Local timezone
Jenkins Status
Jenkins Status < 0.5 for 1 datapoints within 10 seconds
OK
Count
2025-11-18 23:19:00 Local
Jenkins Status 1
In alarm
OK
Insufficient data
Disabled actions


Select metric
Jenkins Restart
1h 3h 12h 1d 3d 1w Custom
Local timezone
Number
1
Jenkins Restart
Browse (2)
Multi source query
Graphed metrics (1)
Options
Source
Add math
Add query
Add dynamic label
Statistic: Sum
Period: 10 secon...
Clear graph
Label Details Statistic Period Y axis Actions
Jenkins Restart Jenkins_Monitoring • Jenkins Restart Sum 10 seco...
Cancel
Select metric


aws
Search [Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
CloudWatch > Alarms > Create alarm
Conditions
Threshold type
Static
Use a value as a threshold
Anomaly detection
Use a band as a threshold
Whenever Jenkins Restart is...
Define the alarm condition.
Greater
> threshold
Greater/Equal
>= threshold
Lower/Equal
<= threshold
Lower
< threshold
than...
Define the threshold value.
1
Must be a number.
Additional configuration
Datapoints to alarm
Define the number of datapoints within the evaluation period that must be breaching to cause the alarm to go to ALARM state.
1 out of 1
CloudShell Feedback Console Mobile App
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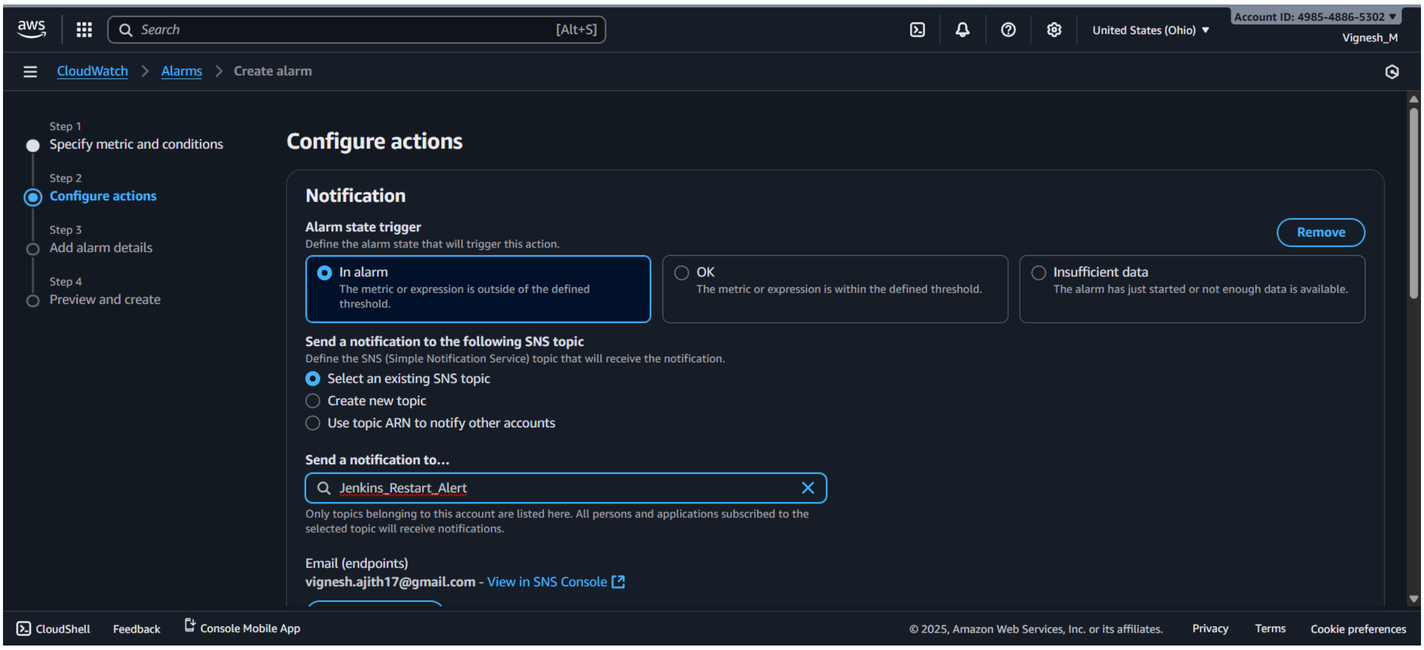
aws
Search [Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
CloudWatch > Alarms > Create alarm
Step 1
Specify metric and conditions
Step 2
Configure actions
Step 3
Add alarm details
Step 4
Preview and create
Configure actions
Notification
Alarm state trigger
Define the alarm state that will trigger this action.
Remove
In alarm
The metric or expression is outside of the defined threshold.
OK
The metric or expression is within the defined threshold.
Insufficient data
The alarm has just started or not enough data is available.
Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.
Select an existing SNS topic
Create new topic
Use topic ARN to notify other accounts
Send a notification to...
Jenkins_Restart_Alert
Only topics belonging to this account are listed here. All persons and applications subscribed to the selected topic will receive notifications.
Email (endpoints)
vignesh.ajith17@gmail.com - View in SNS Console
CloudShell Feedback Console Mobile App
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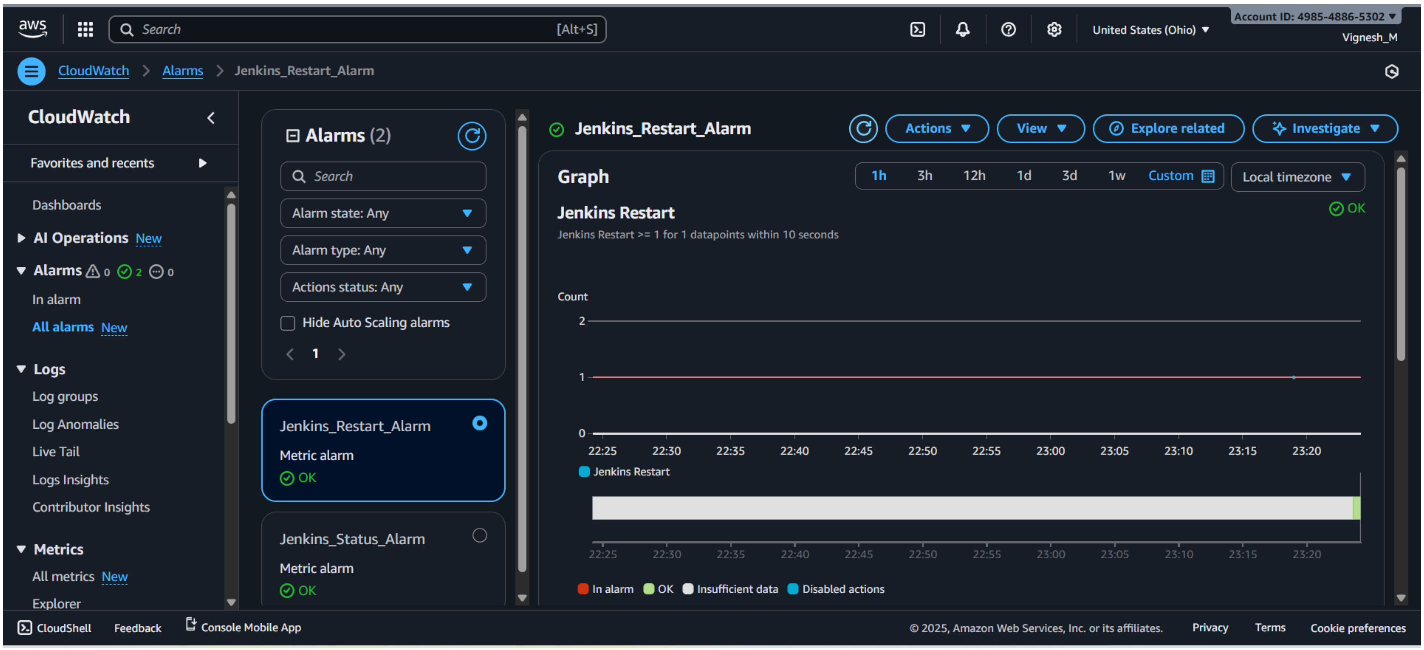
CloudWatch > Alarms > Jenkins_Restart_Alarm
CloudWatch
Favorites and recents
Dashboards
AI Operations New
Alarms 0 2 0
In alarm
All alarms New
Logs
Log groups
Log Anomalies
Live Tail
Logs Insights
Contributor Insights
Metrics
All metrics New
Explorer
Alarms (2)
Search
Alarm state: Any
Alarm type: Any
Actions status: Any
Hide Auto Scaling alarms
Jenkins_Restart_Alarm
Metric alarm
OK
Jenkins_Status_Alarm
Metric alarm
OK
Jenkins_Restart_Alarm
Actions
View
Explore related
Investigate
Graph
1h 3h 12h 1d 3d 1w Custom
Local timezone
Jenkins Restart
OK
Jenkins Restart >= 1 for 1 datapoints within 10 seconds
Count
2
1
0
22:25 22:30 22:35 22:40 22:45 22:50 22:55 23:00 23:05 23:10 23:15 23:20
Jenkins Restart
In alarm
OK
Insufficient data
Disabled actions
CloudShell
Feedback
Console Mobile App
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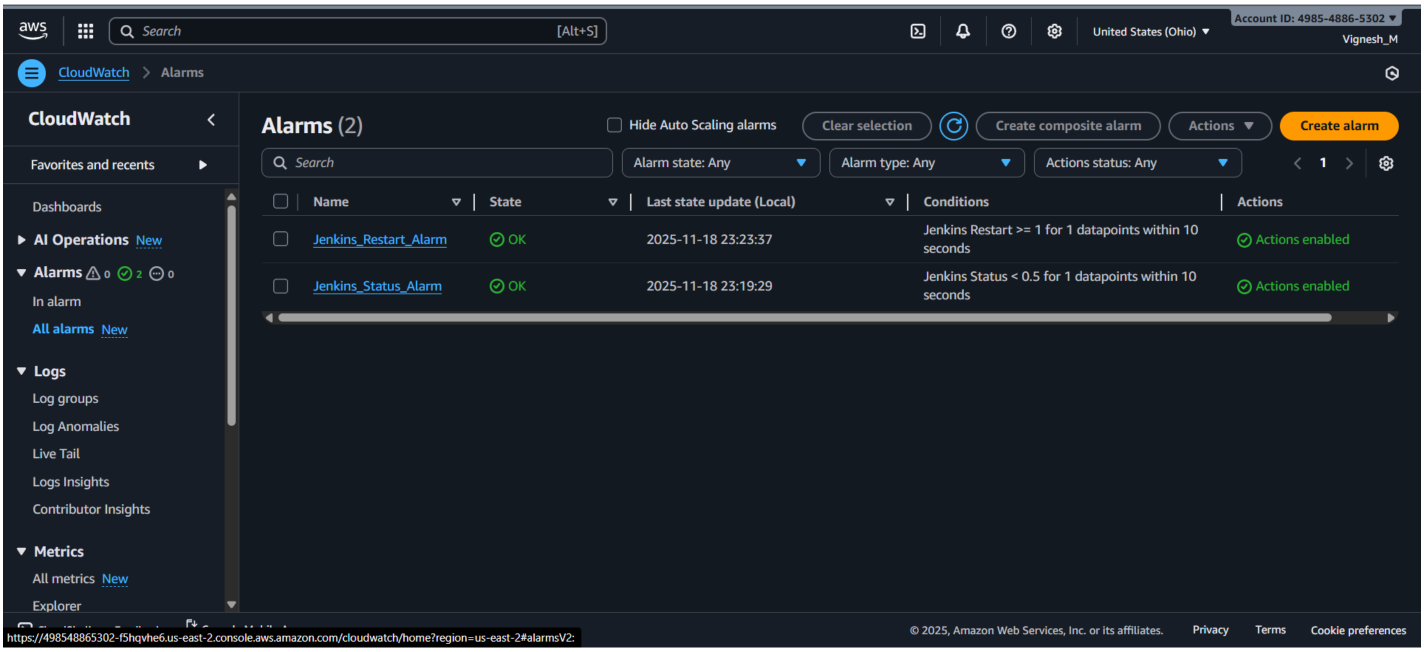
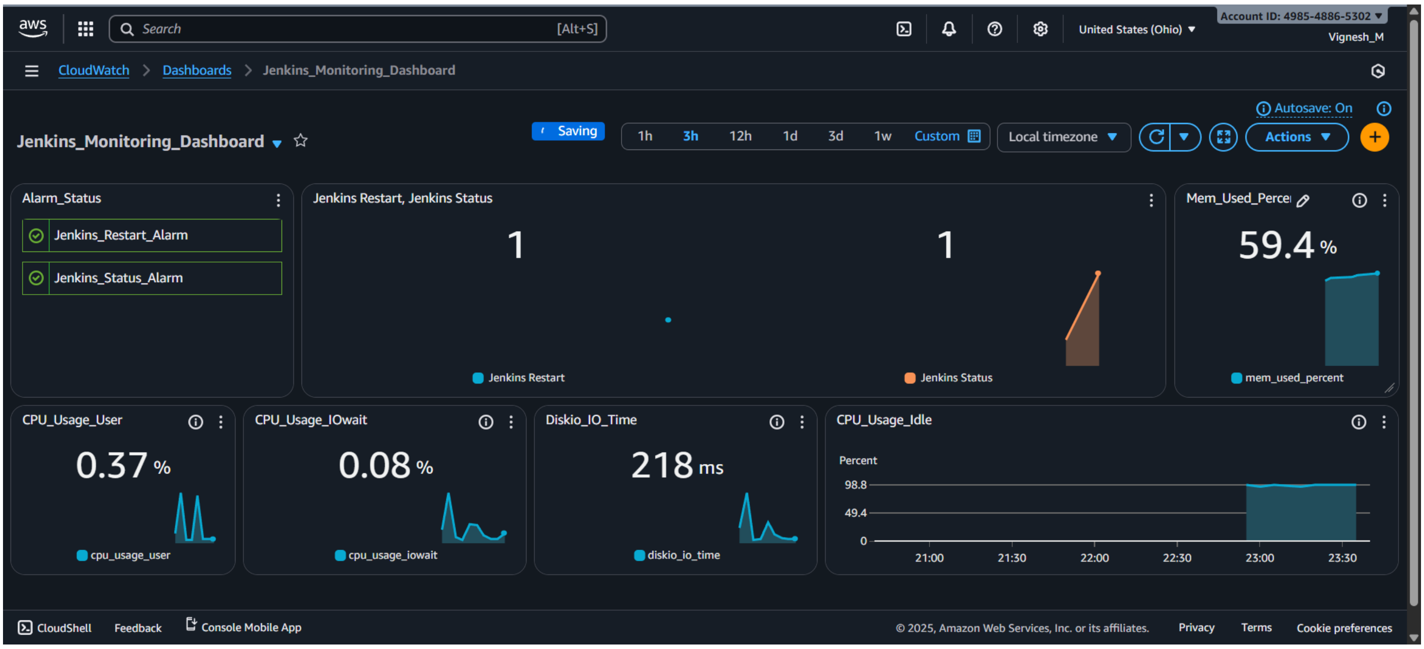
Cookie preferences
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M

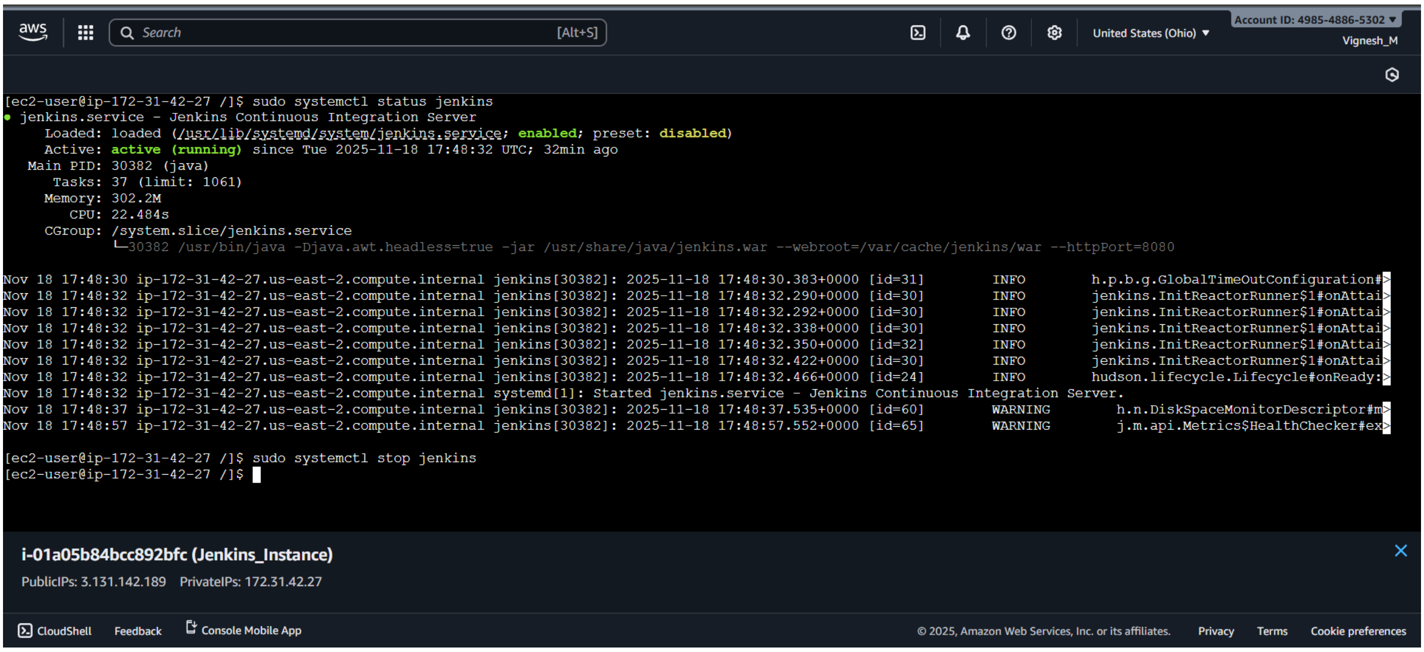

CloudWatch > Alarms
Alarms (2)
Hide Auto Scaling alarms
Clear selection
Create composite alarm
Actions
Create alarm
Search
Alarm state: Any
Alarm type: Any
Actions status: Any
Name
State
Last state update (Local)
Conditions
Actions
Jenkins_Restart_Alarm
OK
2025-11-18 23:23:37
Jenkins Restart >= 1 for 1 datapoints within 10 seconds
Actions enabled
Jenkins_Status_Alarm
OK
2025-11-18 23:19:29
Jenkins Status < 0.5 for 1 datapoints within 10 seconds
Actions enabled
https://498548865302-f5hqvhe6.us-east-2.console.aws.amazon.com/cloudwatch/home?region=us-east-2#alarmsV2:

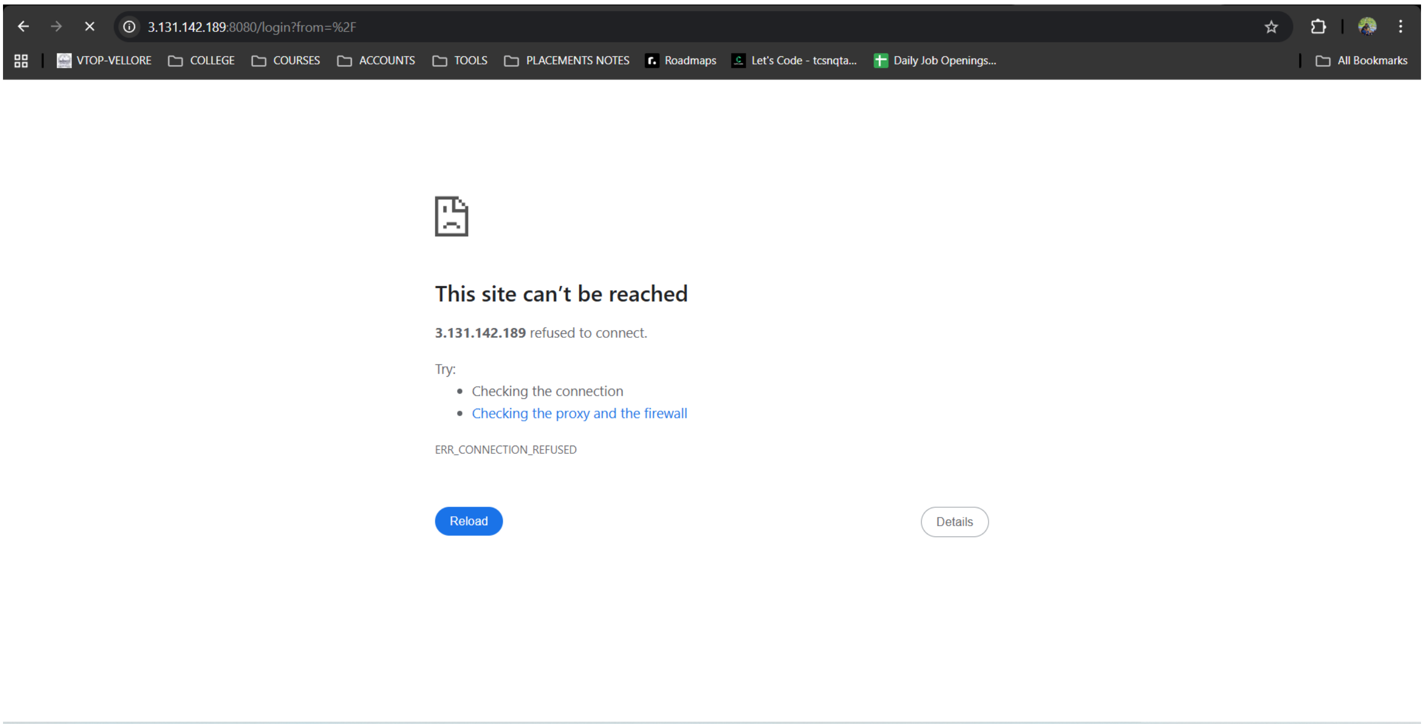

CloudWatch > Dashboards > Jenkins_Monitoring_Dashboard
Jenkins_Monitoring_Dashboard
Saving
Autosave: On
1h 3h 12h 1d 3d 1w Custom
Local timezone
Actions
Alarm_Status
Jenkins_Restart_Alarm
Jenkins_Status_Alarm
Jenkins Restart, Jenkins Status
1
1
Jenkins Restart
Jenkins Status
Mem_Used_Perce
59.4 %
mem_used_percent
CPU_Usage_User
0.37 %
cpu_usage_user
CPU_Usage_IOwait
0.08 %
cpu_usage_iowait
Diskio_IO_Time
218 ms
diskio_io_time
CPU_Usage_Idle
Percent
98.8
49.4
0
21:00 21:30 22:00 22:30 23:00 23:30

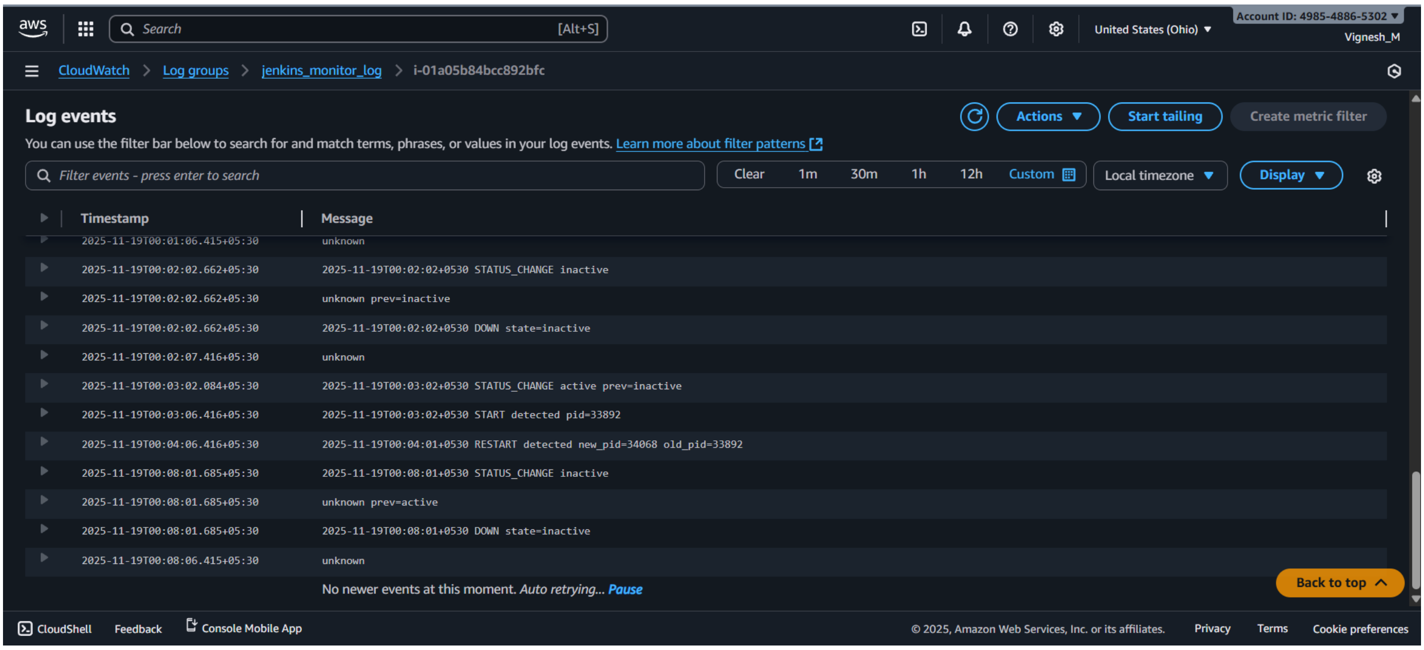
Step 7: Test Alarm Trigger by Manually Shutting Down Jenkins Server:

Procedure:

- In EC2 console check the Jenkins status and down the Jenkins server.
  - **Command 1:** sudo systemctl status jenkins
  - **Command 2:** sudo systemctl stop Jenkins
- After stopping the Jenkins, the web interface will go down.
- After one minute the cronjob will tigger and log will be sent to Jenkins_monitor.log
- Cloud watch agent will fetch that log and send it to the cloud watch console.
- After log comes to cloud watch, after sometimes Jenkins Status metric will goes to 0.
- Jenkins_Status_Alarm get triggered immediately and we will receive the Email notification through SNS.
- **Result:**
  - Jenkins server went down and Web interface not able to access.
  - Cronjob triggered and log was fetched by the cloud watch agent correctly.
  - Jenkins Status metrics changed, and Jenkins_Status_Alarm triggered successfully.
  - Jenkins Alert was sent to Email through SNS correctly.




3.131.142.189:8080/login?from=%2F
VTOP-VELLORE
COLLEGE
COURSES
ACCOUNTS
TOOLS
PLACEMENTS NOTES
Roadmaps
Let's Code - tcsnqta...
Daily Job Openings...
All Bookmarks
This site can't be reached
3.131.142.189 refused to connect.
Try:
Checking the connection
Checking the proxy and the firewall
ERR_CONNECTION_REFUSED
Reload
Details


aws
Search
[Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
CloudWatch
Log groups
jenkins_monitor_log
i-01a05b84bcc892bfc
Log events
Actions
Start tailing
Create metric filter
You can use the filter bar below to search for and match terms, phrases, or values in your log events. Learn more about filter patterns
Filter events - press enter to search
Clear
1m
30m
1h
12h
Custom
Local timezone
Display
Timestamp
Message
2025-11-19T00:01:06.415+05:30 unknown
2025-11-19T00:02:02.662+05:30 2025-11-19T00:02:02+0530 STATUS_CHANGE inactive
2025-11-19T00:02:02.662+05:30 unknown prev=inactive
2025-11-19T00:02:02.662+05:30 2025-11-19T00:02:02+0530 DOWN state=inactive
2025-11-19T00:02:07.416+05:30 unknown
2025-11-19T00:03:02.084+05:30 2025-11-19T00:03:02+0530 STATUS_CHANGE active prev=inactive
2025-11-19T00:03:06.416+05:30 2025-11-19T00:03:02+0530 START detected pid=33892
2025-11-19T00:04:06.416+05:30 2025-11-19T00:04:01+0530 RESTART detected new_pid=34068 old_pid=33892
2025-11-19T00:08:01.685+05:30 2025-11-19T00:08:01+0530 STATUS_CHANGE inactive
2025-11-19T00:08:01.685+05:30 unknown prev=active
2025-11-19T00:08:01.685+05:30 2025-11-19T00:08:01+0530 DOWN state=inactive
2025-11-19T00:08:06.415+05:30 unknown
No newer events at this moment. Auto retrying... Pause
Back to top
CloudShell
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Console Mobile App
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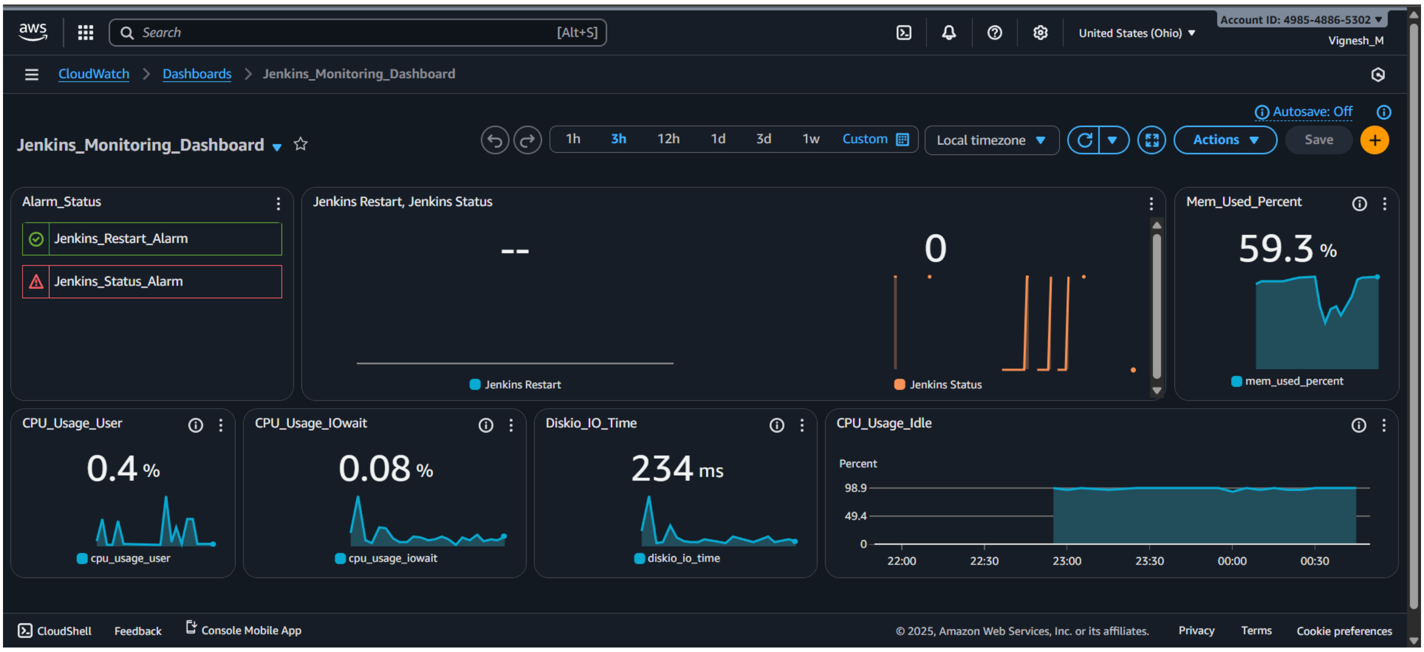
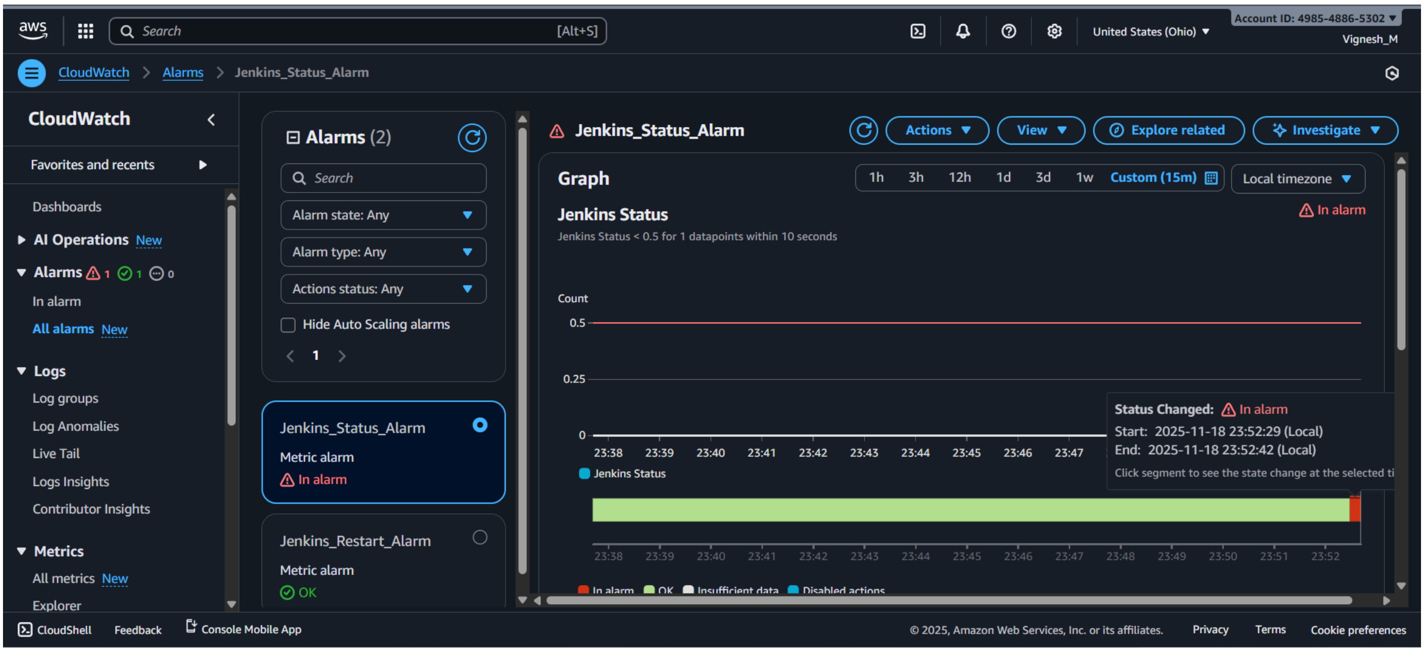




ALARM: "Jenkins_Status_Alarm" in US East (Ohio)

Inbox ×

AWS Notifications <no-reply@sns.amazonaws.com> 00:51 (0 minutes ago)

to me

You are receiving this email because your Amazon CloudWatch Alarm "Jenkins_Status_Alarm" in the US East (Ohio) region has entered the ALARM state, because "Threshold Crossed: 1 out of the last 1 datapoints [0.0 (18/11/25 19:21:00)] was less than the threshold (0.5) (minimum 1 datapoint for OK -> ALARM transition)." at "Tuesday 18 November, 2025 19:21:28 UTC".

View this alarm in the AWS Management Console:
https://us-east-2.console.aws.amazon.com/cloudwatch/deeplink.js?region=us-east-2#alarmsV2:alarm/Jenkins_Status_Alarm

Alarm Details:
- Name: Jenkins_Status_Alarm
- Description: # !! Important !!

Jenkins Server has went down just now. Please look into it and resolve the issue.

Thanks!
- State Change: OK -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [0.0 (18/11/25 19:21:00)] was less than the threshold (0.5) (minimum 1 datapoint for OK -> ALARM transition).
- Timestamp: Tuesday 18 November, 2025 19:21:28 UTC
- AWS Account: 498548865302
- Alarm Arn: arn:aws:cloudwatch:us-east-2:498548865302:alarm:Jenkins_Status_Alarm

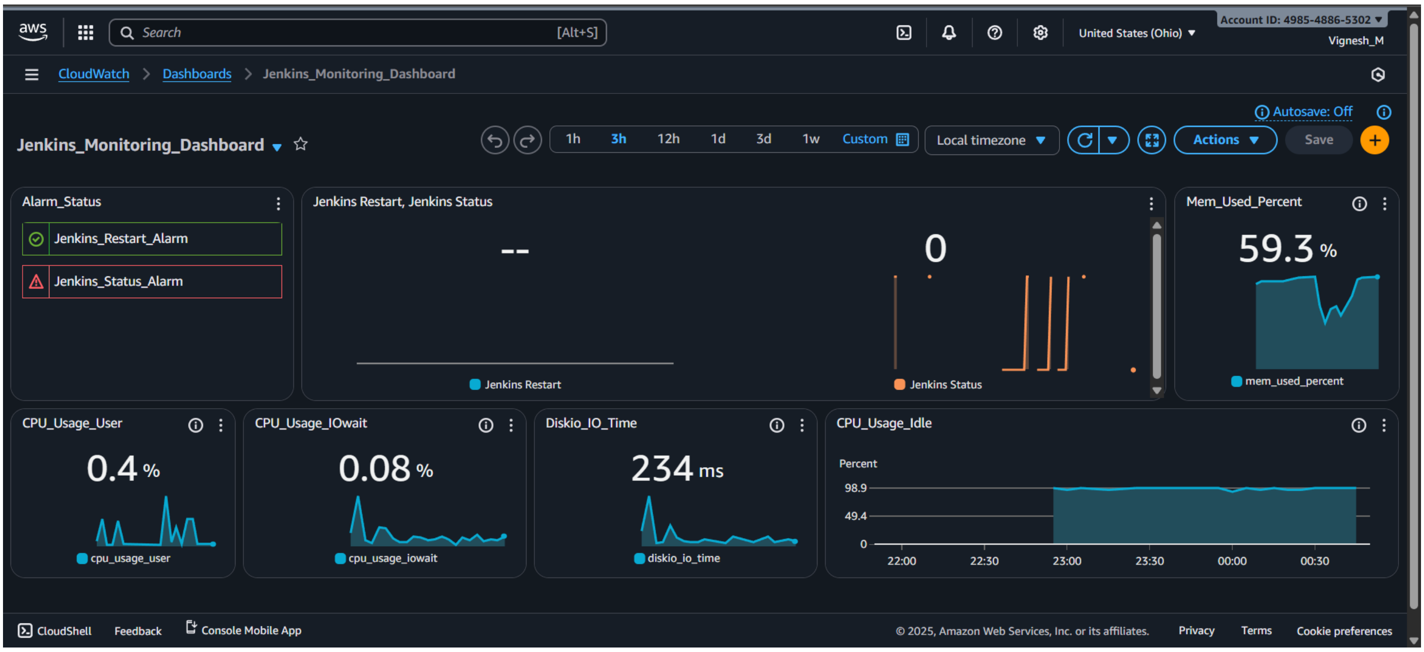
Threshold:
- The alarm is in the ALARM state when the metric is LessThanThreshold 0.5 for at least 1 of the last 1 period(s) of 10 seconds.

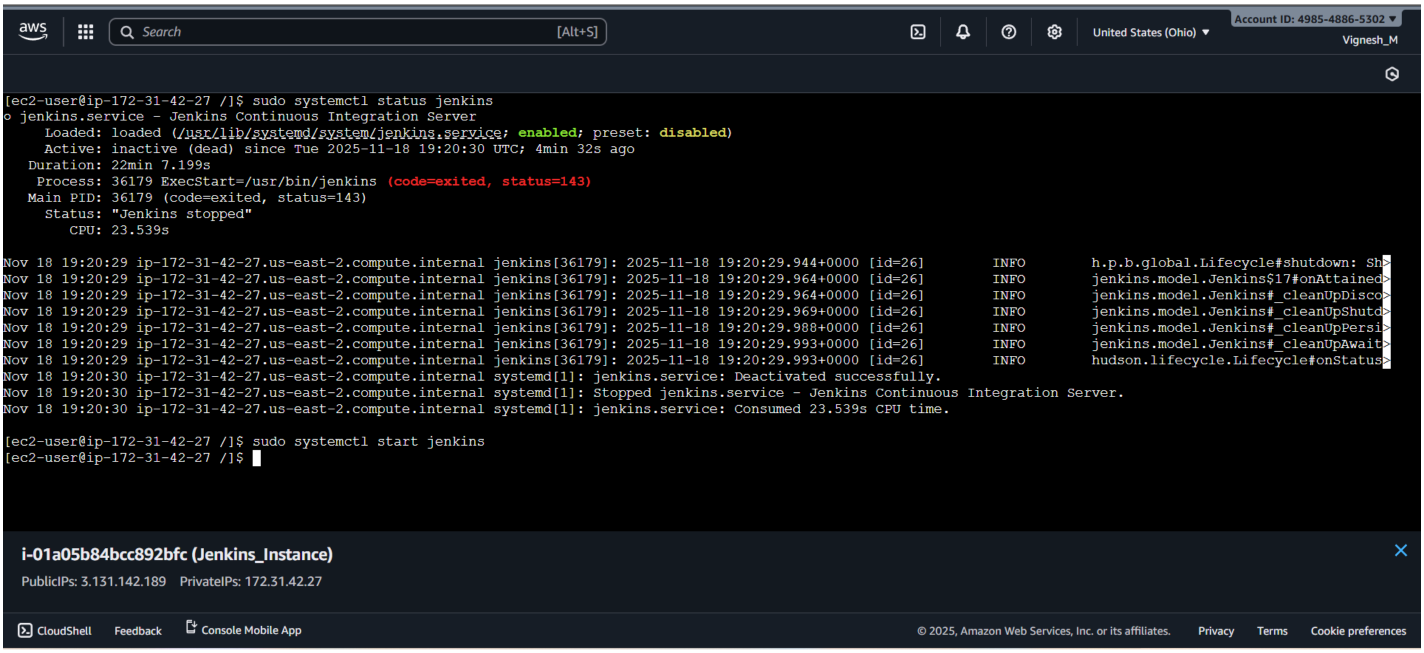
Monitored Metric:

Step 8: Test Alarm Status by Manually Starting Jenkins Server:

Procedure:

- In EC2 console check the Jenkins status and up the Jenkins server.
  - **Command 1:** sudo systemctl status jenkins
  - **Command 2:** sudo systemctl Up Jenkins
- After Starting the Jenkins, the web interface will go work.
- After one minute the cronjob will tigger and log will be sent to Jenkins_monitor.log
- Cloud watch agent will fetch that log and send it to the cloud watch console.
- After log comes to cloud watch, after sometimes Jenkins Status metric will goes to 1.
- Jenkins_Status_Alarm get triggered immediately and it will go back to OK state.
- **Result:**
  - Jenkins server Started and Web interface came and able to access.
  - Cronjob triggered and log was fetched by the cloud watch agent correctly.
  - Jenkins Status metrics changed, and Jenkins_Status_Alarm triggered successfully.
  - Jenkins_Status_Alarm went back to OK state.




aws
Search [Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
[ec2-user@ip-172-31-42-27 /]$ sudo systemctl status jenkins
o jenkins.service - Jenkins Continuous Integration Server
Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: disabled)
Active: inactive (dead) since Tue 2025-11-18 19:20:30 UTC; 4min 32s ago
Duration: 22min 7.199s
Process: 36179 ExecStart=/usr/bin/jenkins (code=exited, status=143)
Main PID: 36179 (code=exited, status=143)
Status: "Jenkins stopped"
CPU: 23.539s
Nov 18 19:20:29 ip-172-31-42-27.us-east-2.compute.internal jenkins[36179]: 2025-11-18 19:20:29.944+0000 [id=26] INFO h.p.b.global.Lifecycle#shutdown: Sh
Nov 18 19:20:29 ip-172-31-42-27.us-east-2.compute.internal jenkins[36179]: 2025-11-18 19:20:29.964+0000 [id=26] INFO jenkins.model.Jenkins$17#onAttained
Nov 18 19:20:29 ip-172-31-42-27.us-east-2.compute.internal jenkins[36179]: 2025-11-18 19:20:29.964+0000 [id=26] INFO jenkins.model.Jenkins#_cleanUpDisco
Nov 18 19:20:29 ip-172-31-42-27.us-east-2.compute.internal jenkins[36179]: 2025-11-18 19:20:29.969+0000 [id=26] INFO jenkins.model.Jenkins#_cleanUpShutd
Nov 18 19:20:29 ip-172-31-42-27.us-east-2.compute.internal jenkins[36179]: 2025-11-18 19:20:29.988+0000 [id=26] INFO jenkins.model.Jenkins#_cleanUpPersi
Nov 18 19:20:29 ip-172-31-42-27.us-east-2.compute.internal jenkins[36179]: 2025-11-18 19:20:29.993+0000 [id=26] INFO jenkins.model.Jenkins#_cleanUpAwait
Nov 18 19:20:29 ip-172-31-42-27.us-east-2.compute.internal jenkins[36179]: 2025-11-18 19:20:29.993+0000 [id=26] INFO hudson.lifecycle.Lifecycle#onStatus
Nov 18 19:20:30 ip-172-31-42-27.us-east-2.compute.internal systemd[1]: jenkins.service: Deactivated successfully.
Nov 18 19:20:30 ip-172-31-42-27.us-east-2.compute.internal systemd[1]: Stopped jenkins.service - Jenkins Continuous Integration Server.
Nov 18 19:20:30 ip-172-31-42-27.us-east-2.compute.internal systemd[1]: jenkins.service: Consumed 23.539s CPU time.
[ec2-user@ip-172-31-42-27 /]$ sudo systemctl start jenkins
[ec2-user@ip-172-31-42-27 /]$
i-01a05b84bcc892bfc (Jenkins_Instance)
PublicIPs: 3.131.142.189 PrivateIPs: 172.31.42.27
CloudShell Feedback Console Mobile App
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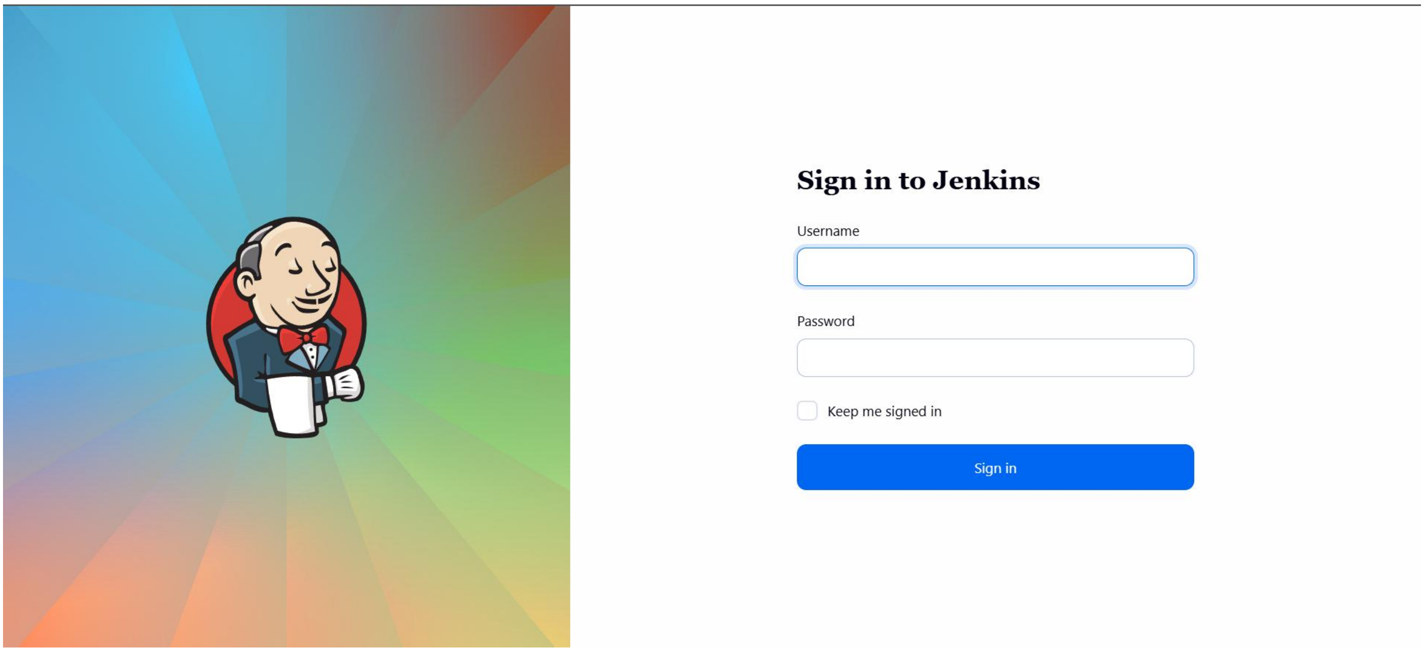
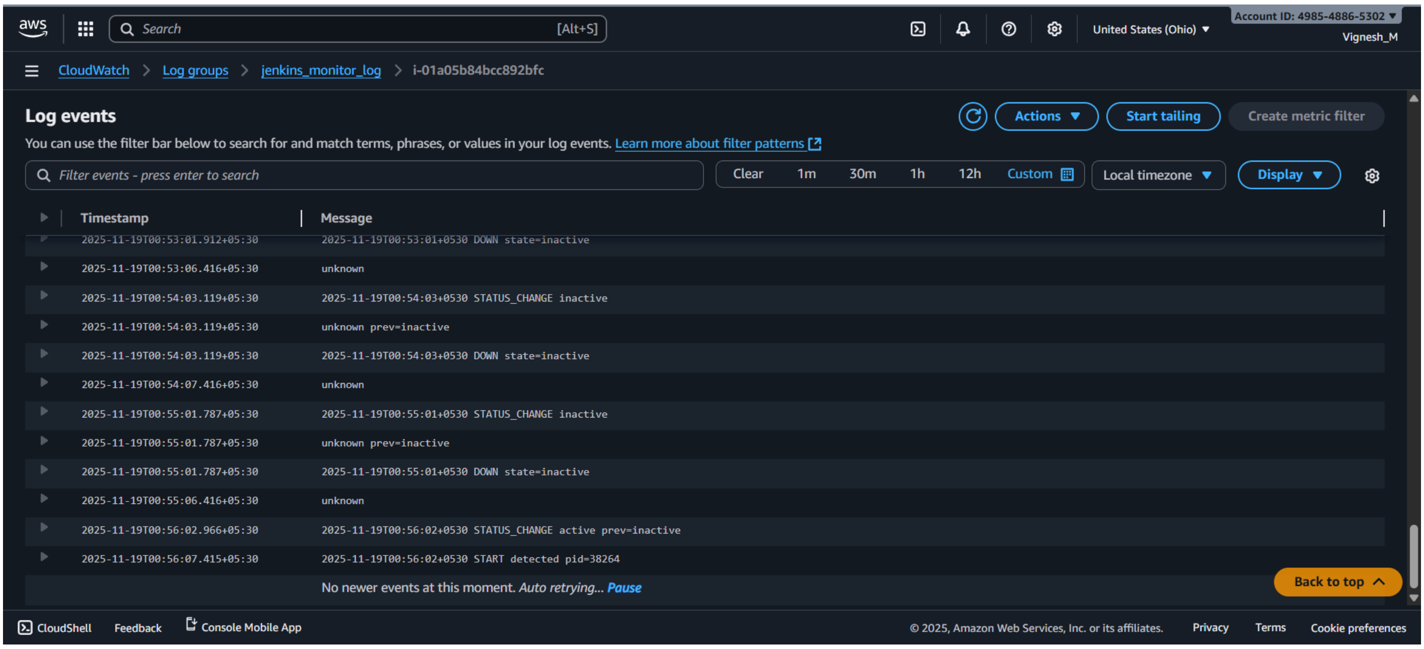

Sign in to Jenkins
Username
Password
Keep me signed in
Sign in


aws
Search [Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
CloudWatch > Log groups > jenkins_monitor_log > i-01a05b84bcc892bfc
Log events
Actions Start tailing Create metric filter
You can use the filter bar below to search for and match terms, phrases, or values in your log events. Learn more about filter patterns
Filter events - press enter to search
Clear 1m 30m 1h 12h Custom Local timezone Display
Timestamp Message
2025-11-19T00:53:01.912+05:30 2025-11-19T00:53:01+0530 DOWN state=inactive
2025-11-19T00:53:06.416+05:30 unknown
2025-11-19T00:54:03.119+05:30 2025-11-19T00:54:03+0530 STATUS_CHANGE inactive
2025-11-19T00:54:03.119+05:30 unknown prev=inactive
2025-11-19T00:54:03.119+05:30 2025-11-19T00:54:03+0530 DOWN state=inactive
2025-11-19T00:54:07.416+05:30 unknown
2025-11-19T00:55:01.787+05:30 2025-11-19T00:55:01+0530 STATUS_CHANGE inactive
2025-11-19T00:55:01.787+05:30 unknown prev=inactive
2025-11-19T00:55:01.787+05:30 2025-11-19T00:55:01+0530 DOWN state=inactive
2025-11-19T00:55:06.416+05:30 unknown
2025-11-19T00:56:02.966+05:30 2025-11-19T00:56:02+0530 STATUS_CHANGE active prev=inactive
2025-11-19T00:56:07.415+05:30 2025-11-19T00:56:02+0530 START detected pid=38264
No newer events at this moment. Auto retrying... Pause
Back to top
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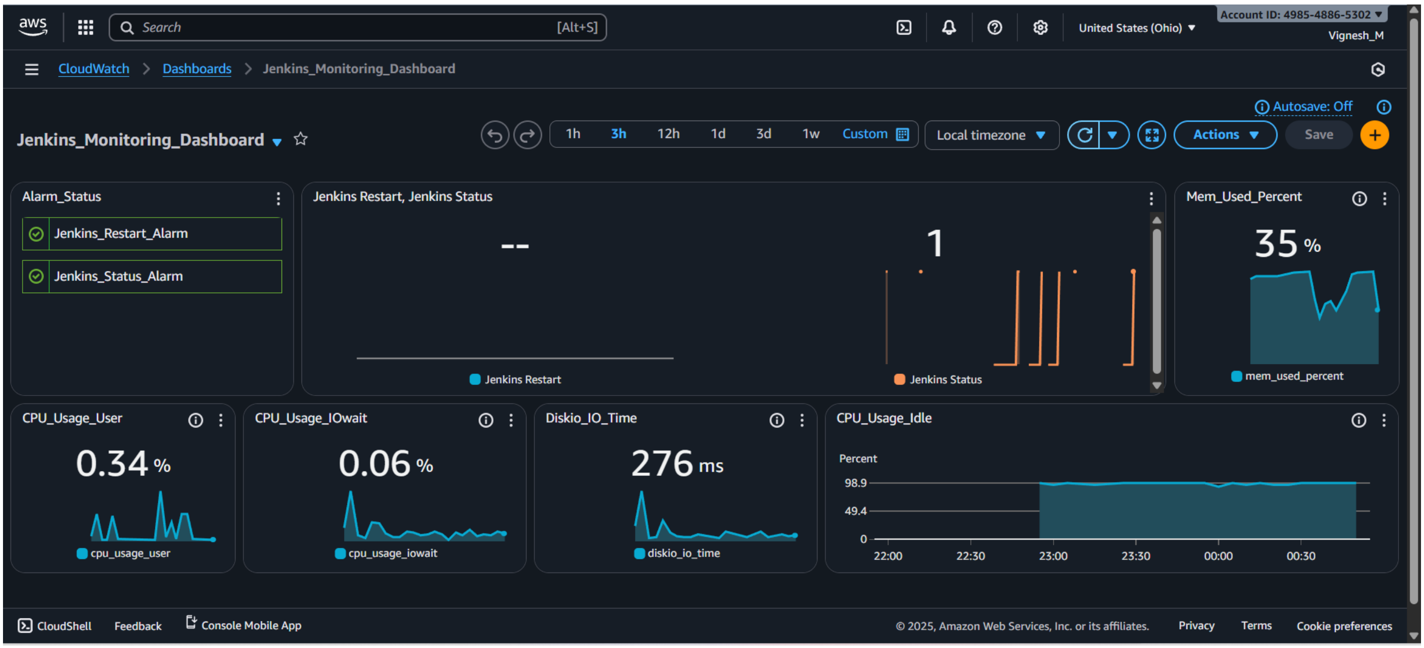

Search
[Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
CloudWatch
Dashboards
Jenkins_Monitoring_Dashboard
Autosave: Off
Jenkins_Monitoring_Dashboard
1h
3h
12h
1d
3d
1w
Custom
Local timezone
Actions
Save
Alarm_Status
Jenkins_Restart_Alarm
Jenkins_Status_Alarm
Jenkins Restart, Jenkins Status
--
1
Jenkins Restart
Jenkins Status
Mem_Used_Percent
35 %
mem_used_percent
CPU_Usage_User
0.34 %
cpu_usage_user
CPU_Usage_IOwait
0.06 %
cpu_usage_iowait
Diskio_IO_Time
276 ms
diskio_io_time
CPU_Usage_Idle
Percent
98.9
49.4
0
22:00
22:30
23:00
23:30
00:00
00:30
CloudShell
Feedback
Console Mobile App
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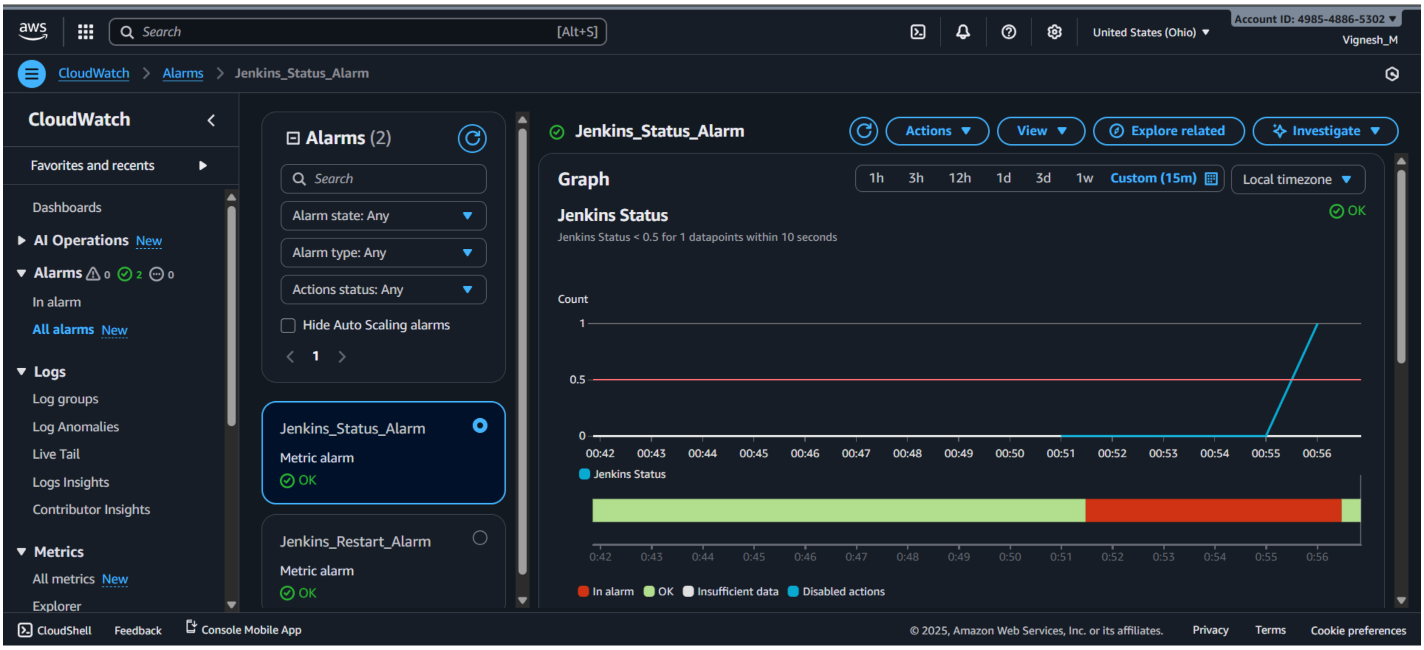

Search
[Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
CloudWatch
Alarms
Jenkins_Status_Alarm
CloudWatch
Favorites and recents
Dashboards
AI Operations New
Alarms 0 2 0
In alarm
All alarms New
Logs
Log groups
Log Anomalies
Live Tail
Logs Insights
Contributor Insights
Metrics
All metrics New
Explorer
Alarms (2)
Search
Alarm state: Any
Alarm type: Any
Actions status: Any
Hide Auto Scaling alarms
Jenkins_Status_Alarm
Metric alarm
OK
Jenkins_Restart_Alarm
Metric alarm
OK
Jenkins_Status_Alarm
Actions
View
Explore related
Investigate
Graph
1h
3h
12h
1d
3d
1w
Custom (15m)
Local timezone
Jenkins Status
OK
Jenkins Status < 0.5 for 1 datapoints within 10 seconds
Count
1
0.5
0
00:42
00:43
00:44
00:45
00:46
00:47
00:48
00:49
00:50
00:51
00:52
00:53
00:54
00:55
00:56
Jenkins Status
In alarm
OK
Insufficient data
Disabled actions
CloudShell
Feedback
Console Mobile App
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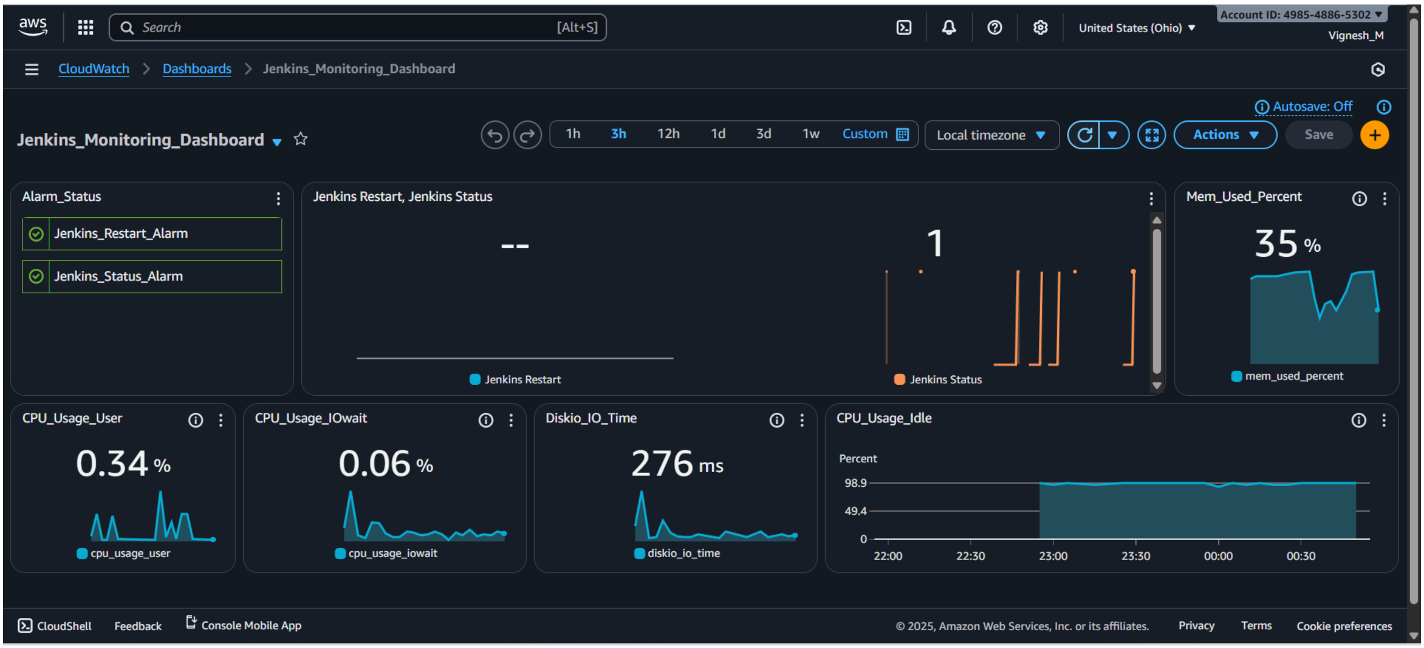
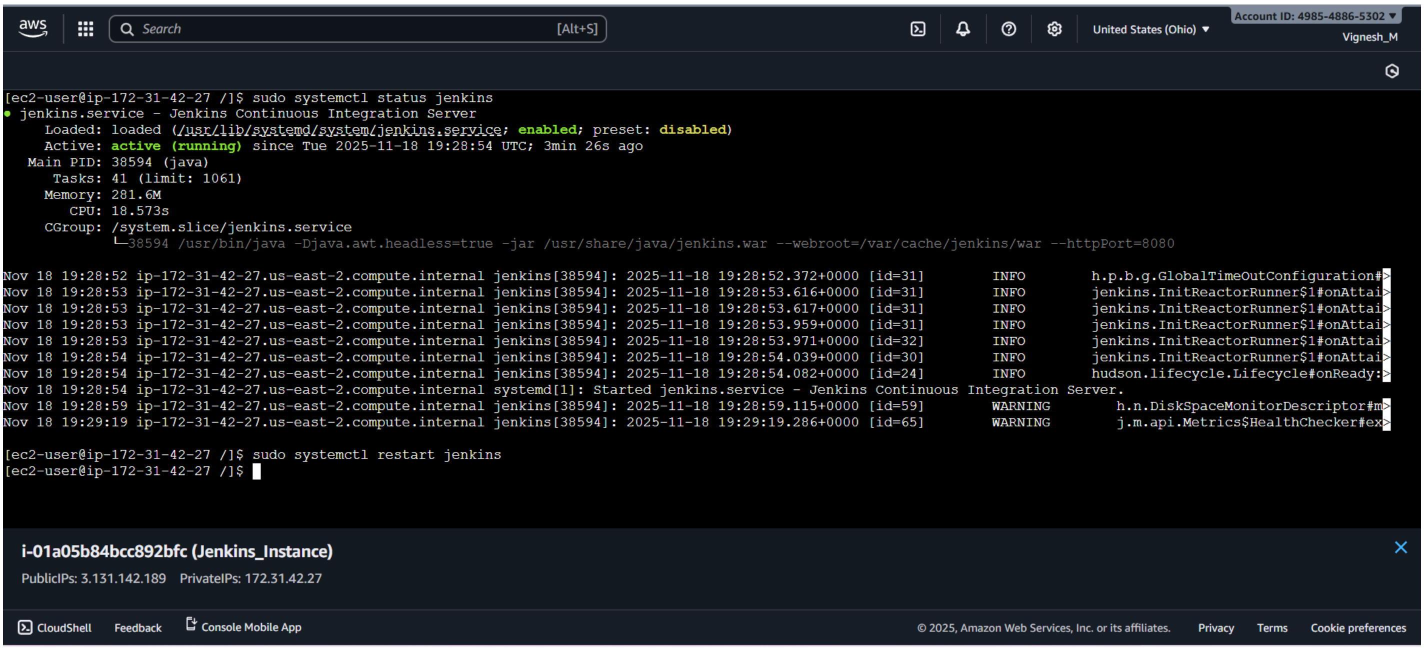
Step 9: Test Alarm Tigger by Manually Restarting Jenkins Server:

Procedure:

- In EC2 console check the Jenkins status and restart the Jenkins server.
  - **Command 1:** sudo systemctl status jenkins
  - **Command 2:** sudo systemctl restart Jenkins
- After restarting the Jenkins, after some time the web interface will go work.
- After one minute the cronjob will tigger and log will be sent to Jenkins_monitor.log
- Cloud watch agent will fetch that log and send it to the cloud watch console.
- After log comes to cloud watch, after sometimes Jenkins Restart metric will goes to 1.
- Jenkins_Restart_Alarm get triggered immediately and it will go to In Alarm state.
- We will receive the Jenkins_Restart_Alarm alert through Email by SNS.
- After 3 Evaluation periods (1 minute), if we did not receive any restart log then the Jenkins_Restart_Alarm will automatically go back to OK state.
- **Result:**
  - Jenkins server Restarted and Web interface came and able to access.
  - Cronjob triggered and log was fetched by the cloud watch agent correctly.
  - Jenkins Restart metrics changed, and Jenkins_Restart_Alarm triggered successfully.
  - Jenkins Restart Alert was sent to Email through SNS correctly.
  - After completion of evaluation periods, Jenkins_Restart_Alarm state went back to OK state successfully.





aws
Search
[Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
[ec2-user@ip-172-31-42-27 /]$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: disabled)
Active: active (running) since Tue 2025-11-18 19:28:54 UTC; 3min 26s ago
Main PID: 38594 (java)
Tasks: 41 (limit: 1061)
Memory: 281.6M
CPU: 18.573s
CGroup: /system.slice/jenkins.service
└─38594 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080
Nov 18 19:28:52 ip-172-31-42-27.us-east-2.compute.internal jenkins[38594]: 2025-11-18 19:28:52.372+0000 [id=31] INFO h.p.b.g.GlobalTimeOutConfiguration#
Nov 18 19:28:53 ip-172-31-42-27.us-east-2.compute.internal jenkins[38594]: 2025-11-18 19:28:53.616+0000 [id=31] INFO jenkins.InitReactorRunner$1#onAttai
Nov 18 19:28:53 ip-172-31-42-27.us-east-2.compute.internal jenkins[38594]: 2025-11-18 19:28:53.617+0000 [id=31] INFO jenkins.InitReactorRunner$1#onAttai
Nov 18 19:28:53 ip-172-31-42-27.us-east-2.compute.internal jenkins[38594]: 2025-11-18 19:28:53.959+0000 [id=31] INFO jenkins.InitReactorRunner$1#onAttai
Nov 18 19:28:53 ip-172-31-42-27.us-east-2.compute.internal jenkins[38594]: 2025-11-18 19:28:53.971+0000 [id=32] INFO jenkins.InitReactorRunner$1#onAttai
Nov 18 19:28:54 ip-172-31-42-27.us-east-2.compute.internal jenkins[38594]: 2025-11-18 19:28:54.039+0000 [id=30] INFO jenkins.InitReactorRunner$1#onAttai
Nov 18 19:28:54 ip-172-31-42-27.us-east-2.compute.internal jenkins[38594]: 2025-11-18 19:28:54.082+0000 [id=24] INFO hudson.lifecycle.Lifecycle#onReady:
Nov 18 19:28:54 ip-172-31-42-27.us-east-2.compute.internal systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Nov 18 19:28:59 ip-172-31-42-27.us-east-2.compute.internal jenkins[38594]: 2025-11-18 19:28:59.115+0000 [id=59] WARNING h.n.DiskSpaceMonitorDescriptor#m
Nov 18 19:29:19 ip-172-31-42-27.us-east-2.compute.internal jenkins[38594]: 2025-11-18 19:29:19.286+0000 [id=65] WARNING j.m.api.Metrics$HealthChecker#ex
[ec2-user@ip-172-31-42-27 /]$ sudo systemctl restart jenkins
[ec2-user@ip-172-31-42-27 /]$
i-01a05b84bcc892bfc (Jenkins_Instance)
PublicIPs: 3.131.142.189 PrivateIPs: 172.31.42.27
CloudShell
Feedback
Console Mobile App
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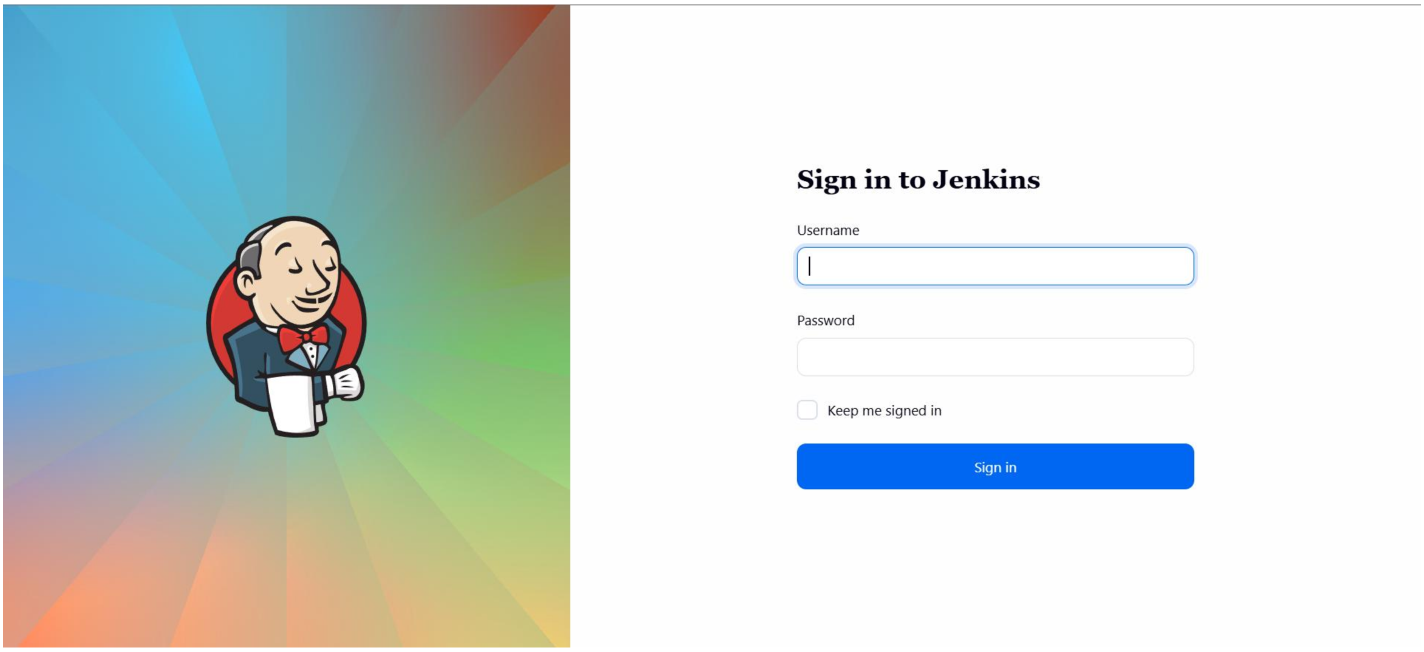
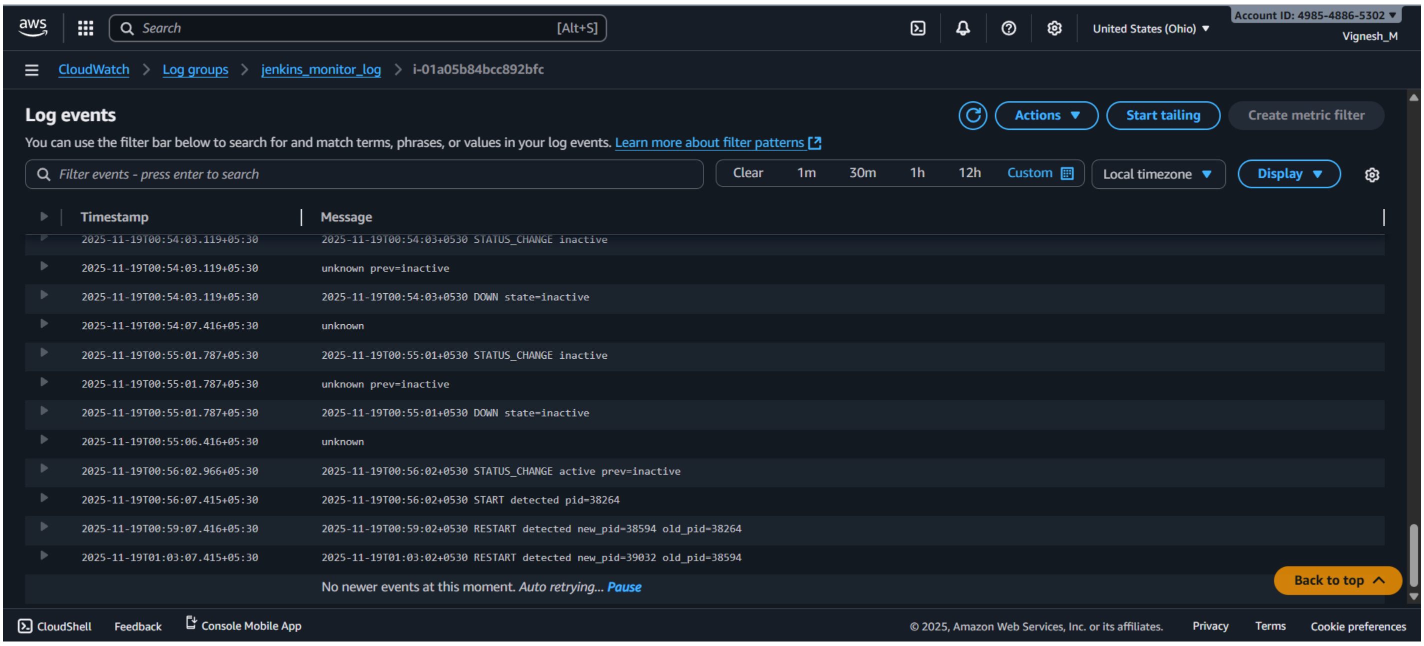
Privacy
Terms
Cookie preferences



Sign in to Jenkins
Username
Password
Keep me signed in
Sign in



aws
Search
[Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
CloudWatch > Log groups > jenkins_monitor_log > i-01a05b84bcc892bfc
Log events
Actions
Start tailing
Create metric filter
You can use the filter bar below to search for and match terms, phrases, or values in your log events. Learn more about filter patterns
Filter events - press enter to search
Clear 1m 30m 1h 12h Custom
Local timezone
Display
Timestamp
Message
2025-11-19T00:54:03.119+05:30 2025-11-19T00:54:03+0530 STATUS_CHANGE inactive
2025-11-19T00:54:03.119+05:30 unknown prev=inactive
2025-11-19T00:54:03.119+05:30 2025-11-19T00:54:03+0530 DOWN state=inactive
2025-11-19T00:54:07.416+05:30 unknown
2025-11-19T00:55:01.787+05:30 2025-11-19T00:55:01+0530 STATUS_CHANGE inactive
2025-11-19T00:55:01.787+05:30 unknown prev=inactive
2025-11-19T00:55:01.787+05:30 2025-11-19T00:55:01+0530 DOWN state=inactive
2025-11-19T00:55:06.416+05:30 unknown
2025-11-19T00:56:02.966+05:30 2025-11-19T00:56:02+0530 STATUS_CHANGE active prev=inactive
2025-11-19T00:56:07.415+05:30 2025-11-19T00:56:02+0530 START detected pid=38264
2025-11-19T00:59:07.416+05:30 2025-11-19T00:59:02+0530 RESTART detected new_pid=38594 old_pid=38264
2025-11-19T01:03:07.415+05:30 2025-11-19T01:03:02+0530 RESTART detected new_pid=39032 old_pid=38594
No newer events at this moment. Auto retrying... Pause
Back to top
CloudShell
Feedback
Console Mobile App
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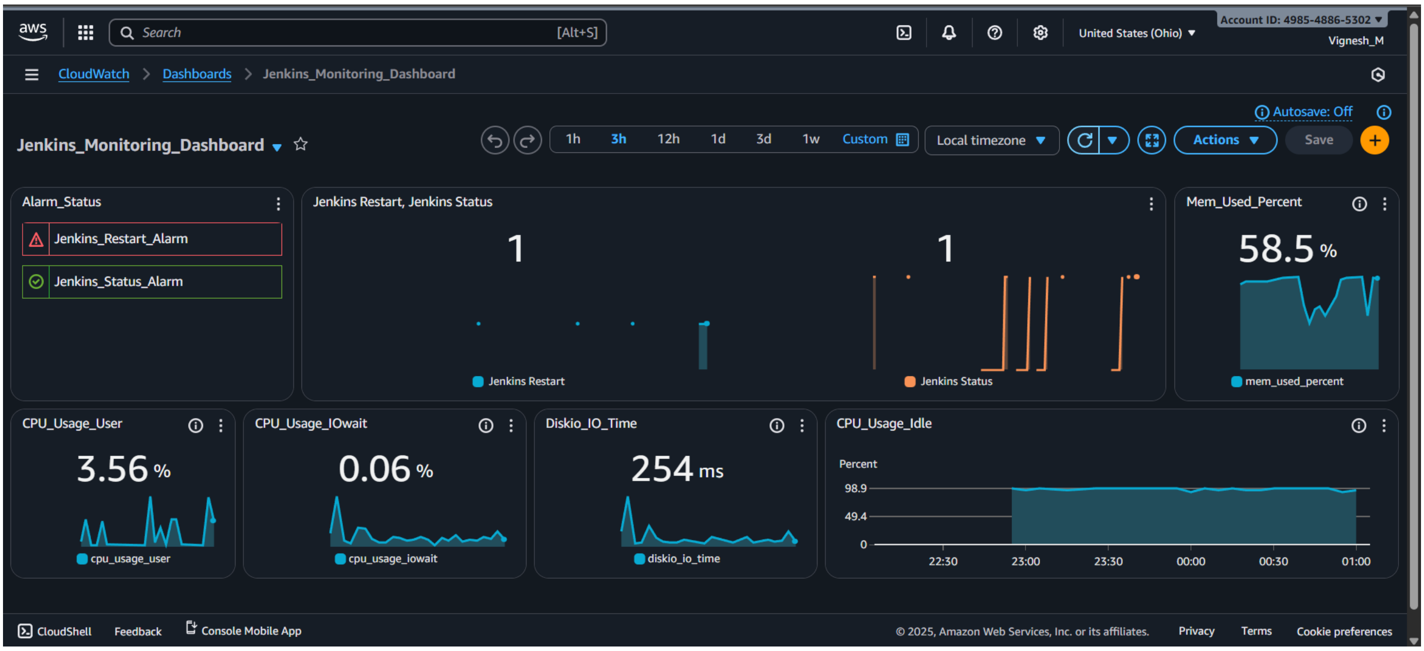
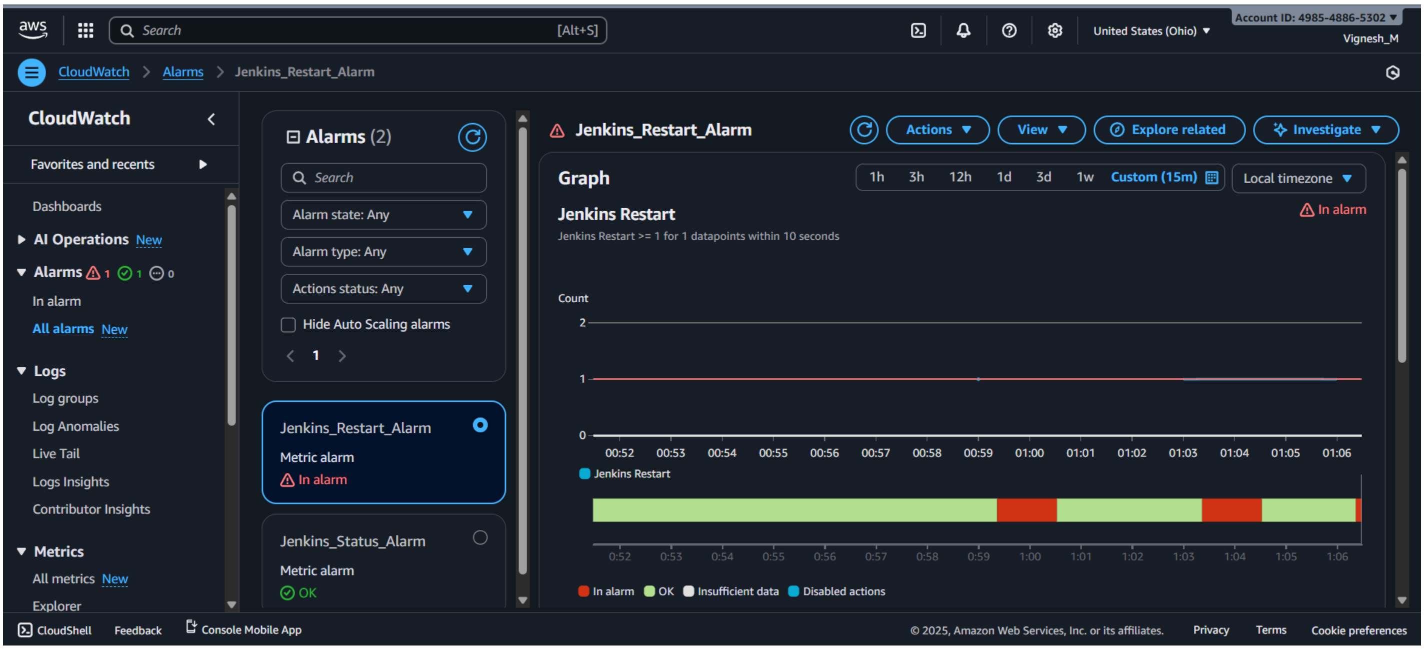
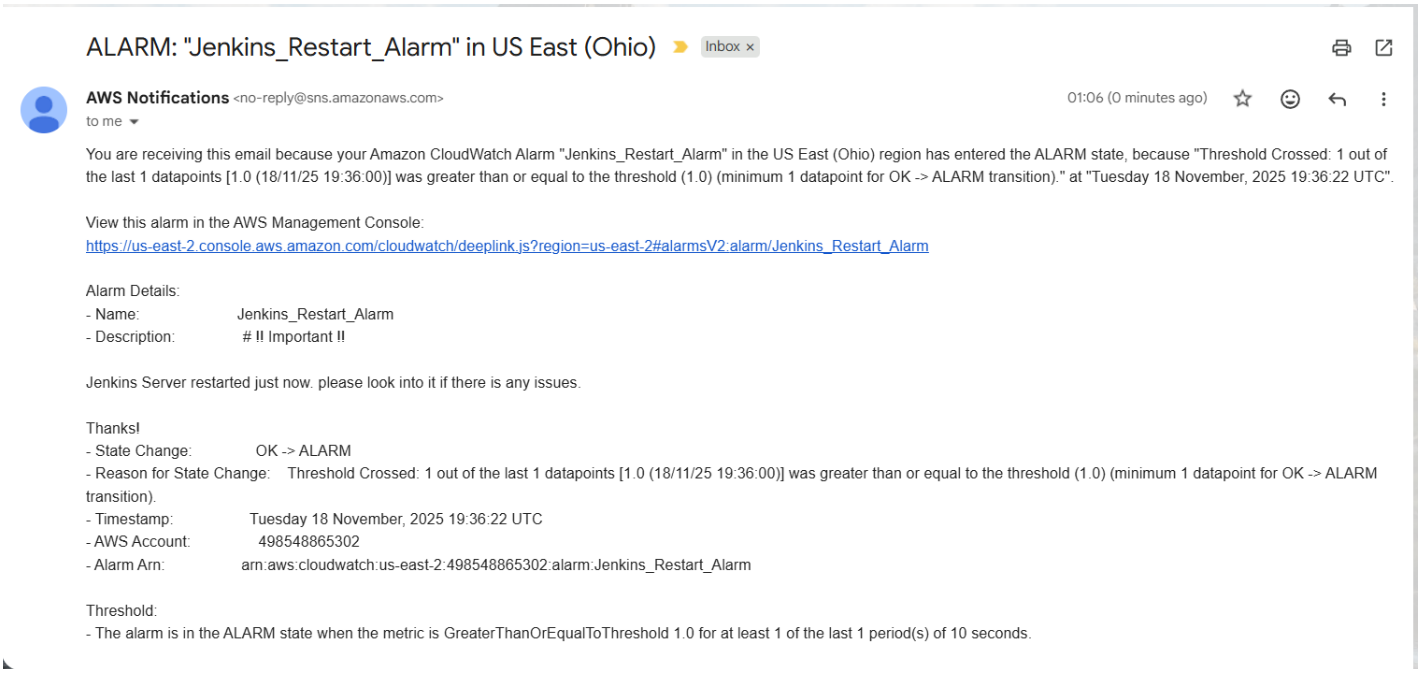
Privacy
Terms
Cookie preferences







ALARM: "Jenkins_Restart_Alarm" in US East (Ohio) Inbox

AWS Notifications <no-reply@sns.amazonaws.com> 01:06 (0 minutes ago)
to me

You are receiving this email because your Amazon CloudWatch Alarm "Jenkins_Restart_Alarm" in the US East (Ohio) region has entered the ALARM state, because "Threshold Crossed: 1 out of the last 1 datapoints [1.0 (18/11/25 19:36:00)] was greater than or equal to the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition)." at "Tuesday 18 November, 2025 19:36:22 UTC".

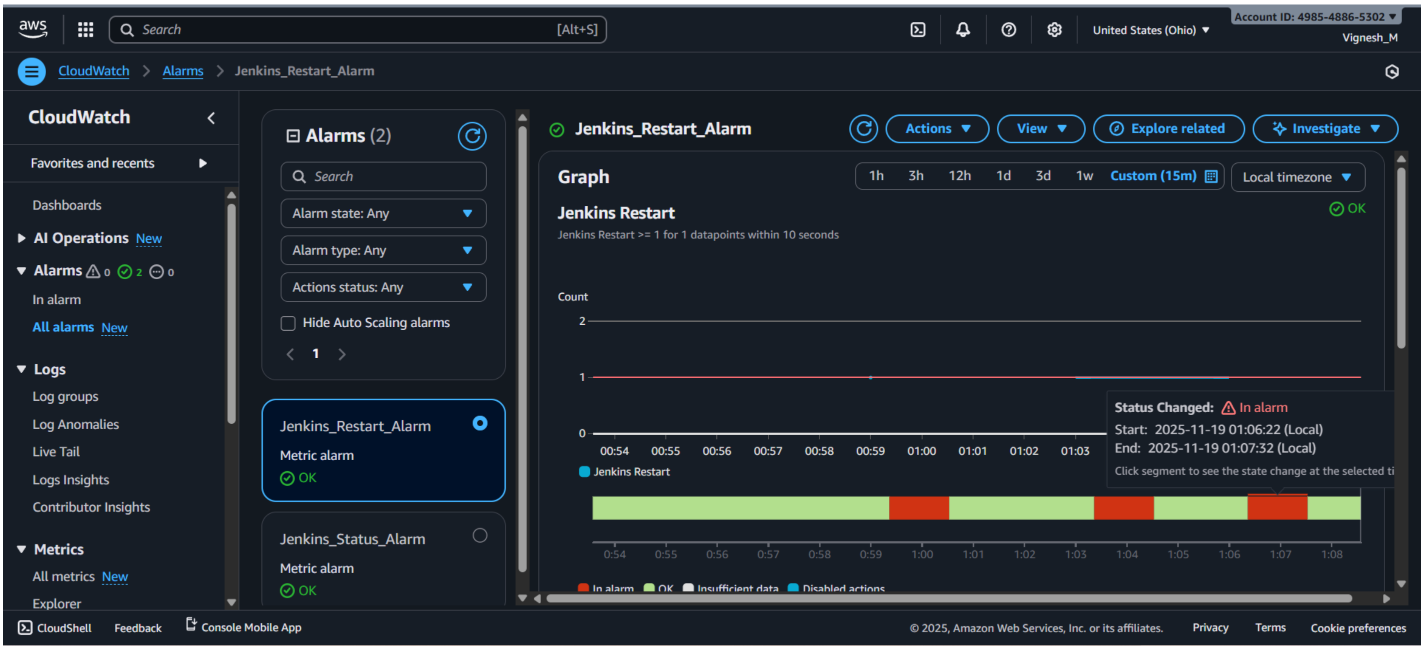
View this alarm in the AWS Management Console:
https://us-east-2.console.aws.amazon.com/cloudwatch/deeplink.js?region=us-east-2#alarmsV2:alarm/Jenkins_Restart_Alarm

Alarm Details:
- Name: Jenkins_Restart_Alarm
- Description: # !! Important !!

Jenkins Server restarted just now. please look into it if there is any issues.

Thanks!
- State Change: OK -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [1.0 (18/11/25 19:36:00)] was greater than or equal to the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition).
- Timestamp: Tuesday 18 November, 2025 19:36:22 UTC
- AWS Account: 498548865302
- Alarm Arn: arn:aws:cloudwatch:us-east-2:498548865302:alarm:Jenkins_Restart_Alarm

Threshold:
- The alarm is in the ALARM state when the metric is GreaterThanOrEqualToThreshold 1.0 for at least 1 of the last 1 period(s) of 10 seconds.



Search [Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
CloudWatch > Alarms > Jenkins_Restart_Alarm
CloudWatch
Favorites and recents
Dashboards
AI Operations New
Alarms 0 2 0
In alarm
All alarms New
Logs
Log groups
Log Anomalies
Live Tail
Logs Insights
Contributor Insights
Metrics
All metrics New
Explorer
Alarms (2)
Search
Alarm state: Any
Alarm type: Any
Actions status: Any
Hide Auto Scaling alarms
Jenkins_Restart_Alarm
Metric alarm
OK
Jenkins_Status_Alarm
Metric alarm
OK
Jenkins_Restart_Alarm
Actions
View
Explore related
Investigate
Graph
1h 3h 12h 1d 3d 1w Custom (15m)
Local timezone
Jenkins Restart
OK
Jenkins Restart >= 1 for 1 datapoints within 10 seconds
Count
Jenkins Restart
Status Changed: In alarm
Start: 2025-11-19 01:06:22 (Local)
End: 2025-11-19 01:07:32 (Local)
Click segment to see the state change at the selected ti
In alarm
OK
Insufficient data
Disabled actions
CloudShell
Feedback
Console Mobile App
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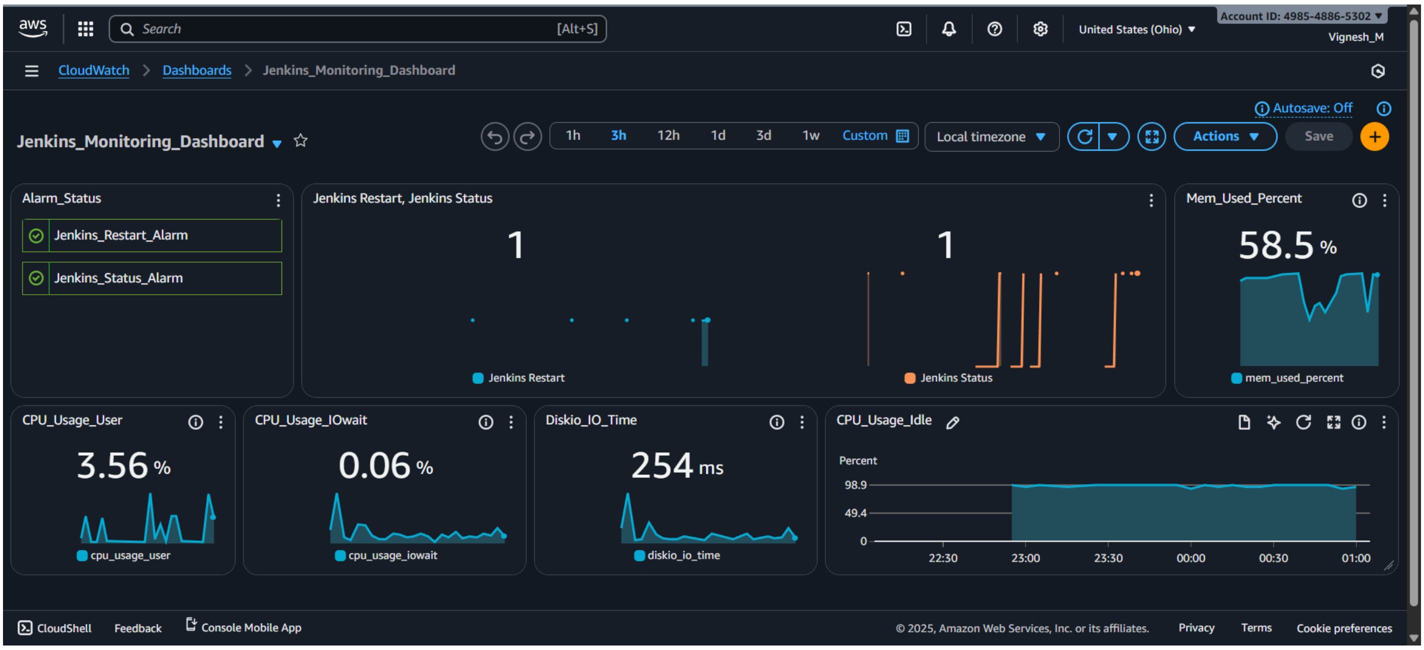
Terms
Cookie preferences



Search [Alt+S]
United States (Ohio)
Account ID: 4985-4886-5302
Vignesh_M
CloudWatch > Dashboards > Jenkins_Monitoring_Dashboard
Autosave: Off
Jenkins_Monitoring_Dashboard
1h 3h 12h 1d 3d 1w Custom
Local timezone
Actions
Save
Alarm_Status
Jenkins_Restart_Alarm
Jenkins_Status_Alarm
Jenkins Restart, Jenkins Status
1
1
Jenkins Restart
Jenkins Status
Mem_Used_Percent
58.5 %
mem_used_percent
CPU_Usage_User
3.56 %
cpu_usage_user
CPU_Usage_IOwait
0.06 %
cpu_usage_iowait
Diskio_IO_Time
254 ms
diskio_io_time
CPU_Usage_Idle
Percent
98.9
49.4
0
22:30 23:00 23:30 00:00 00:30 01:00
CloudShell
Feedback
Console Mobile App
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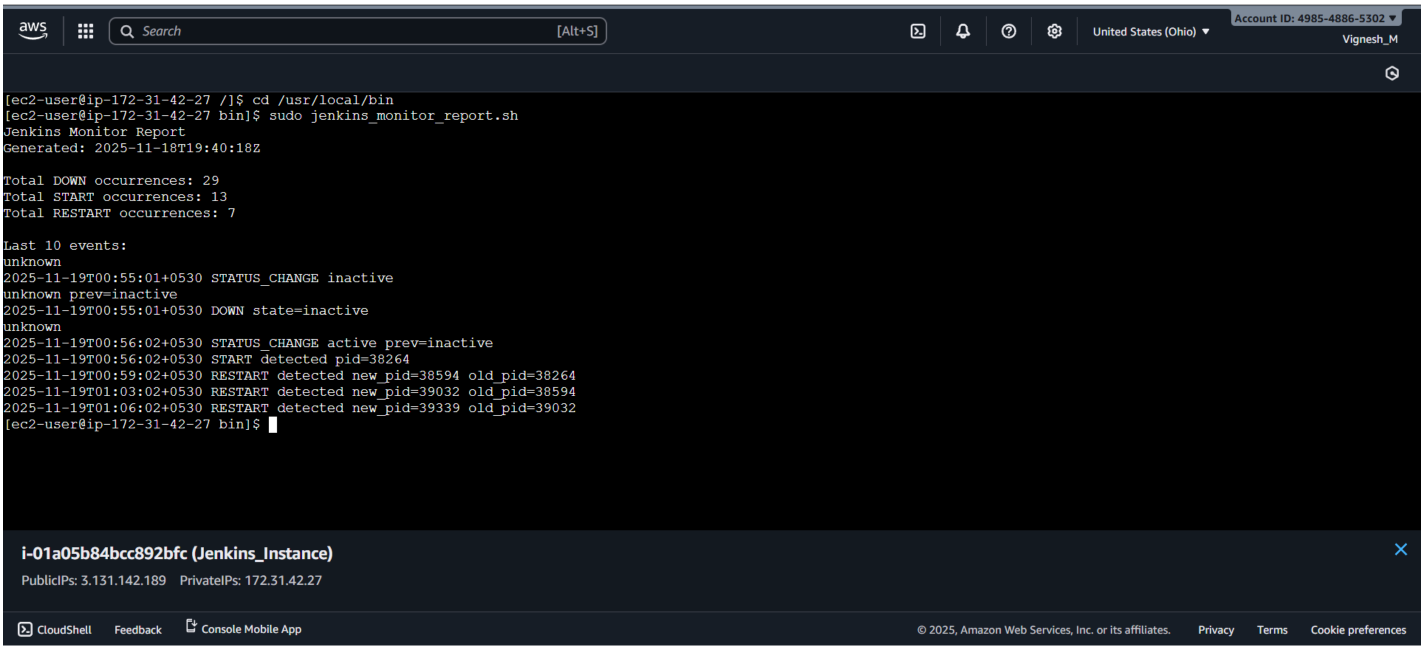
Step 10: Run jenkins_monitor_report.sh to Generate Final Alert Report:

Procedure:

- Now everything was successfully installed and configured to monitor the Jenkins status (Up, Down, Restart)
- Now we want the final report of number of occurrences of Jenkins Up, Jenkins Down and Jenkins restart. For that we have created a script (Jenkins_monitor_report.sh) and we are going to run that script now.
    - **Command:** sudo /usr/local/bin/Jenkins_monitor_report.sh
- After running that script we will get the final report.
- **Result:**
    - jenkins_monitor_report.sh script file was created successfully.
    - jenkins_monitor_report.sh file was executed, and report was generated successfully.



Final Use Case Outcome:

- **Continuous Service Uptime Verification:** Automated monitoring is in place, verifying the real-time operational status of the Jenkins service running on the EC2 instance, ensuring immediate detection of any Downtime or unexpected restarts.
- **Instant Alerting on Failure:** CloudWatch Alarms are configured to trigger automatically based on the monitored Jenkins status metric, initiating near-instantaneous notifications via the configured SNS Topic (Email/SMS) when the service goes DOWN or restarts.
- **Auditable and Traceable Event Logging:** A structured, timestamped log file is maintained on the EC2 instance, creating an unalterable record of all Jenkins UP and DOWN events, which is periodically parsed to generate reports for audit and compliance.
- **Comprehensive Operational Reporting:** The system provides the capability to generate detailed downtime/uptime reports by parsing the log data. This helps operational teams quantify the frequency and duration of service interruptions, supporting RCA and SLA verification.
- **Enhanced System Reliability:** The combination of monitoring, logging, and instant alerting has significantly reduced Mean Time To Detect (MTTD) service failures, enable a faster response and contribute to overall increased reliability and availability of the critical Jenkins CI/CD platform.